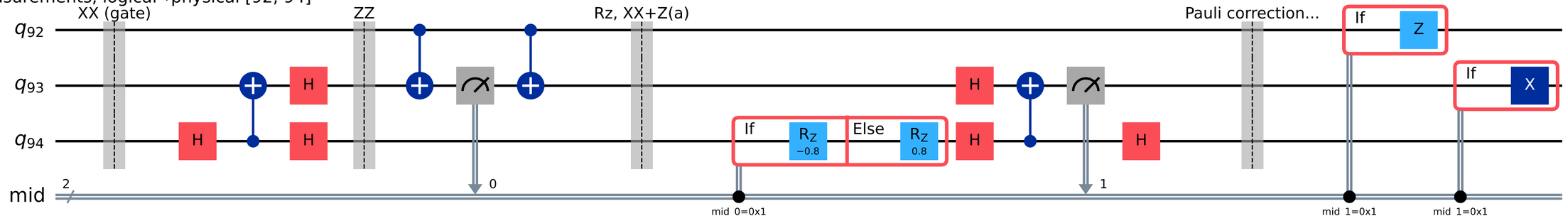
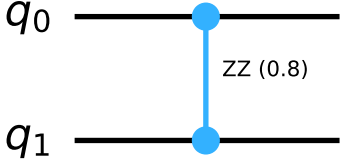
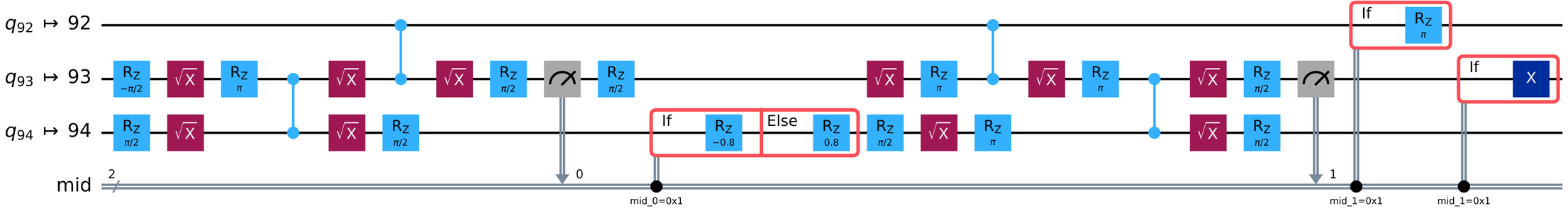
[Z2] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 (2 logical qubits)
2q depth 4, 2 mid-circuit measurements, logical→physical [92, 94]



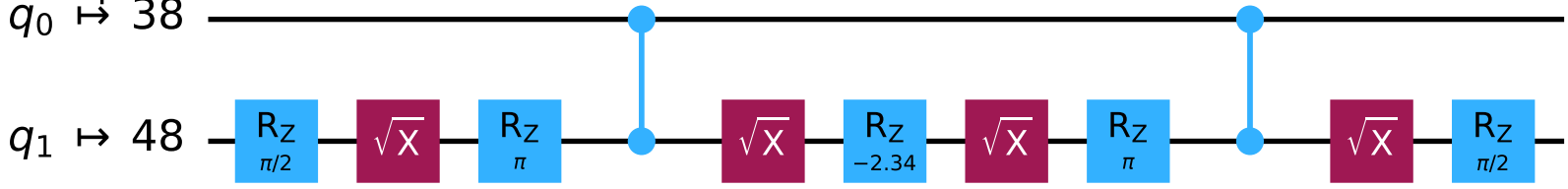
[Z2] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 1

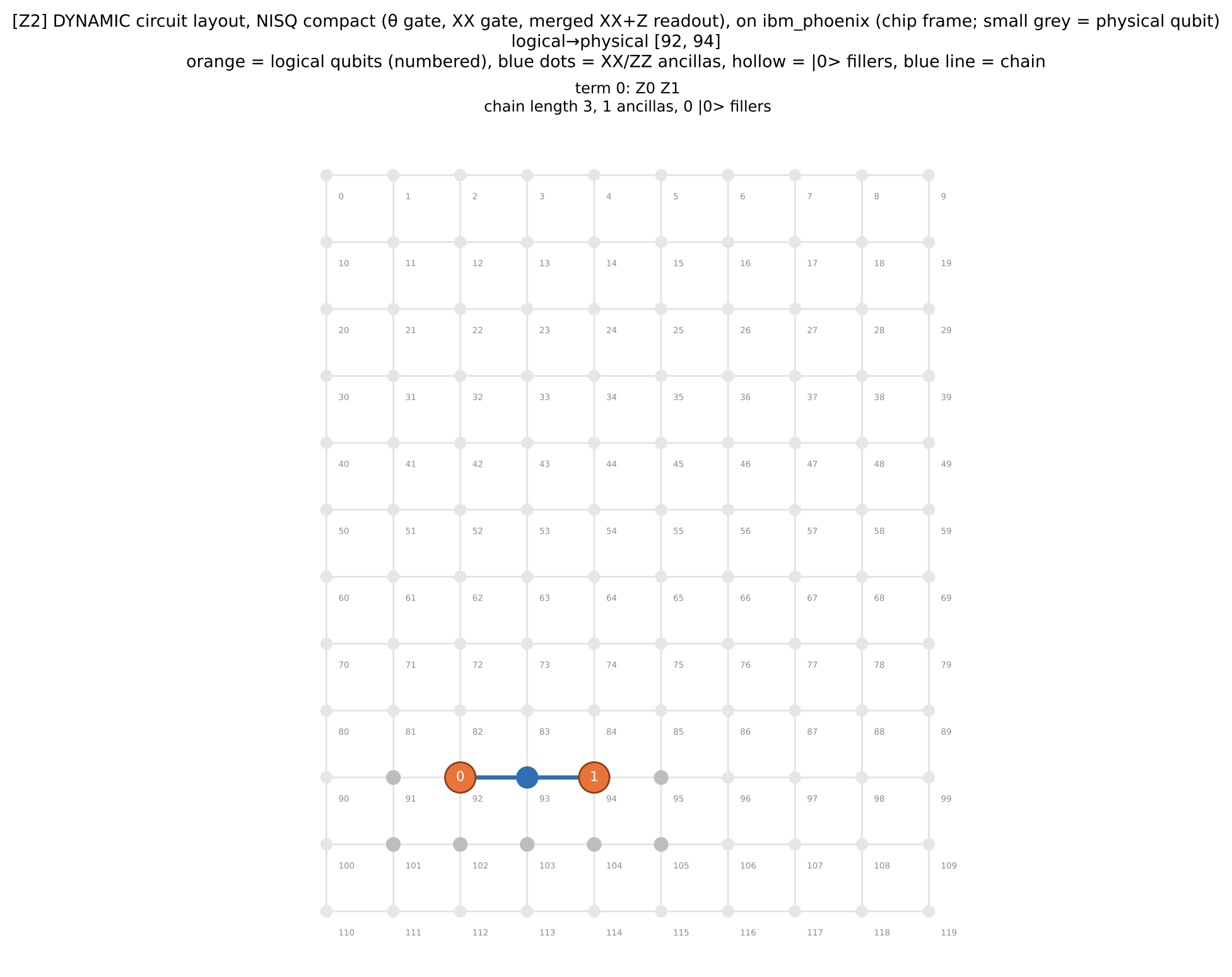


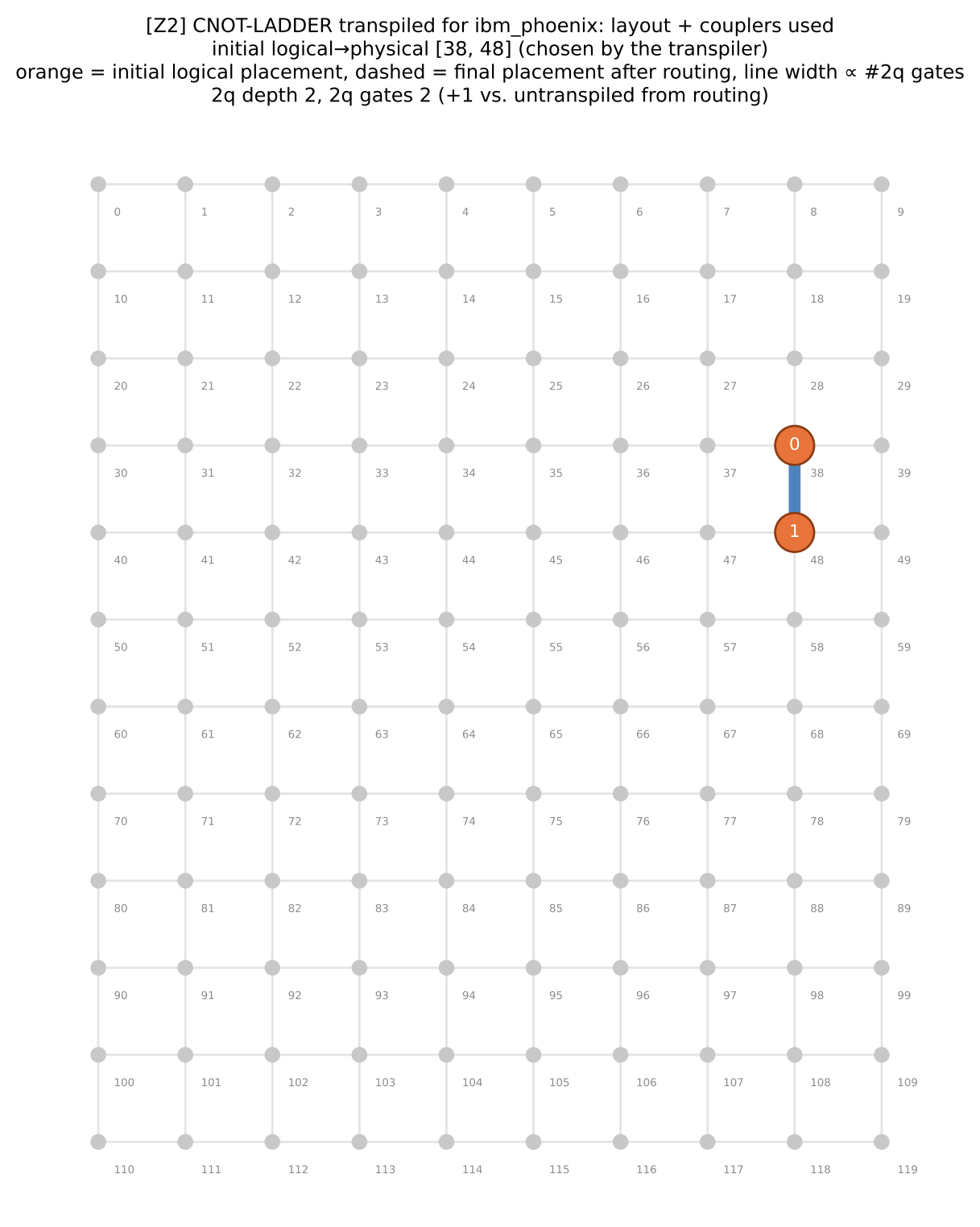
[Z2] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: rz:12 sx:10 cz:4 if_else:3 measure:2)
exp(-i θ P), θ=0.4: P = Z0 Z1 (2 logical qubits)
2q depth 4, total depth 21



[Z2] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11




[Z2] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [92, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1


[Z2] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+1 vs. untranspiled from routing)
0
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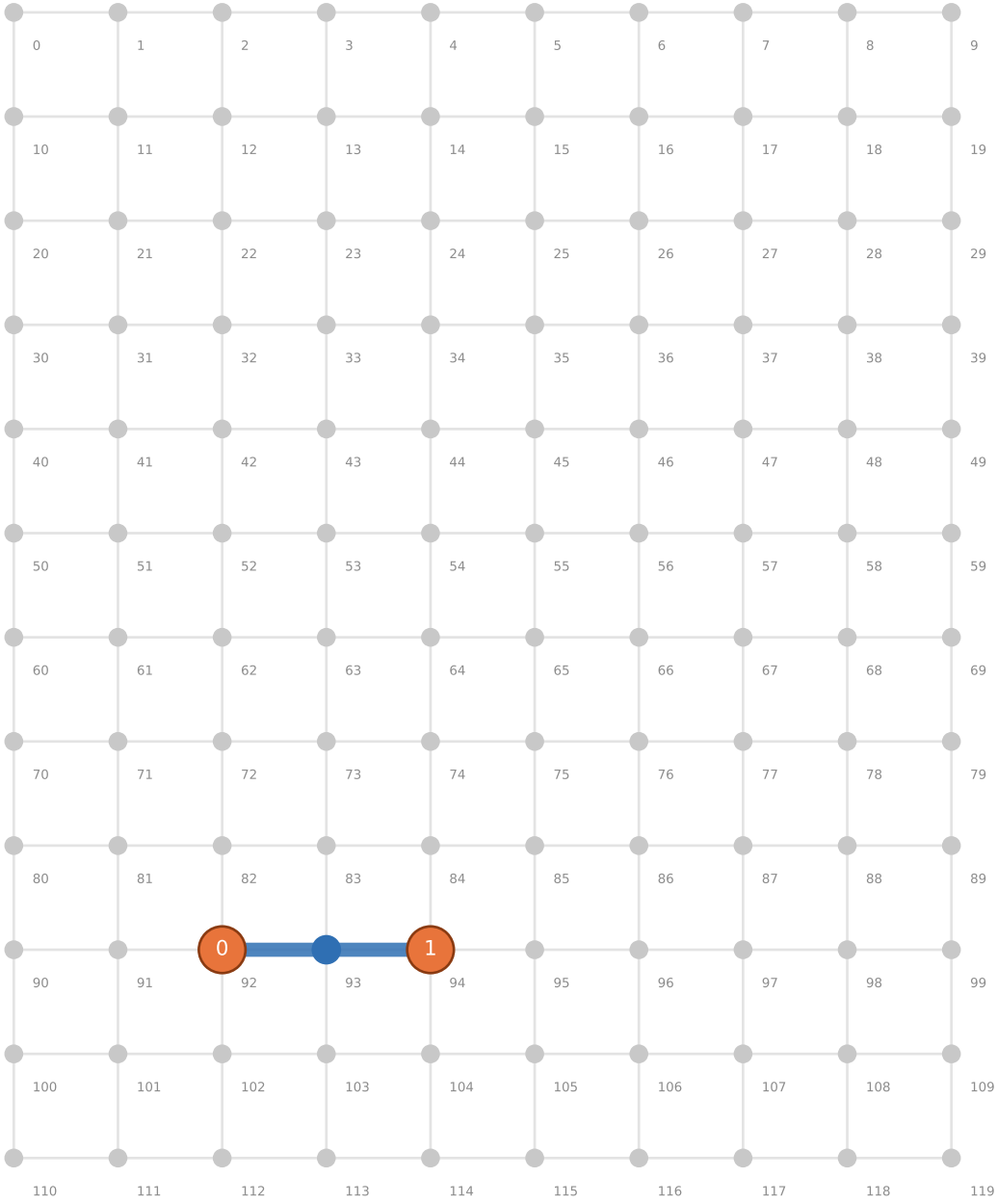
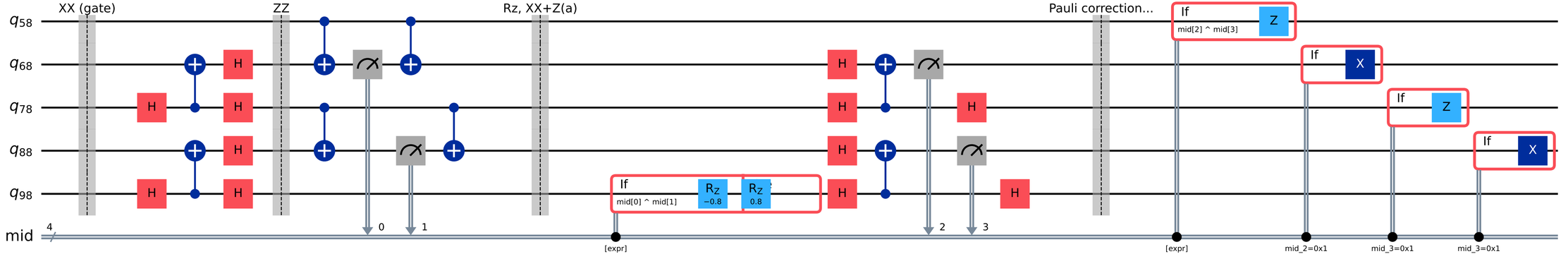
116
117
118
119
0
1

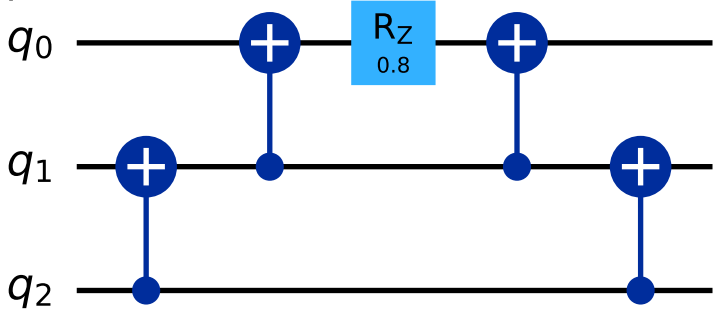
[Z2] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [92, 94]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 4



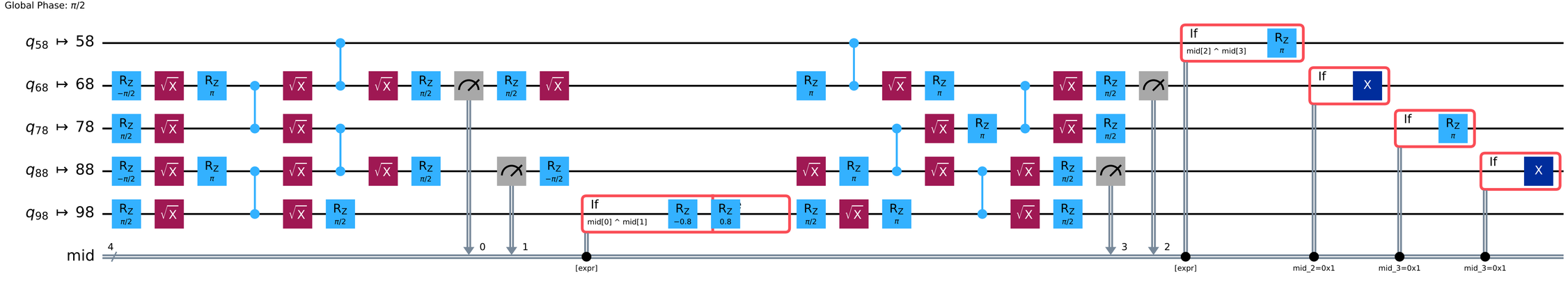
[Z3] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4, 4 mid-circuit measurements, logical→physical [58, 78, 98]



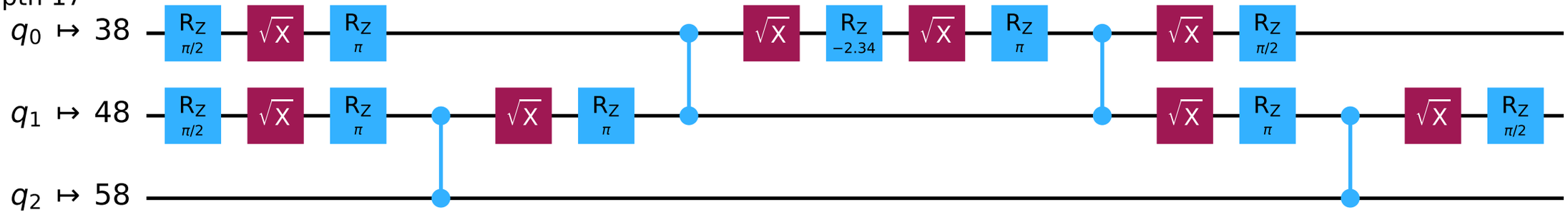
[Z3] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



[Z3] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: rz:21 sx:20 cz:8 if_else:5 measure:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 4, total depth 22

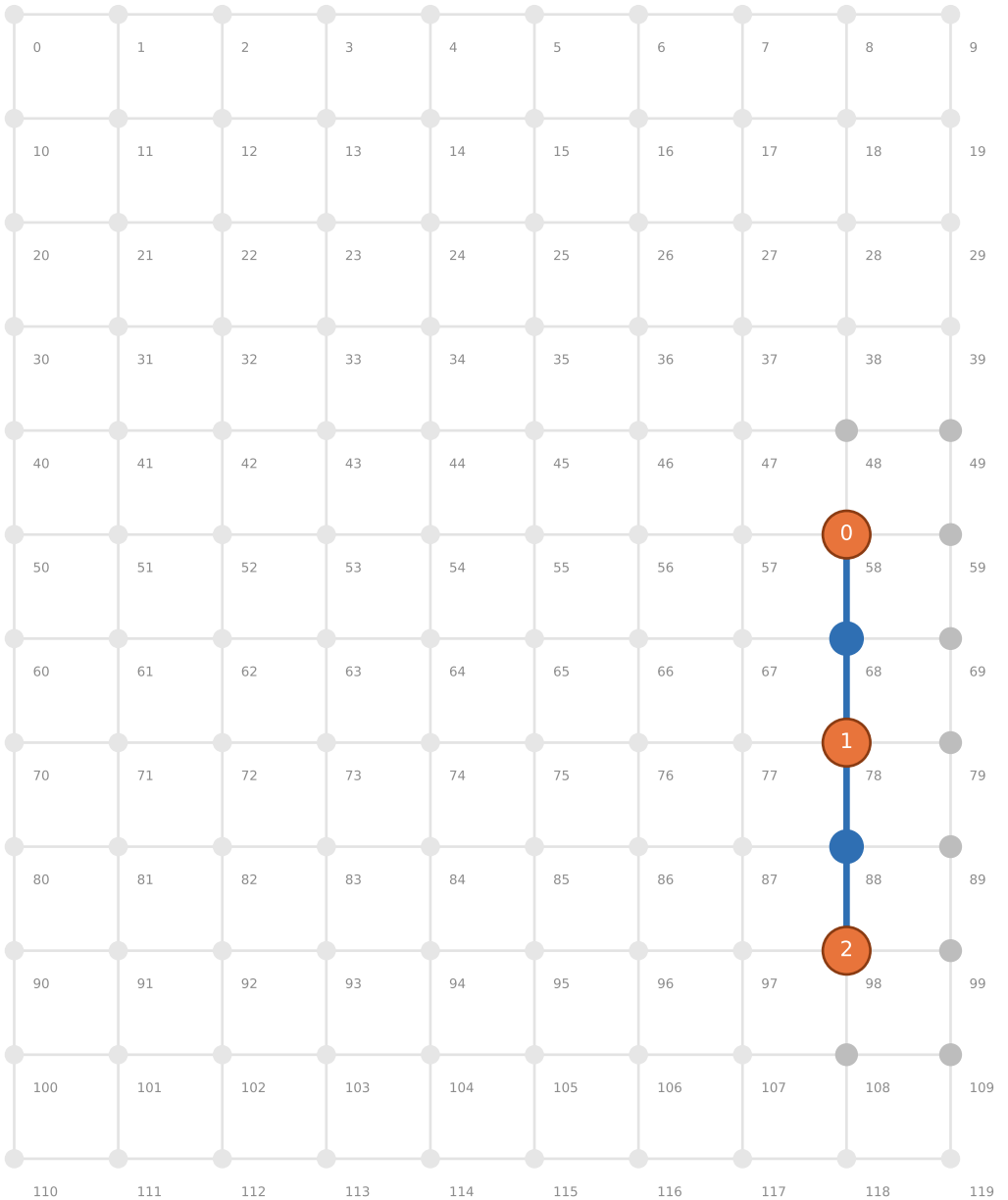


[Z3] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4, total depth 17

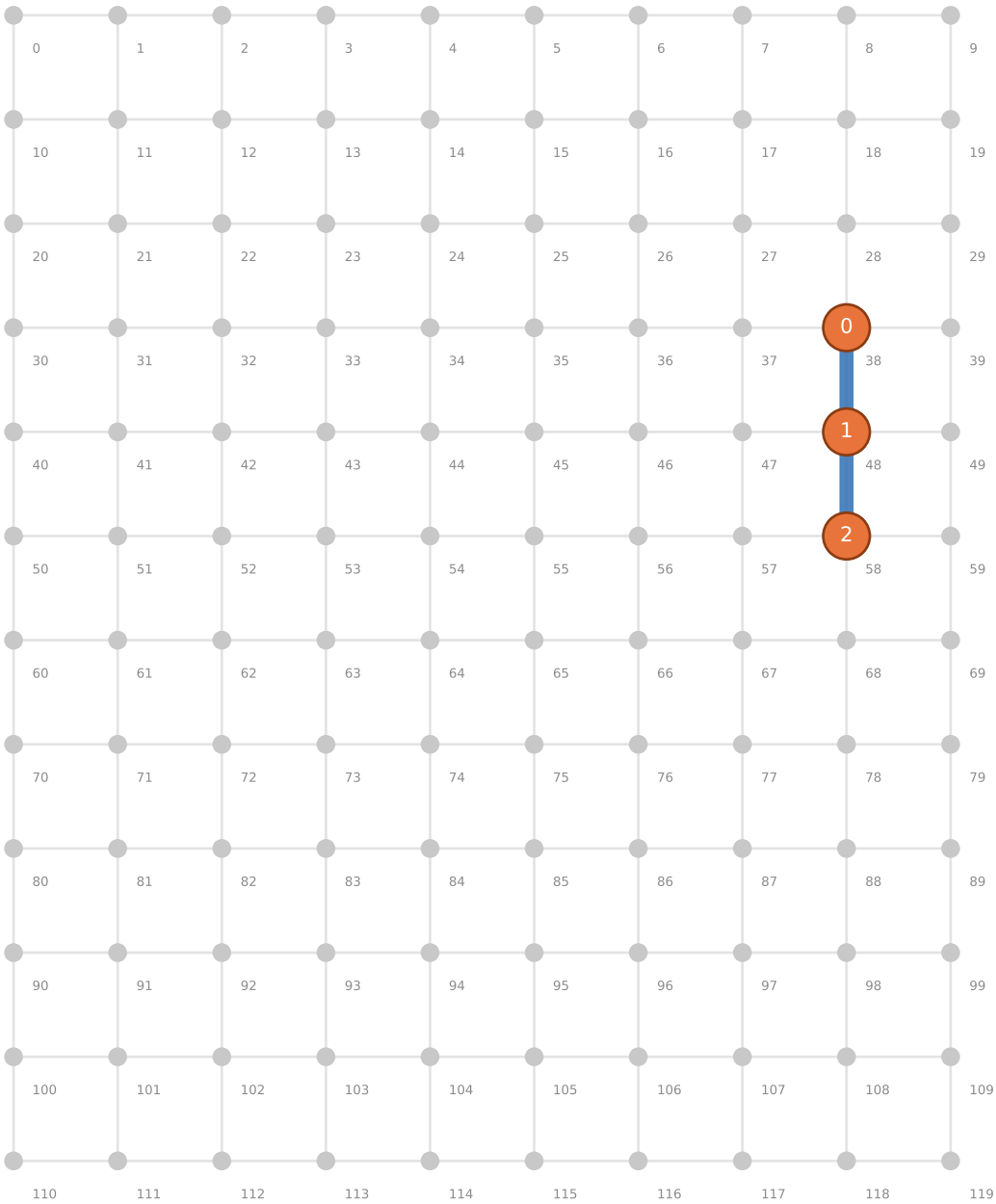


[Z3] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

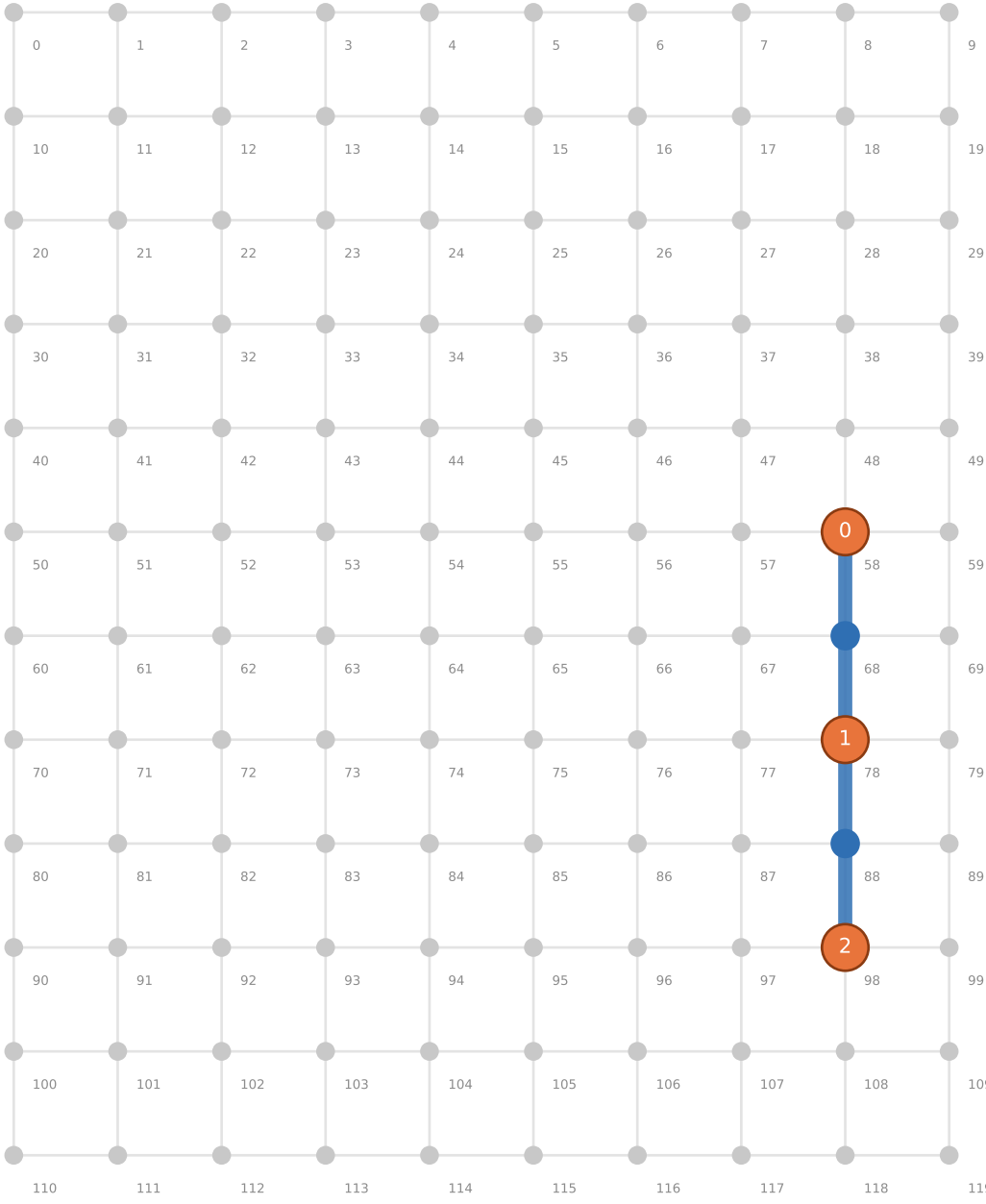
term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers



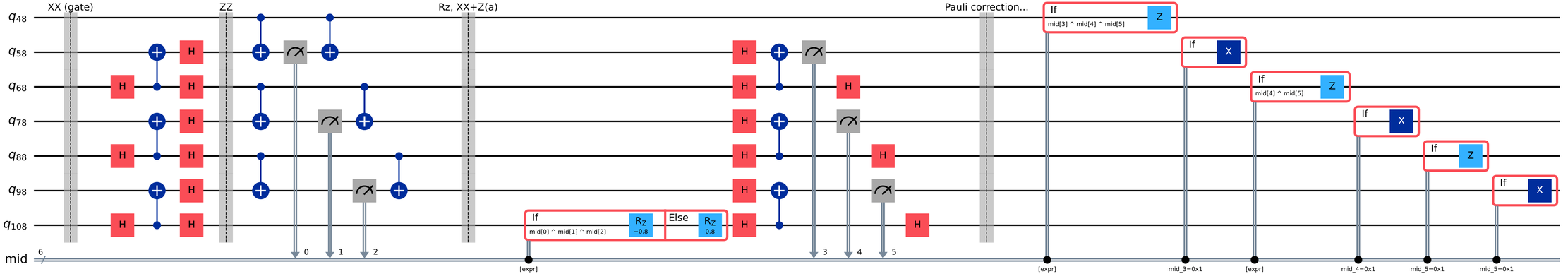
[Z3] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)



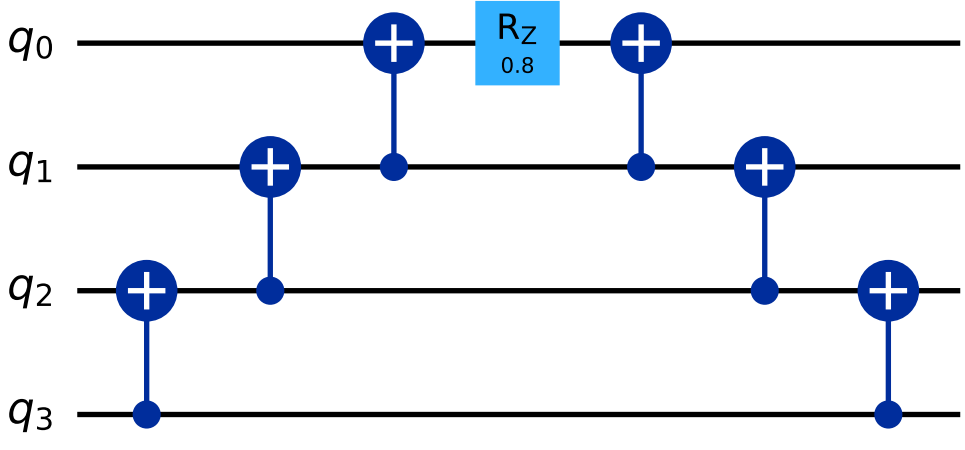
[Z3] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 8



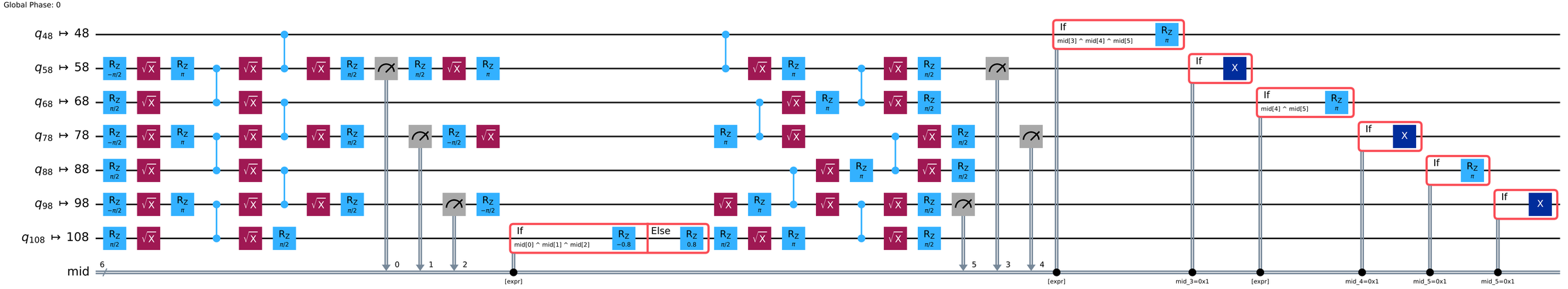
[Z4] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 4, 6 mid-circuit measurements, logical→physical [48, 68, 88, 108]



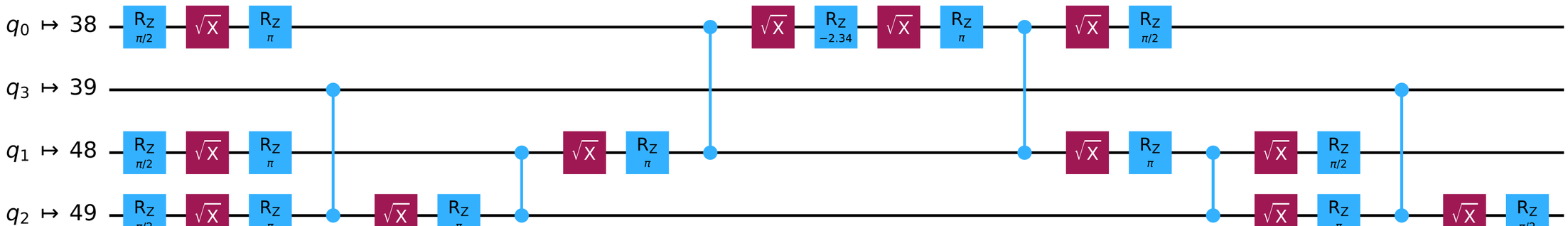
[Z4] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6

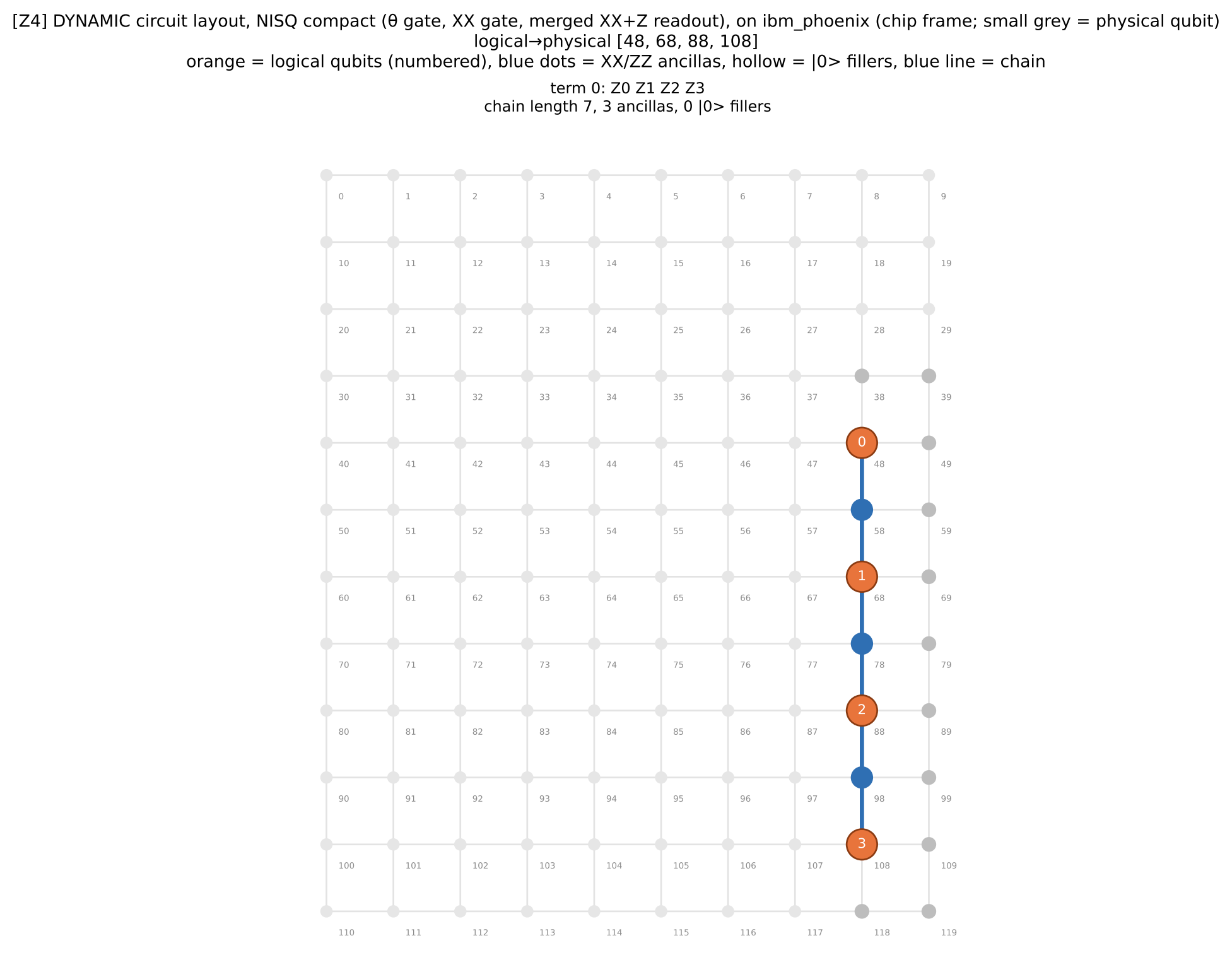


[Z4] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: rz:30 sx:30 cz:12 if_else:7 measure:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 4, total depth 23



[Z4] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23




[Z4] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
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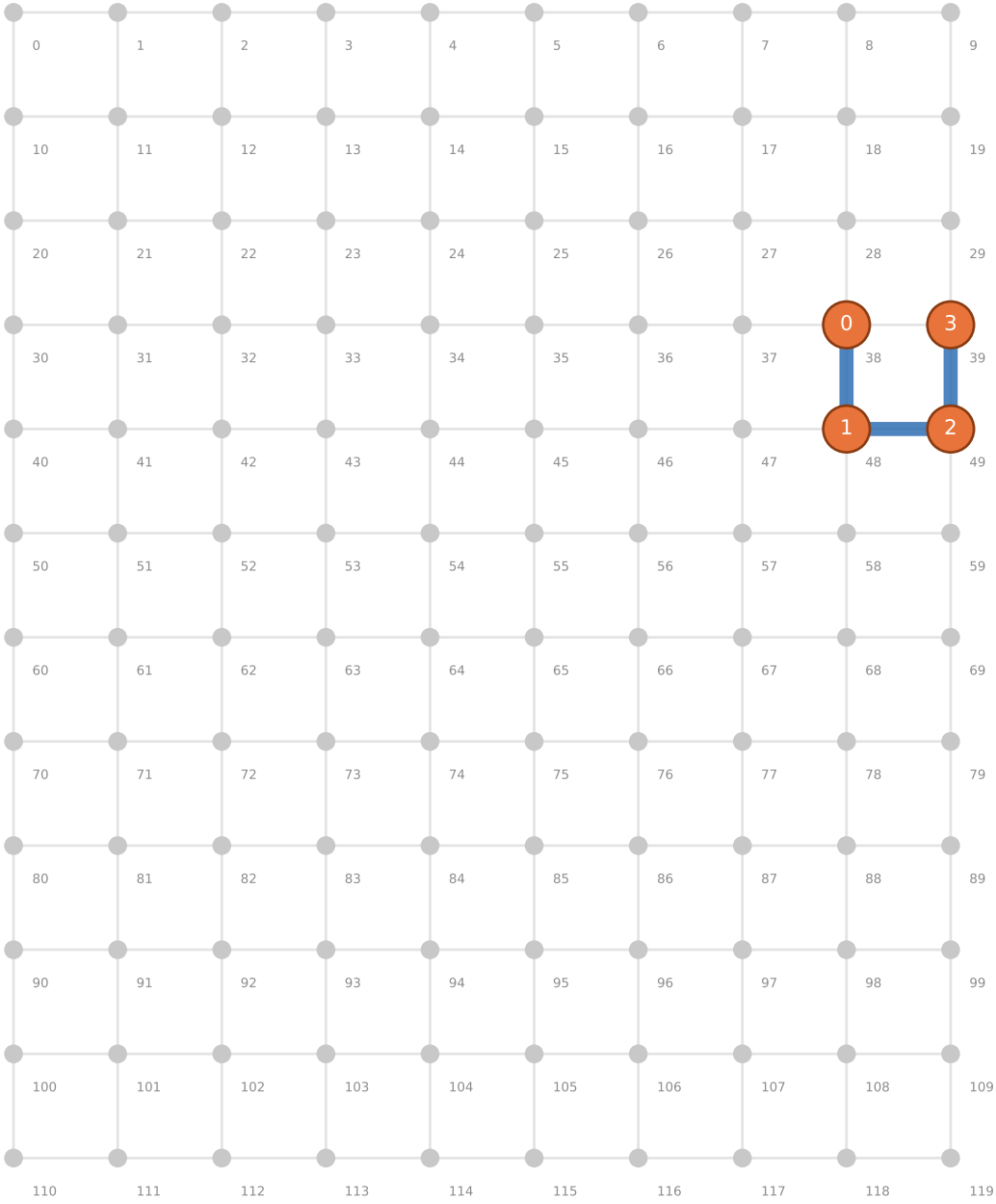
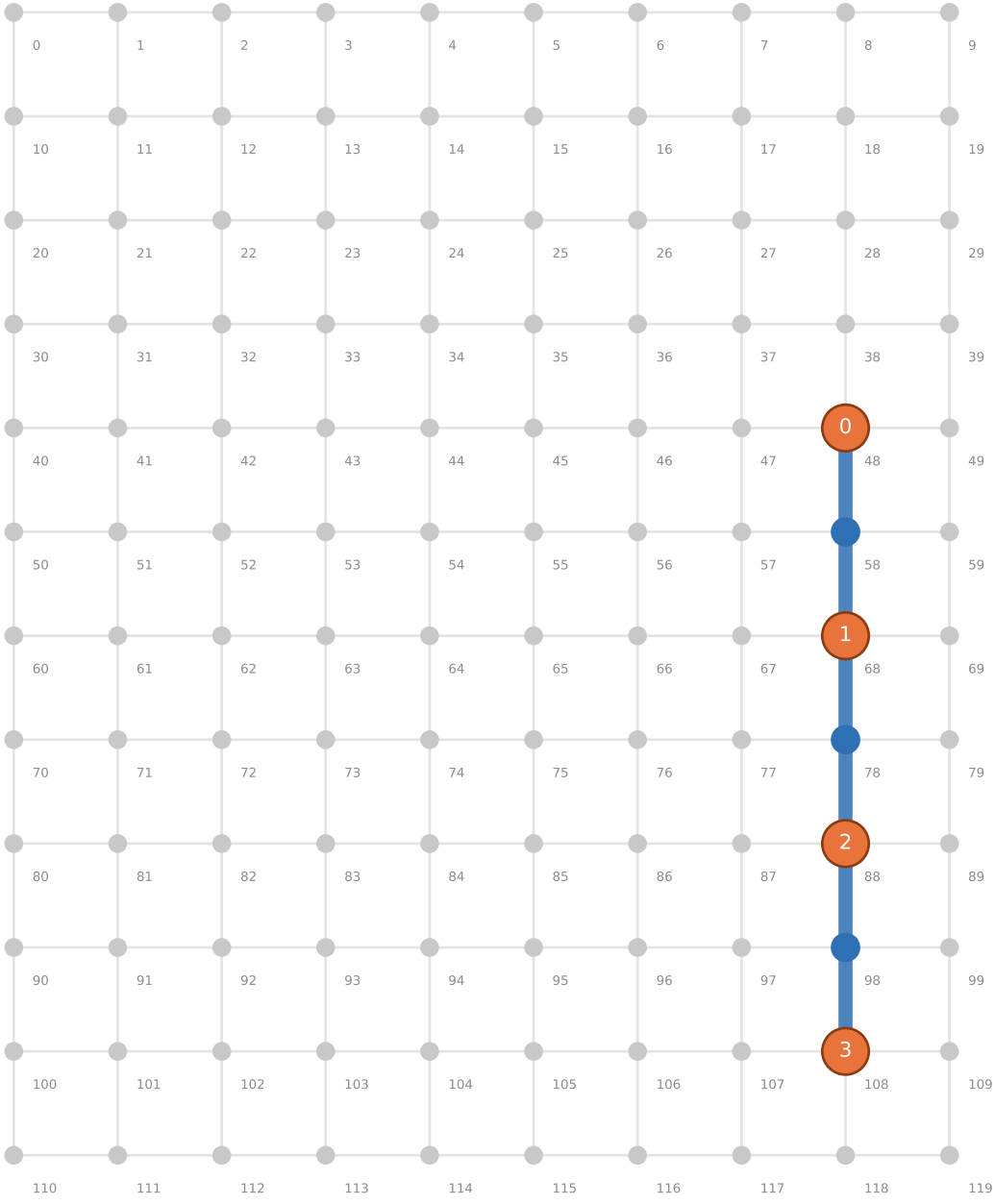
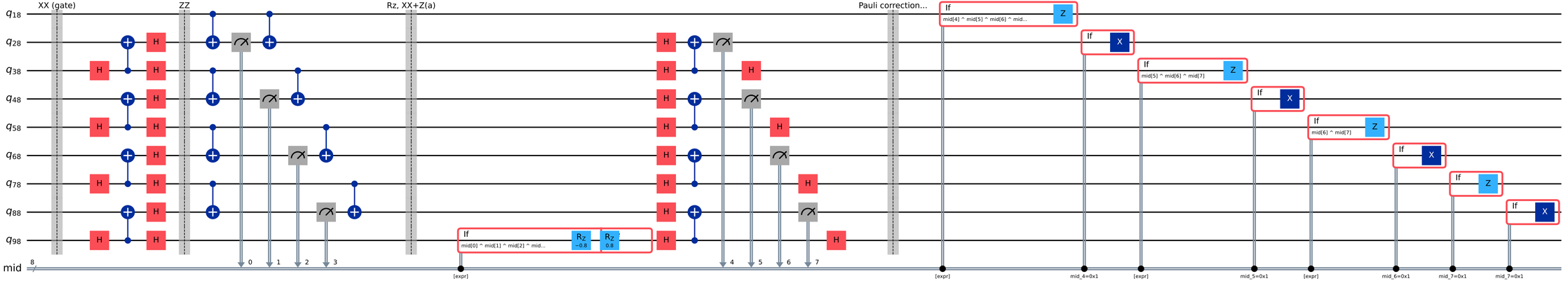
[Z4] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 49, 39] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)



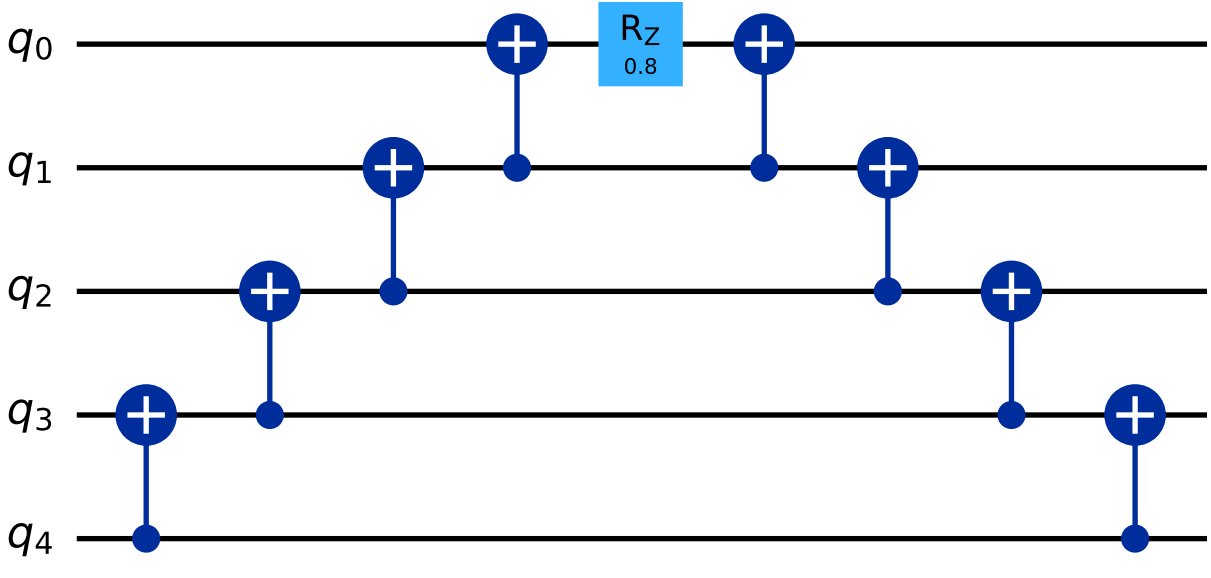
[Z4] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 12



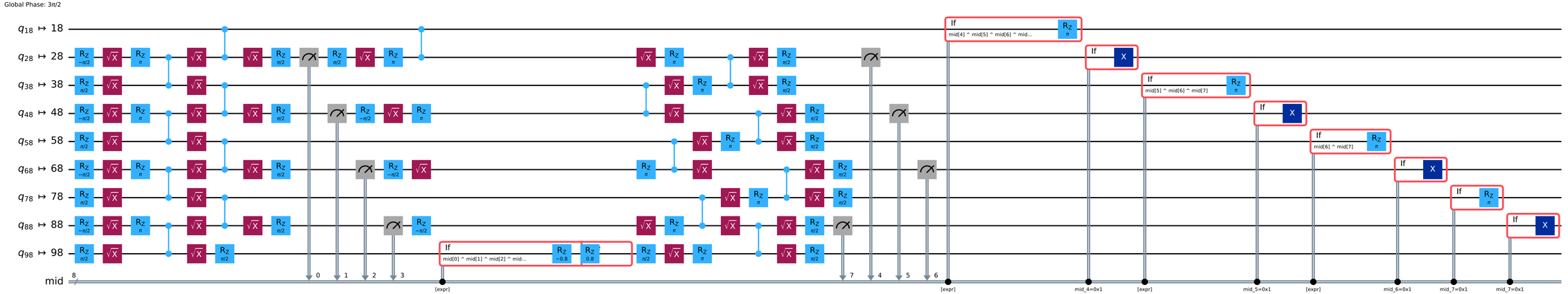
[Z5] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 4, 8 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



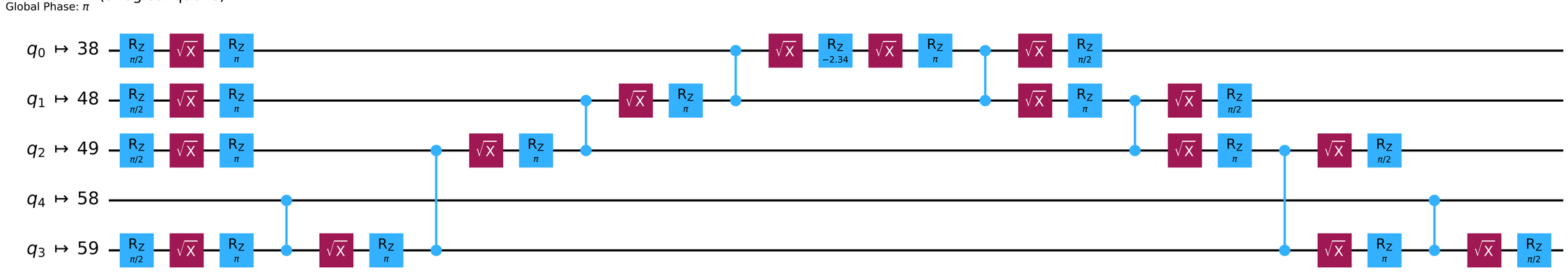
[Z5] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8

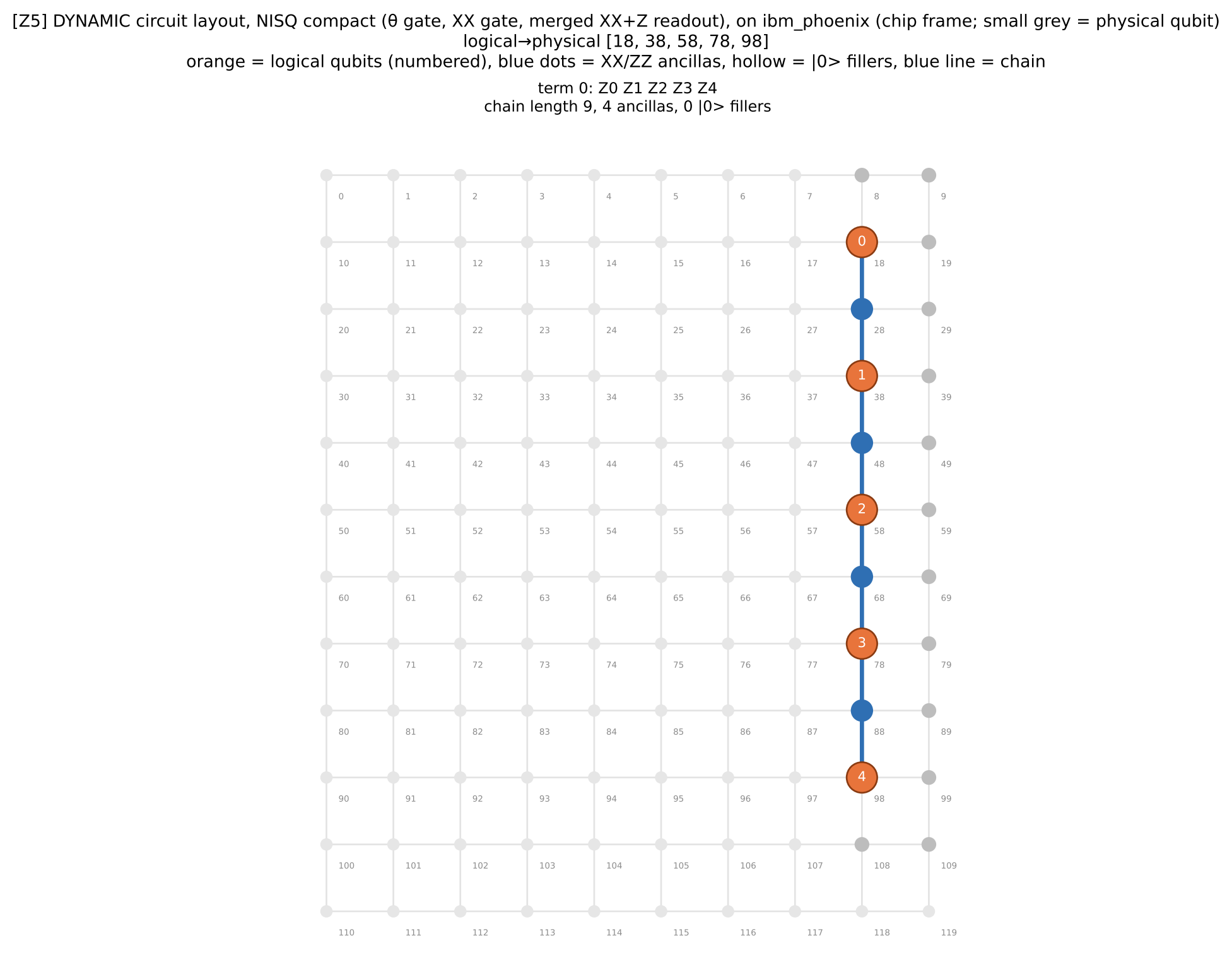


[Z5] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:40 rz:39 cz:16 if_else:9 measure:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 4, total depth 24



[Z5] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29




[Z5] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
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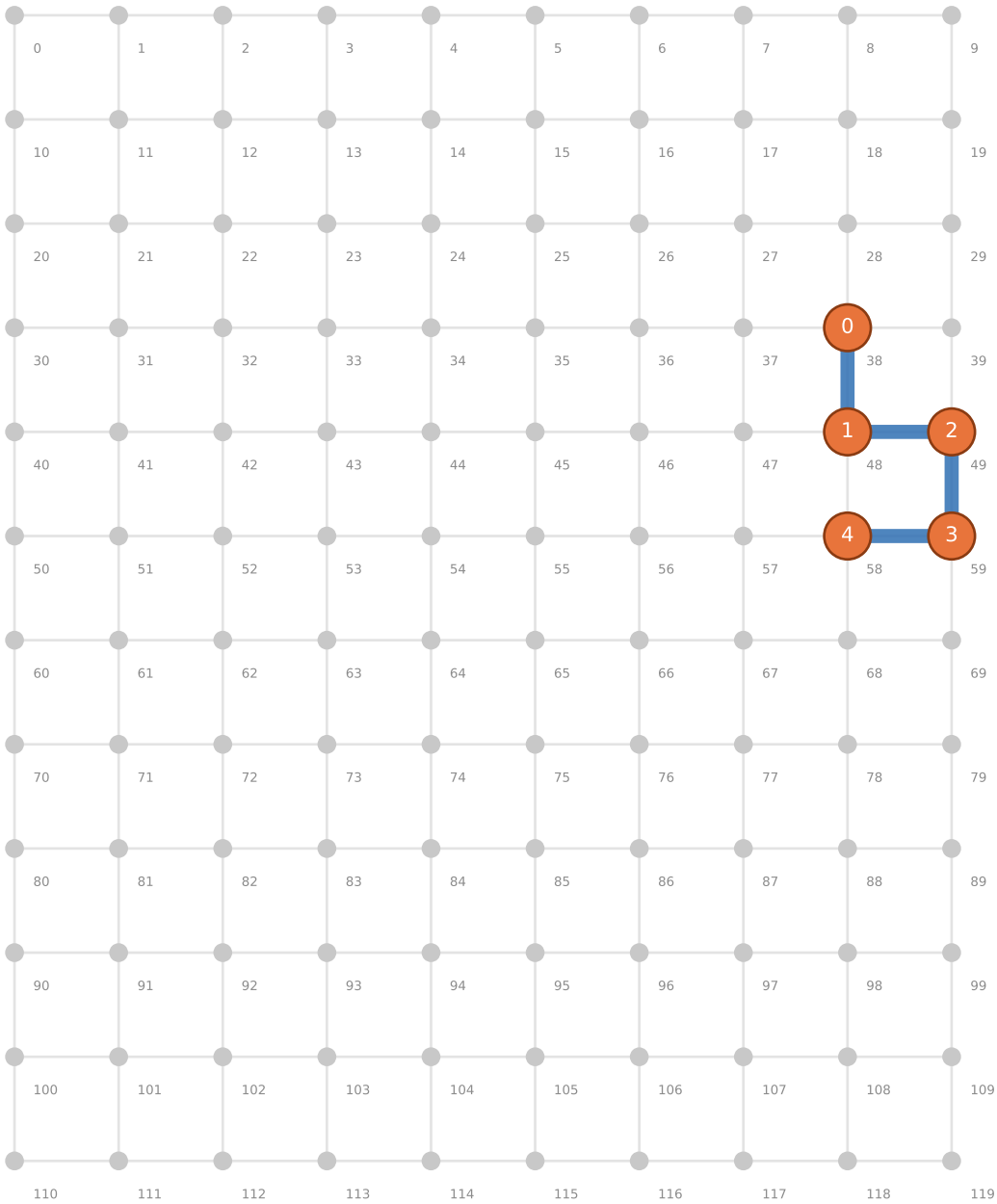
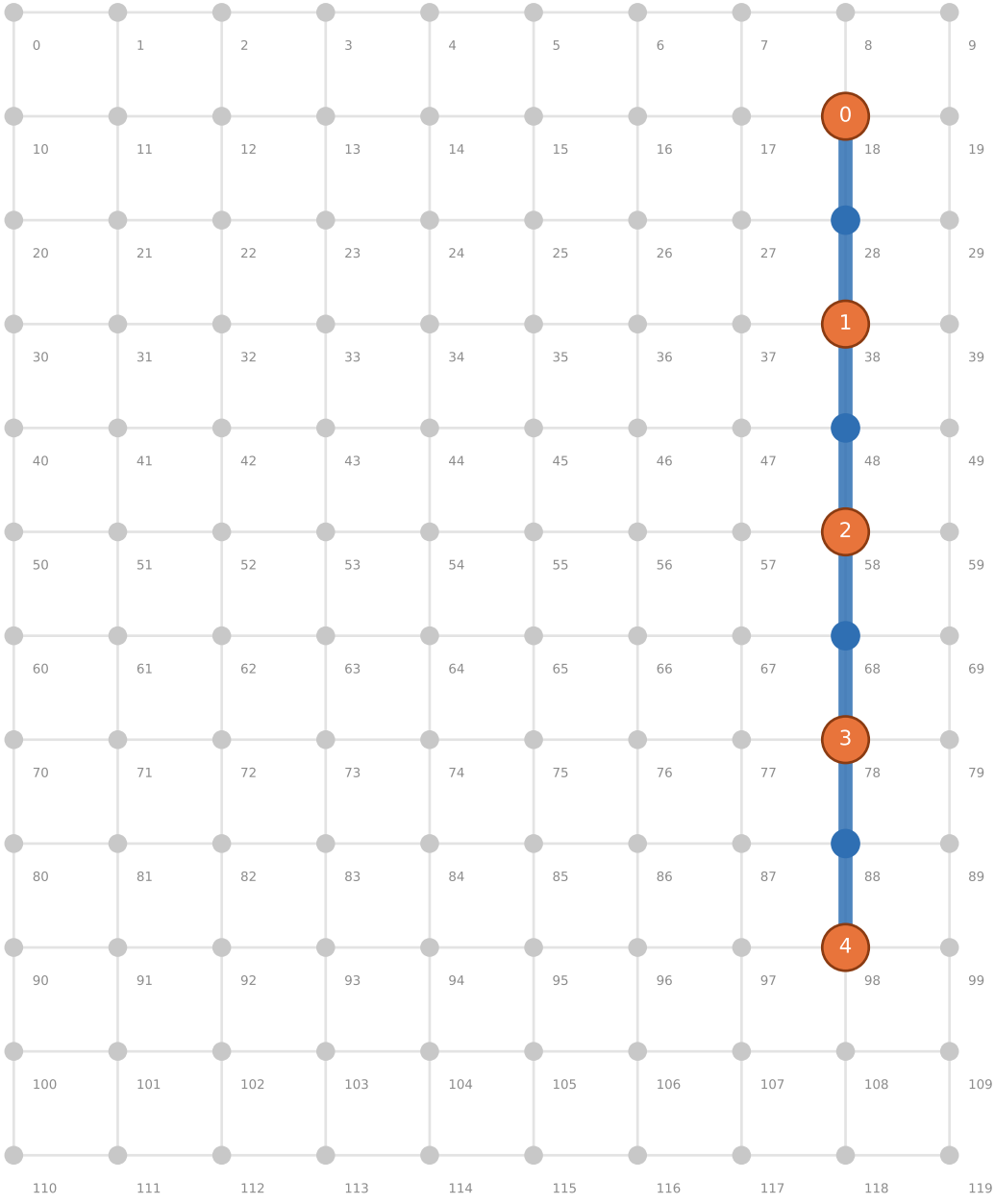
2
3
4

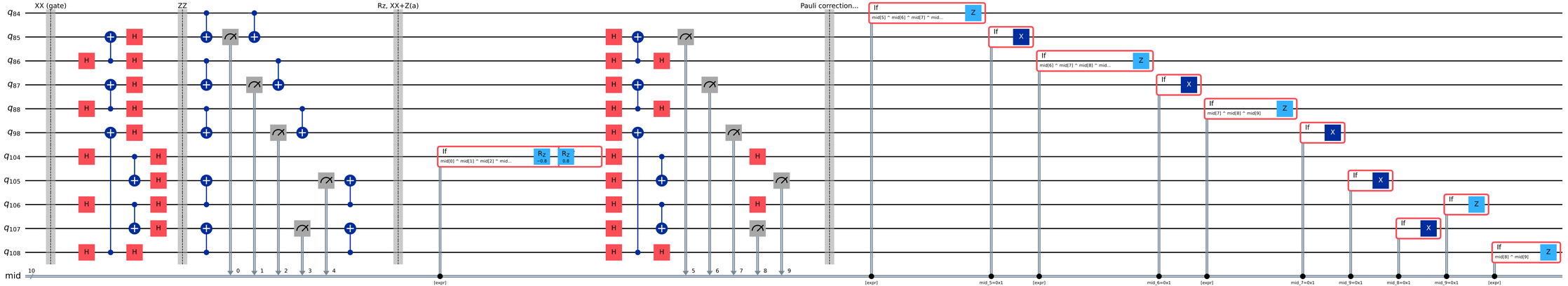
[Z5] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 49, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)



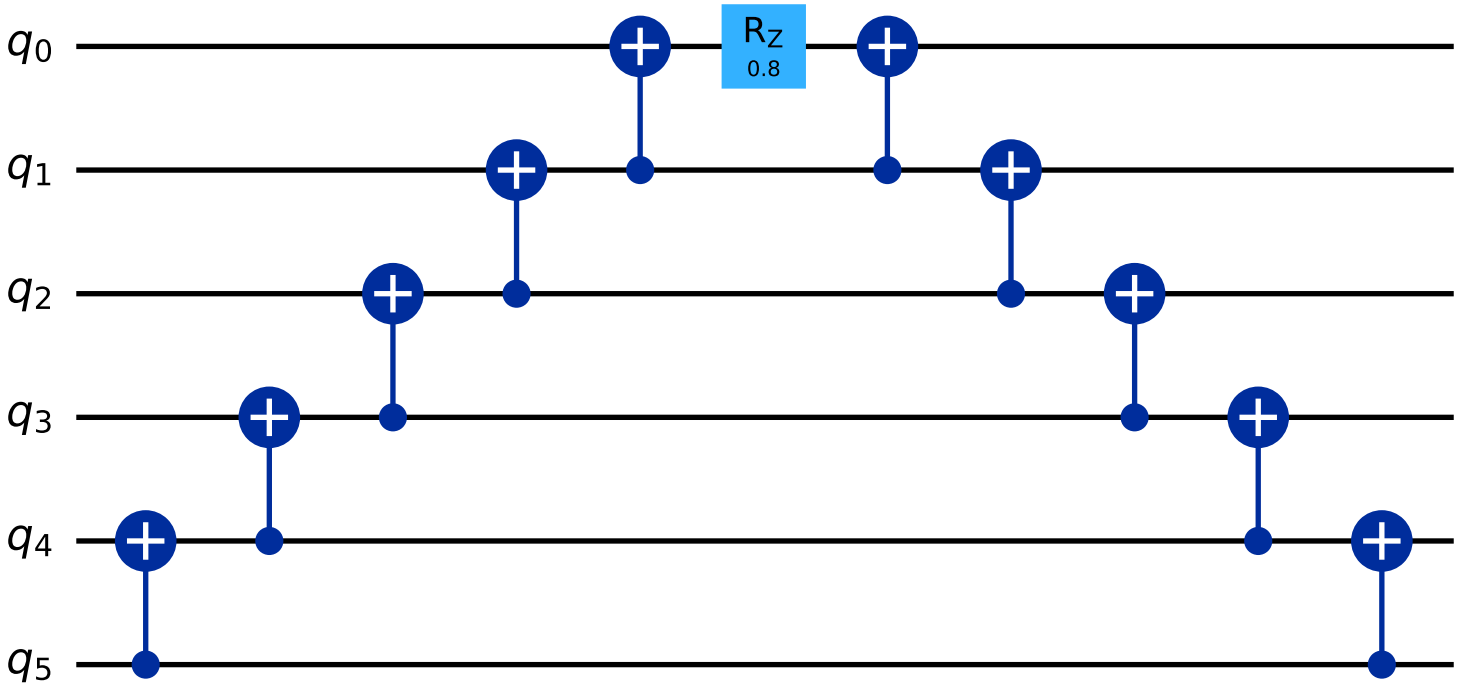
[Z5] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 16



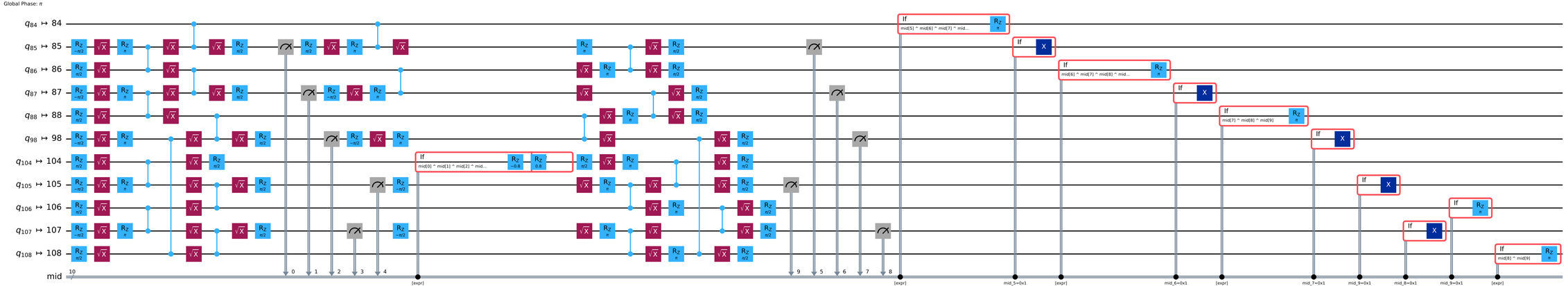
[Z6] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 4, 10 mid-circuit measurements, logical→physical [84, 86, 88, 108, 106, 104]



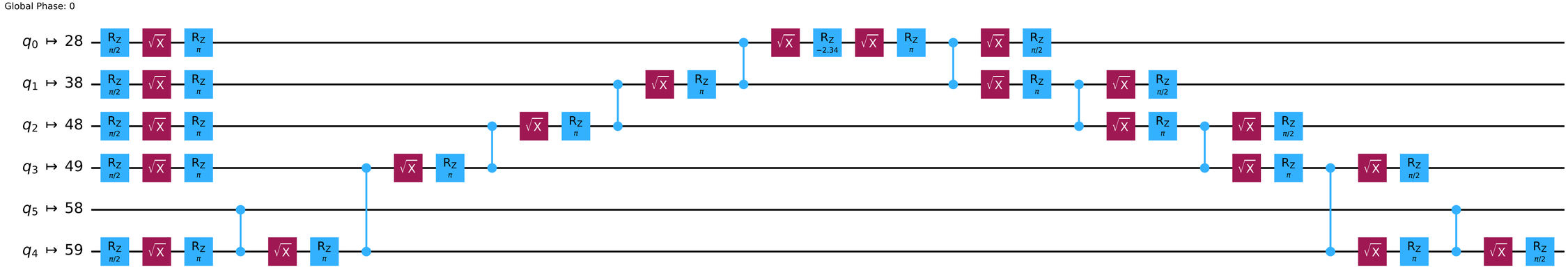
[Z6] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10

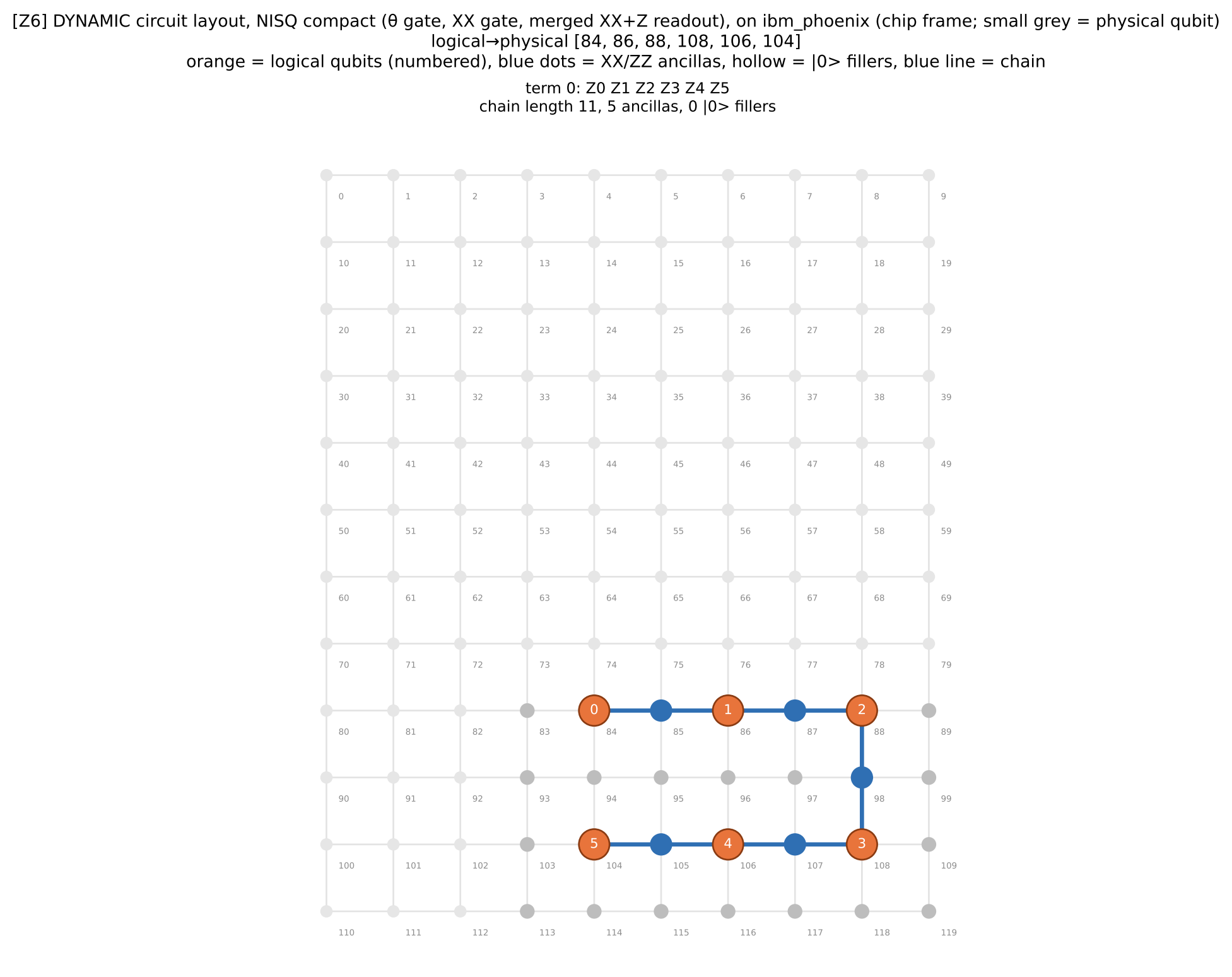


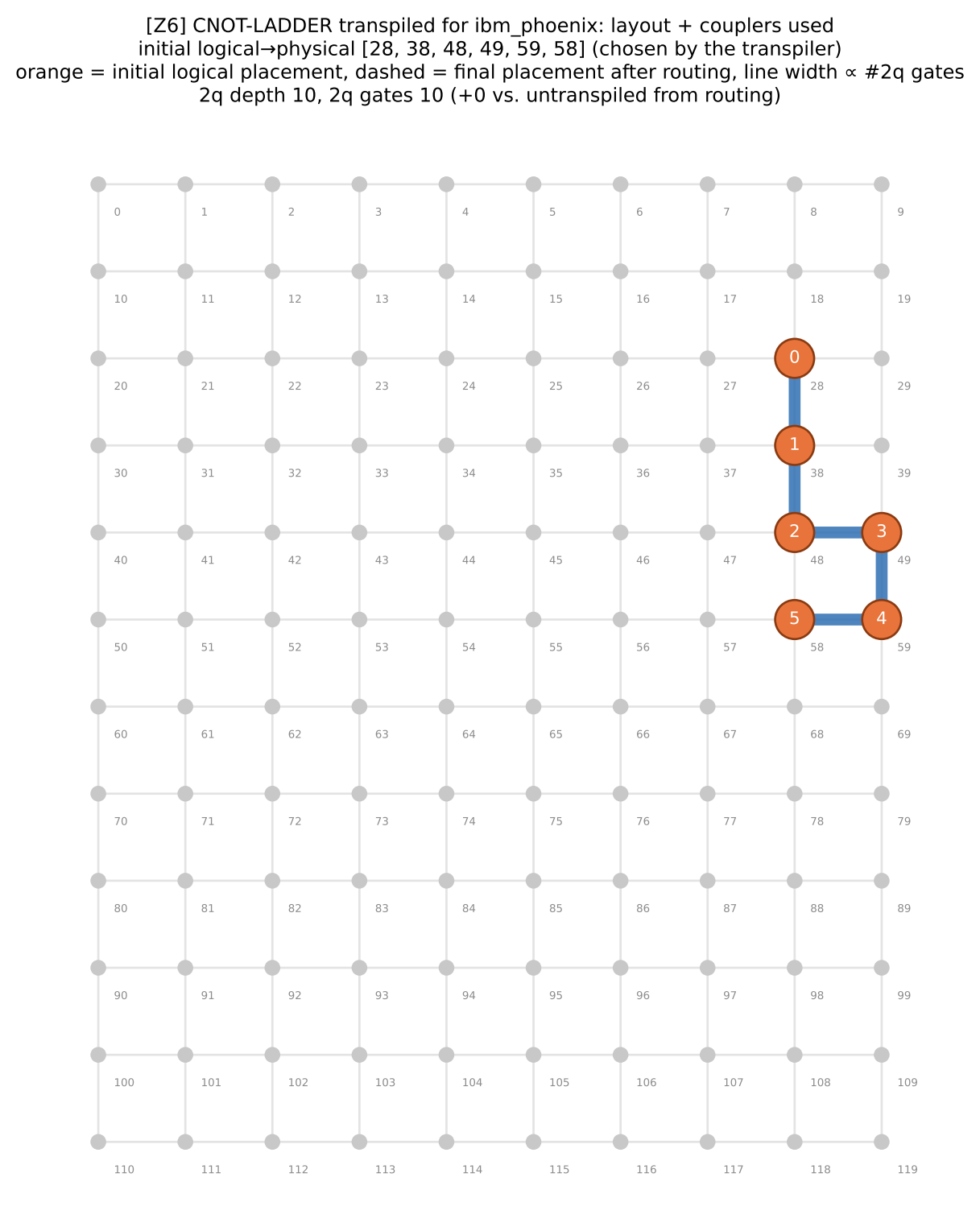
[Z6] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:50 rz:48 cz:20 if_else:11 measure:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 4, total depth 25



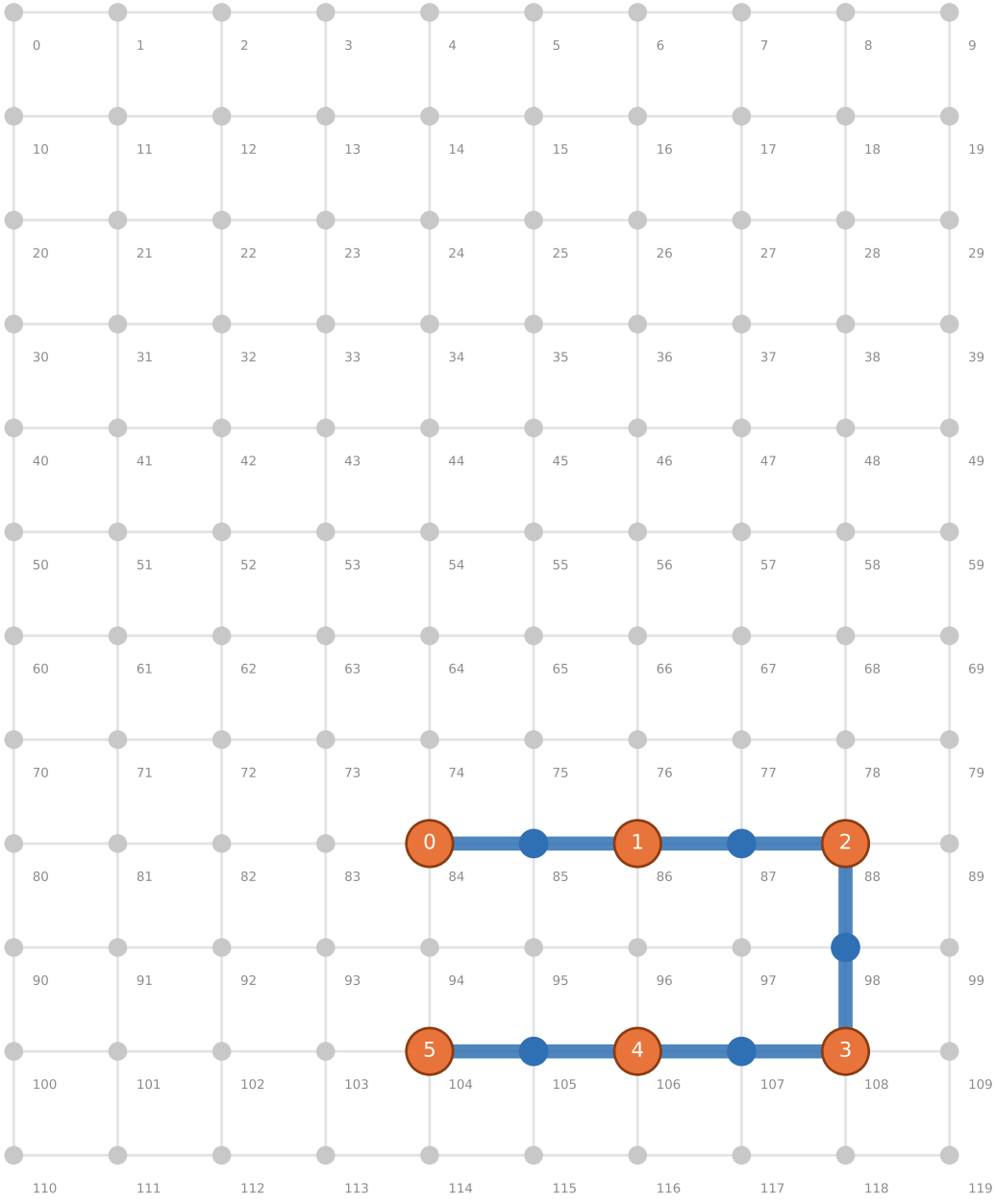
[Z6] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10, total depth 35



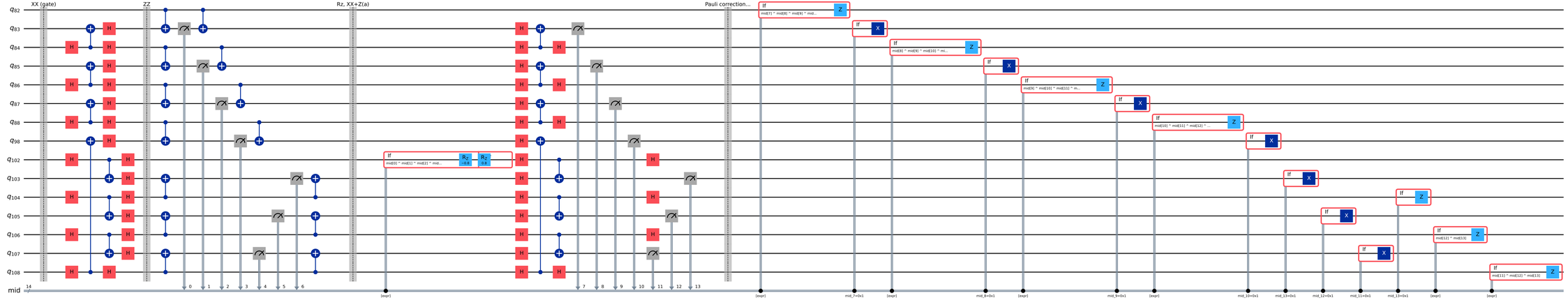

[Z6] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [84, 86, 88, 108, 106, 104]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5


[Z6] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [28, 38, 48, 49, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5

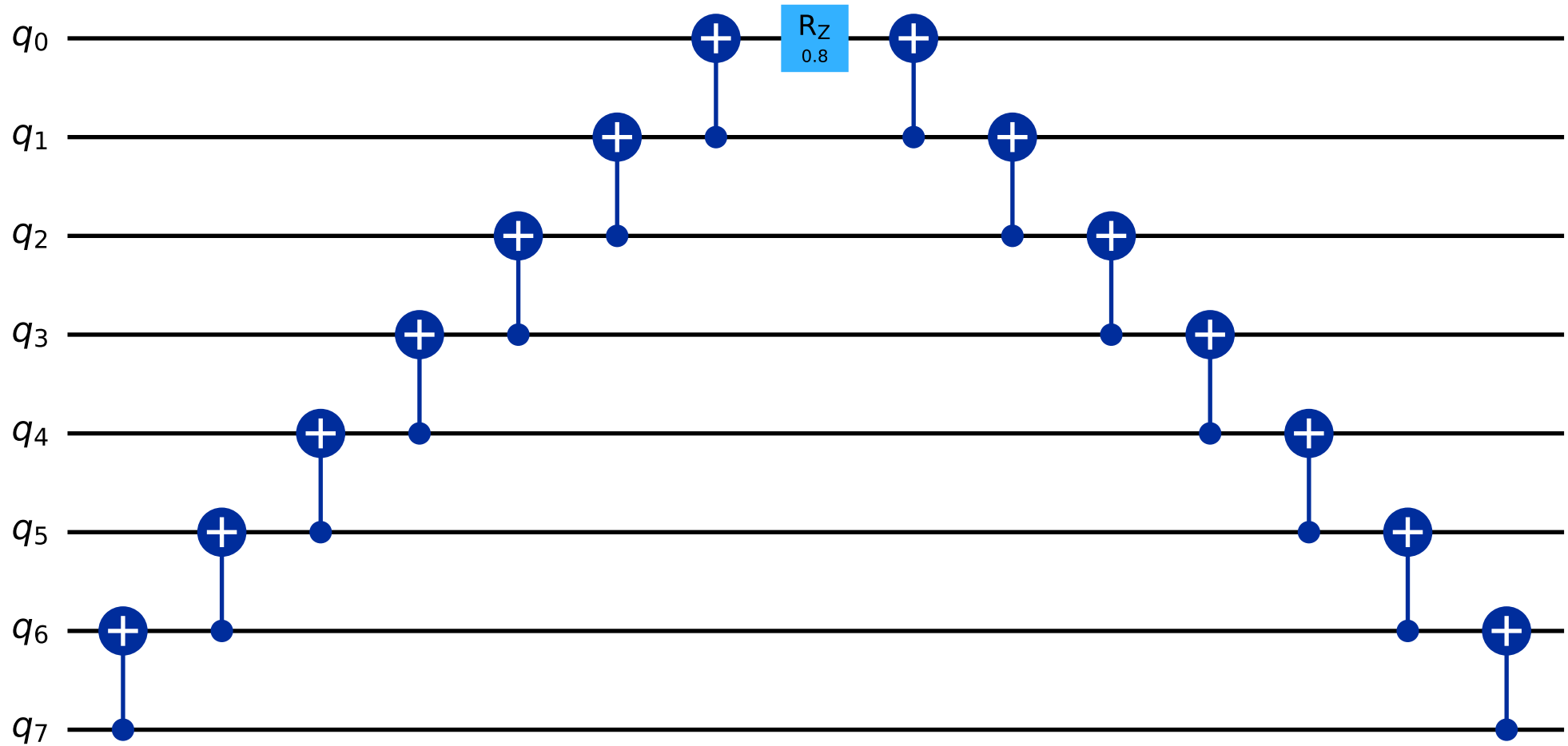
[Z6] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [84, 86, 88, 108, 106, 104]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 20



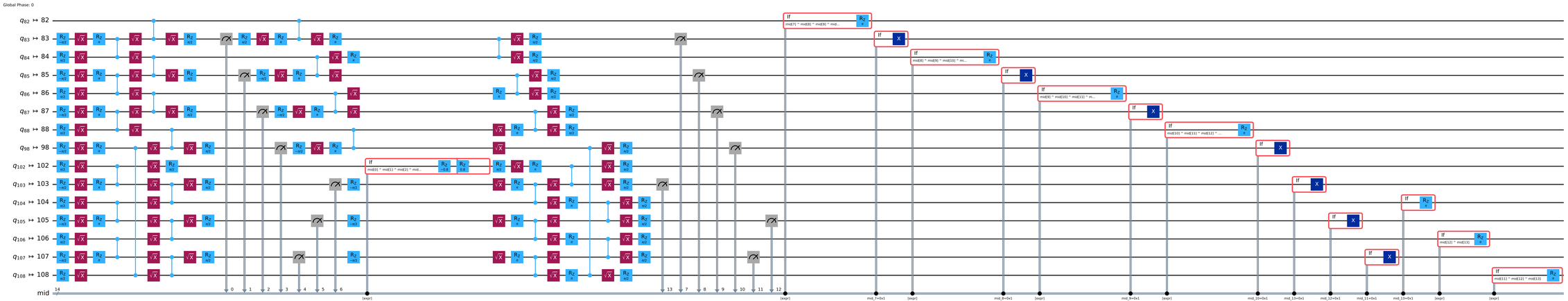
[Z8] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 4, 14 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]



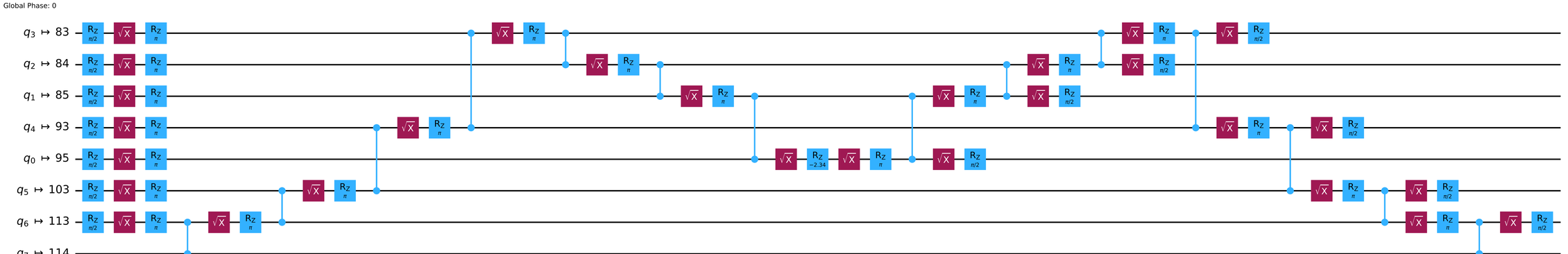
[Z8] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14



[Z8] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:70 rz:66 cz:28 if_else:15 measure:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 4, total depth 27

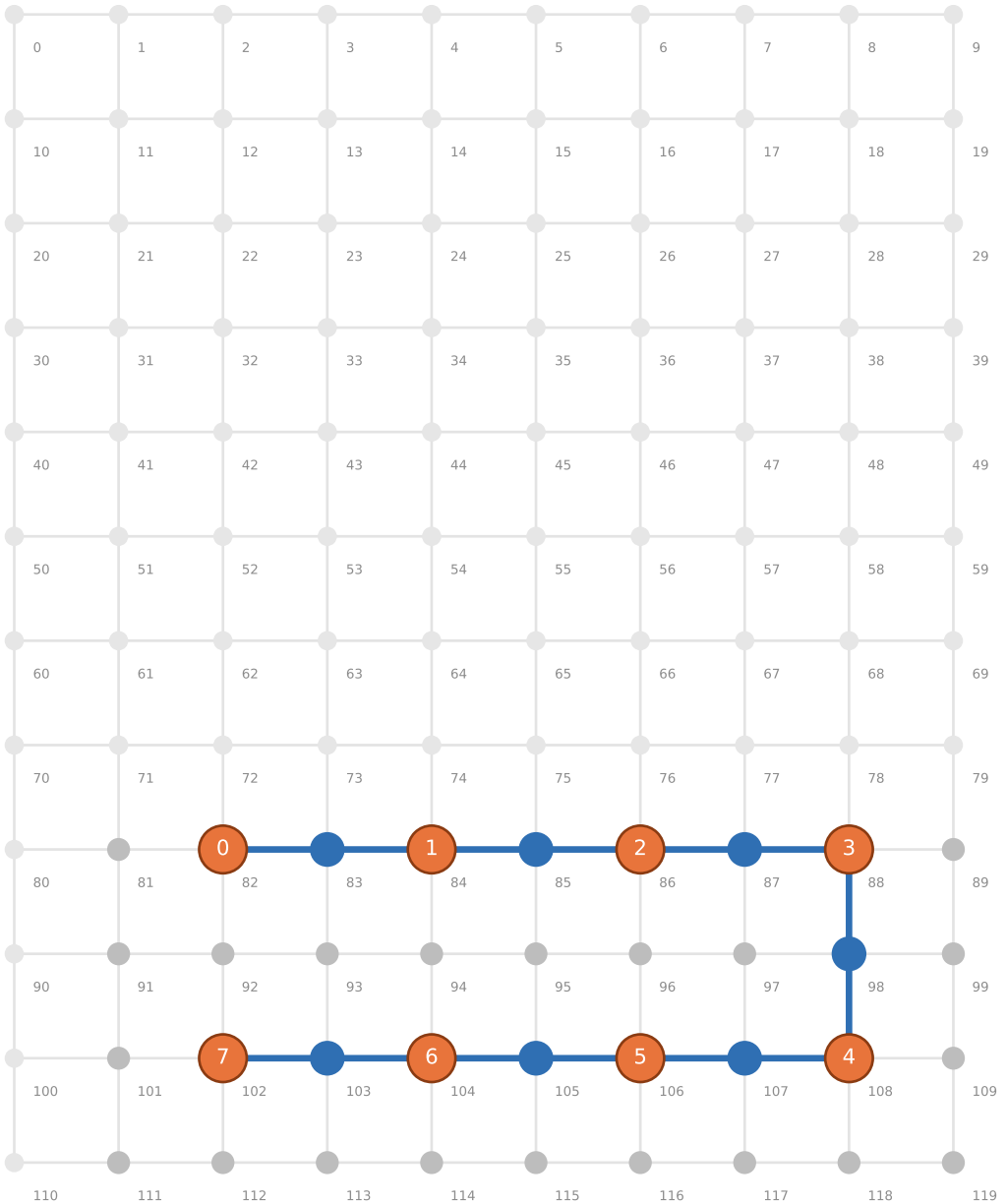


[Z8] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

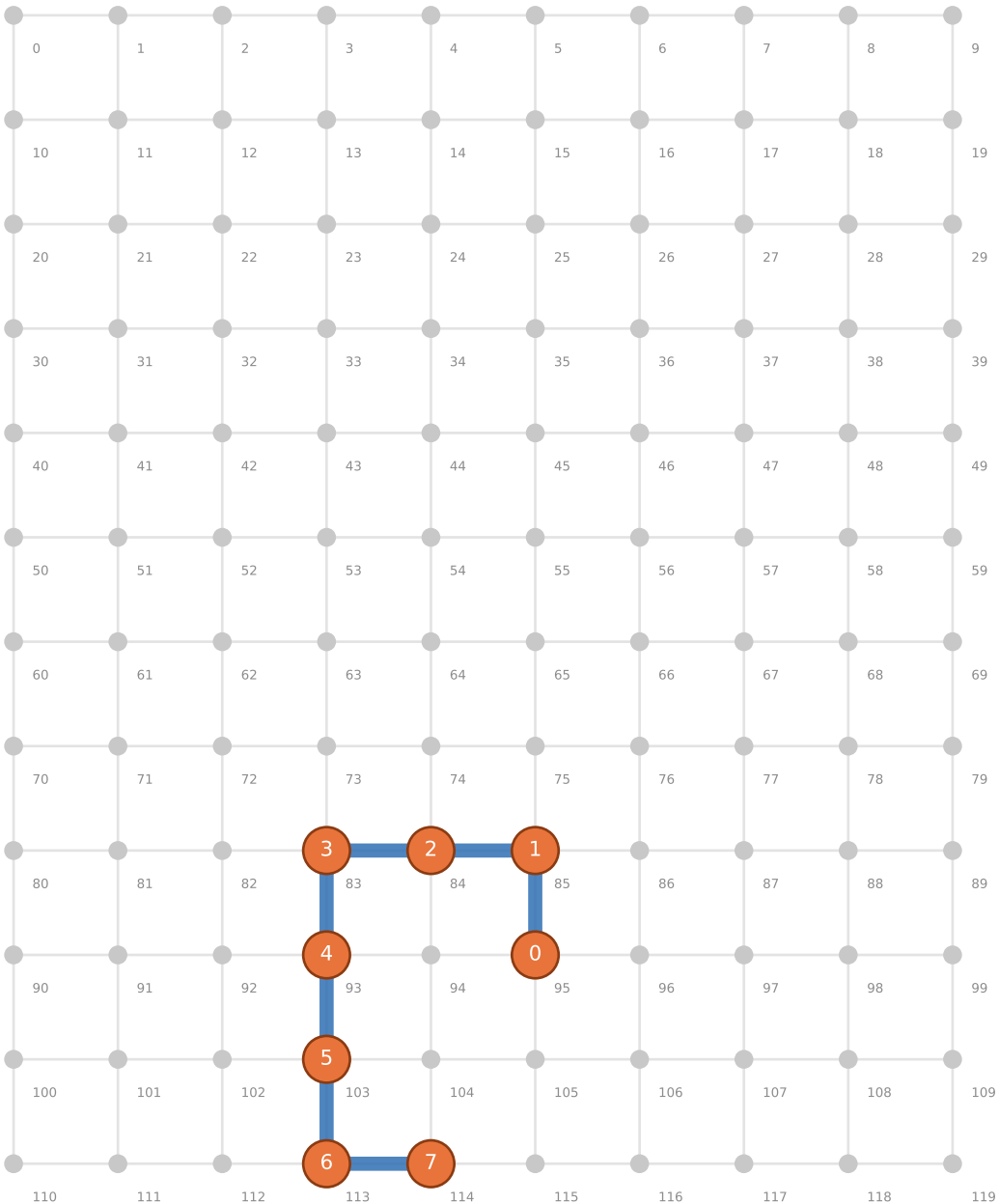


[Z8] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

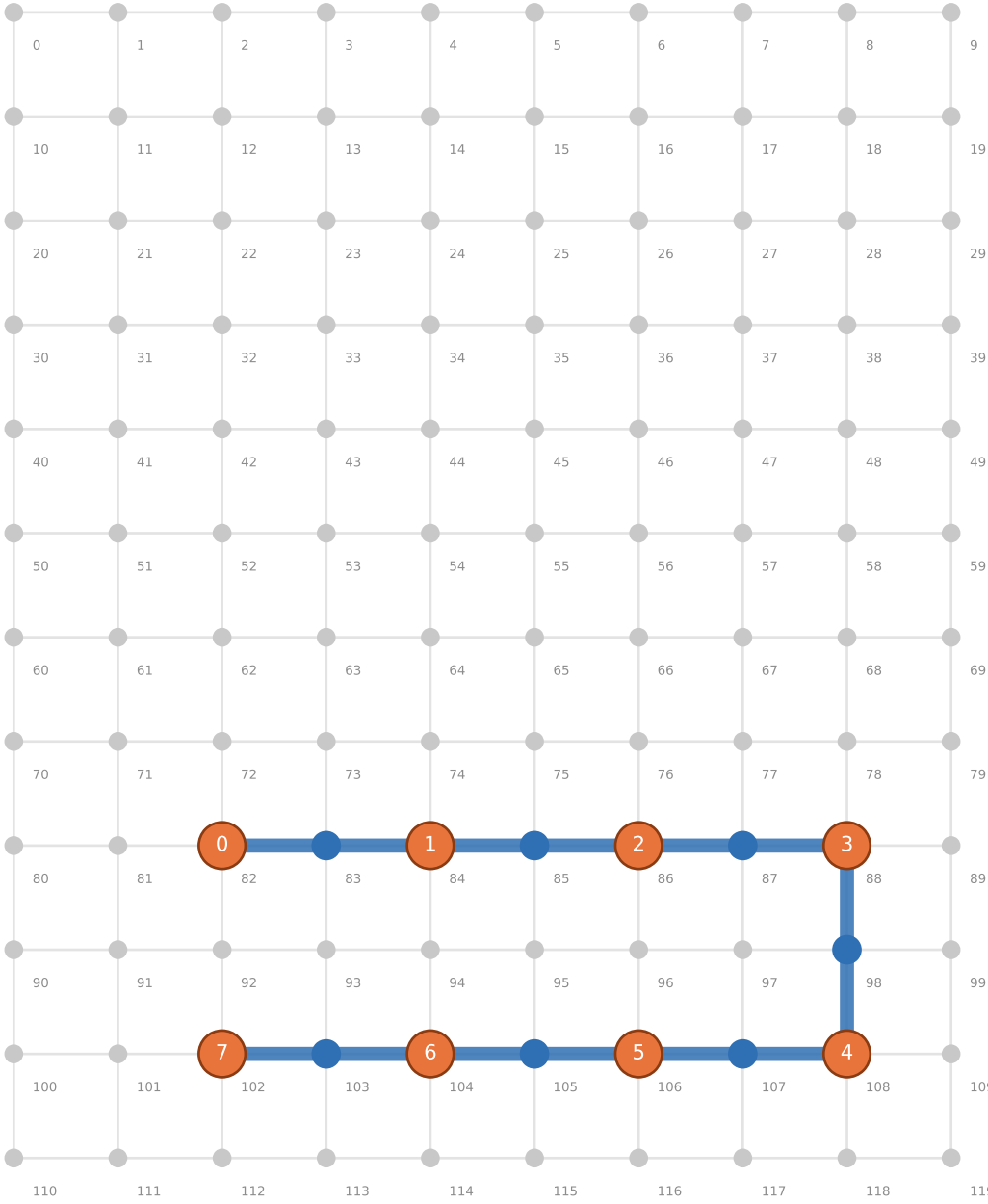
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers



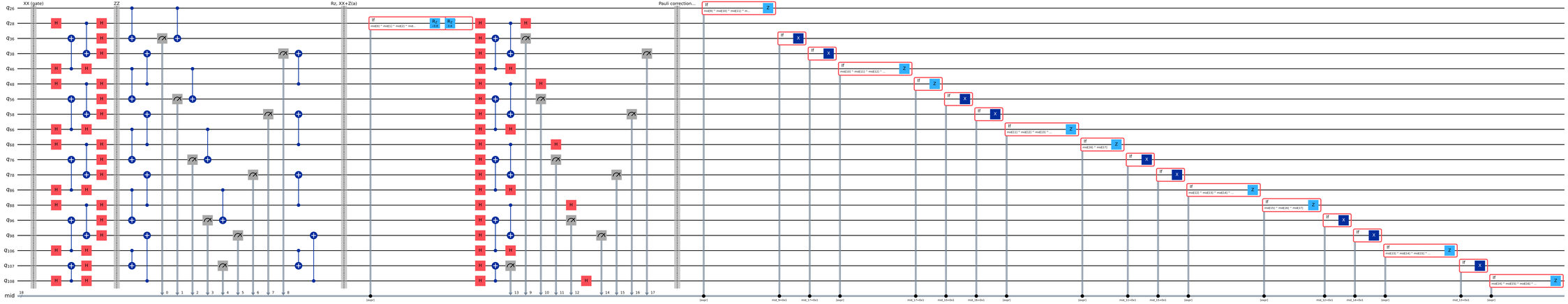
[Z8] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 83, 93, 103, 113, 114] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



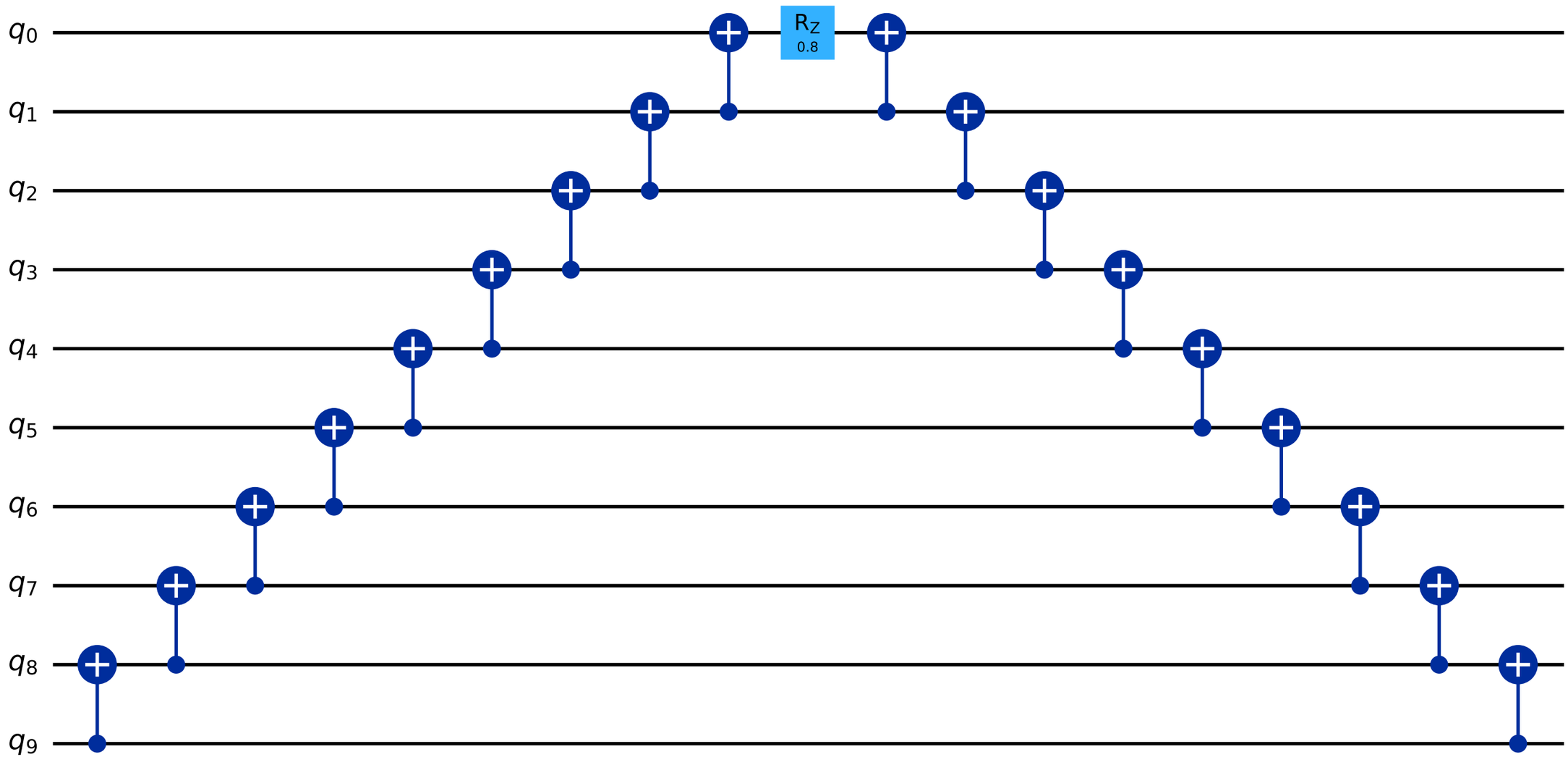
[Z8] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 28



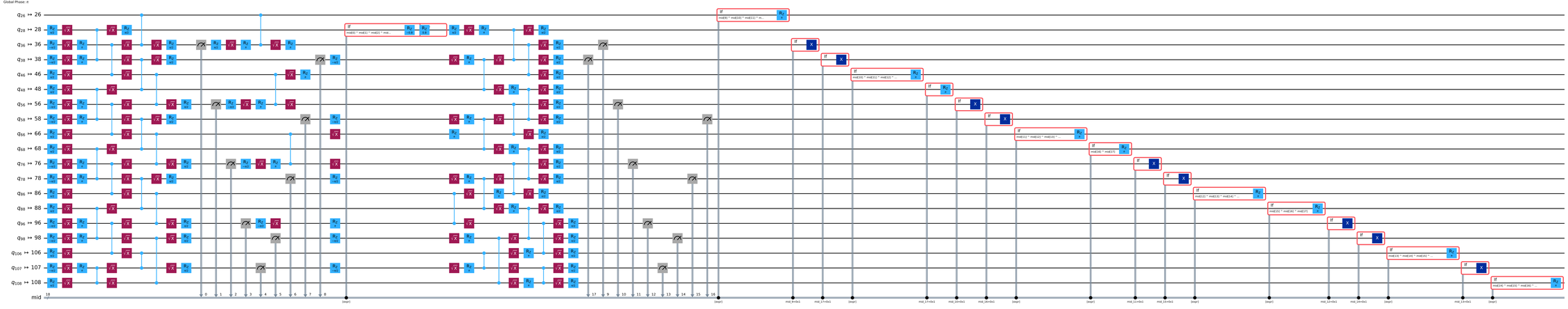
[Z10] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 4, 18 mid-circuit measurements, logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]



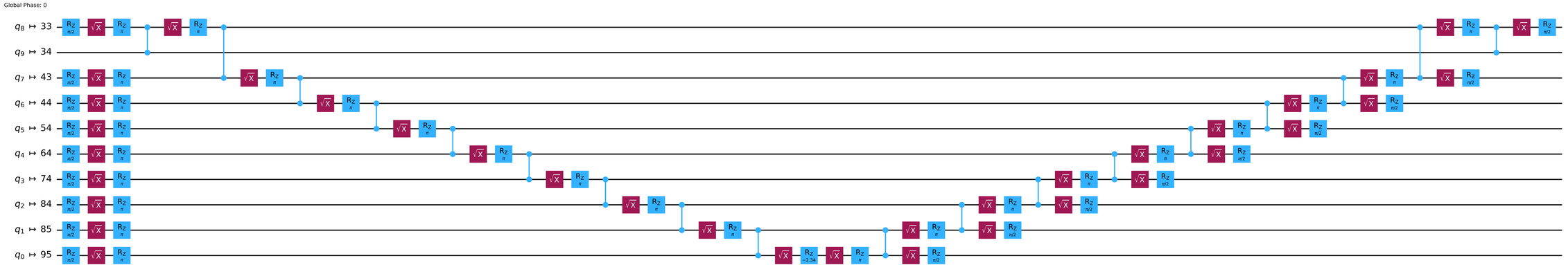
[Z10] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18



[Z10] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:90 rz:84 cz:36 if_else:19 measure:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 4, total depth 29

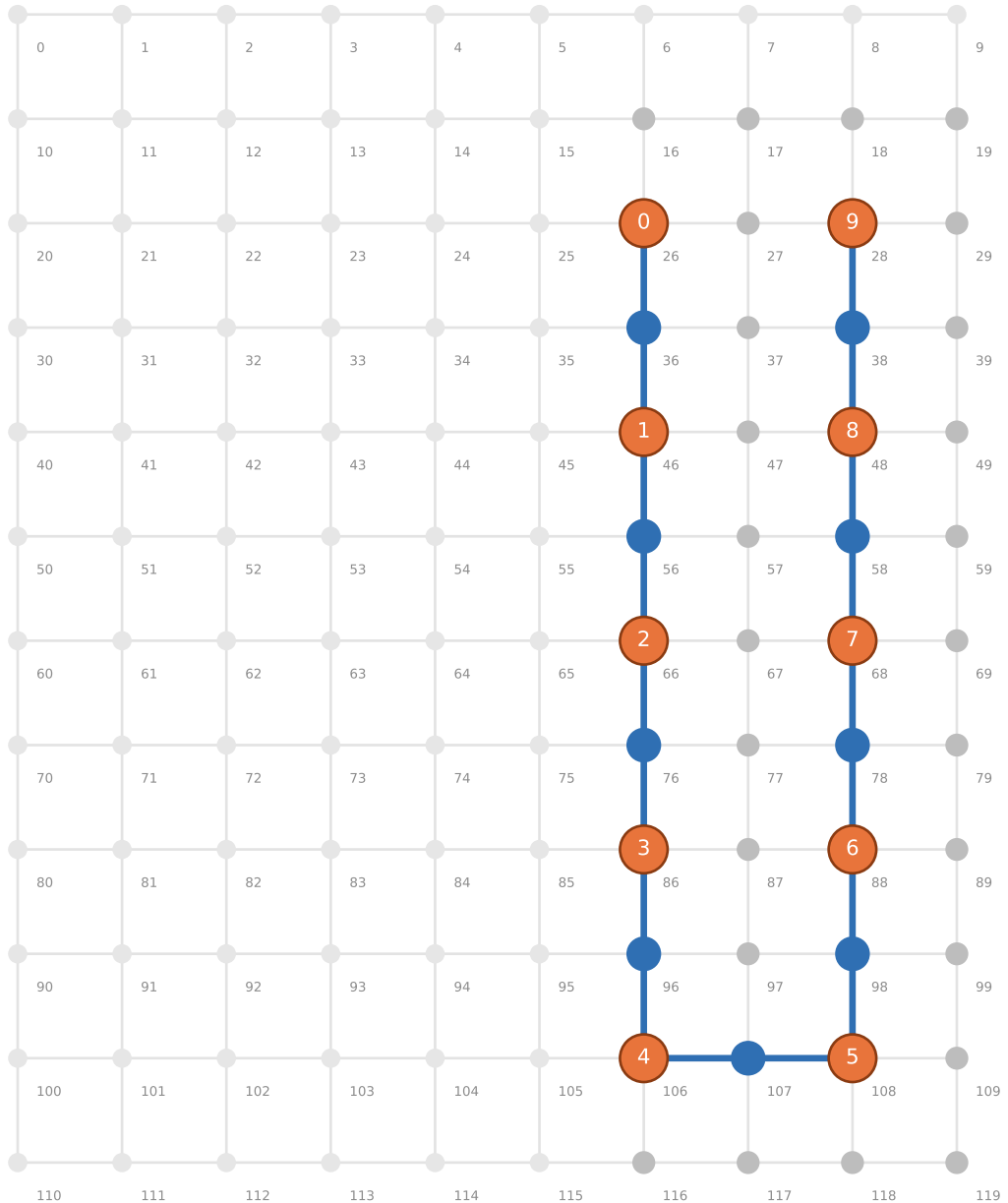


[Z10] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 18, total depth 59

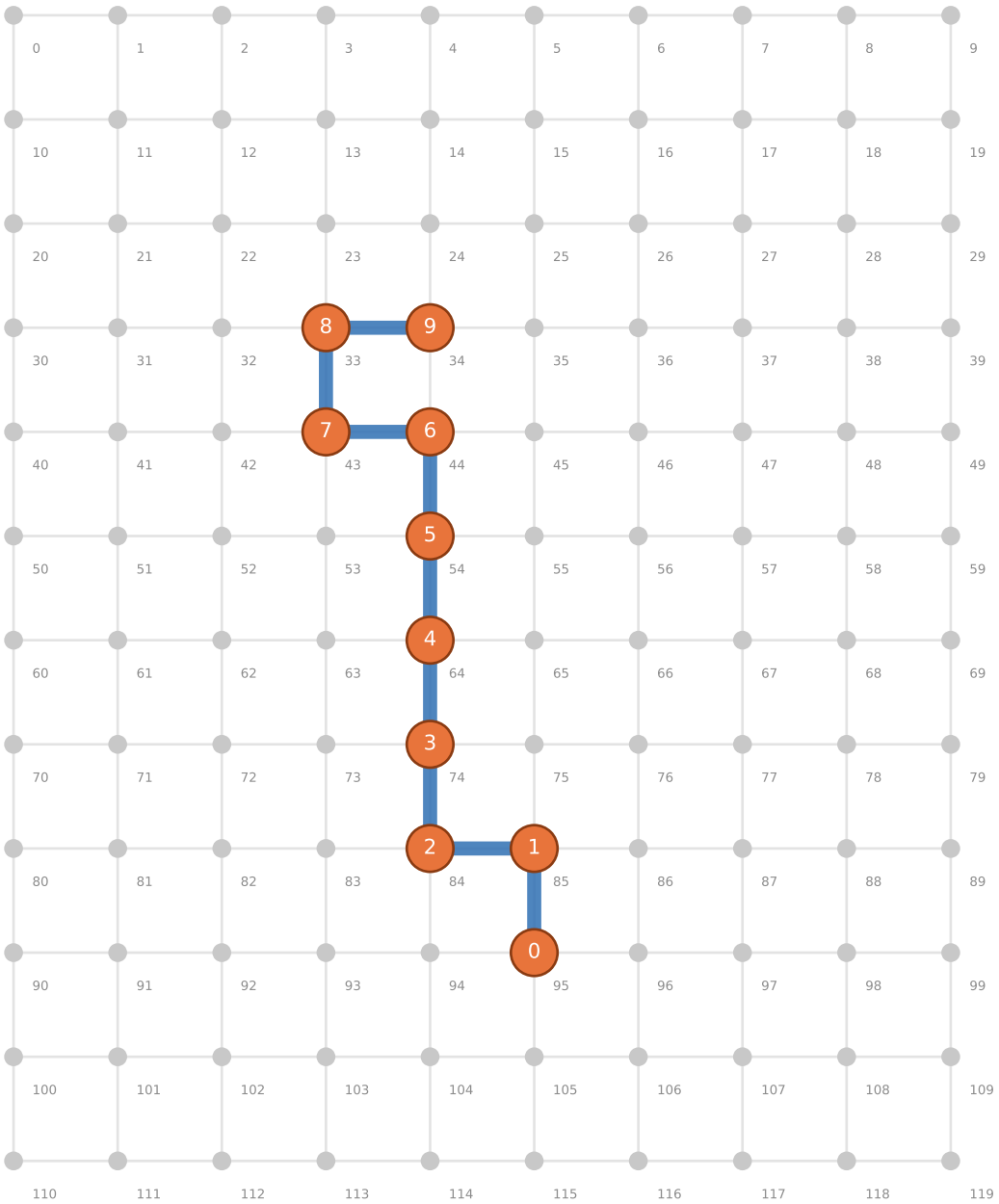


[Z10] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

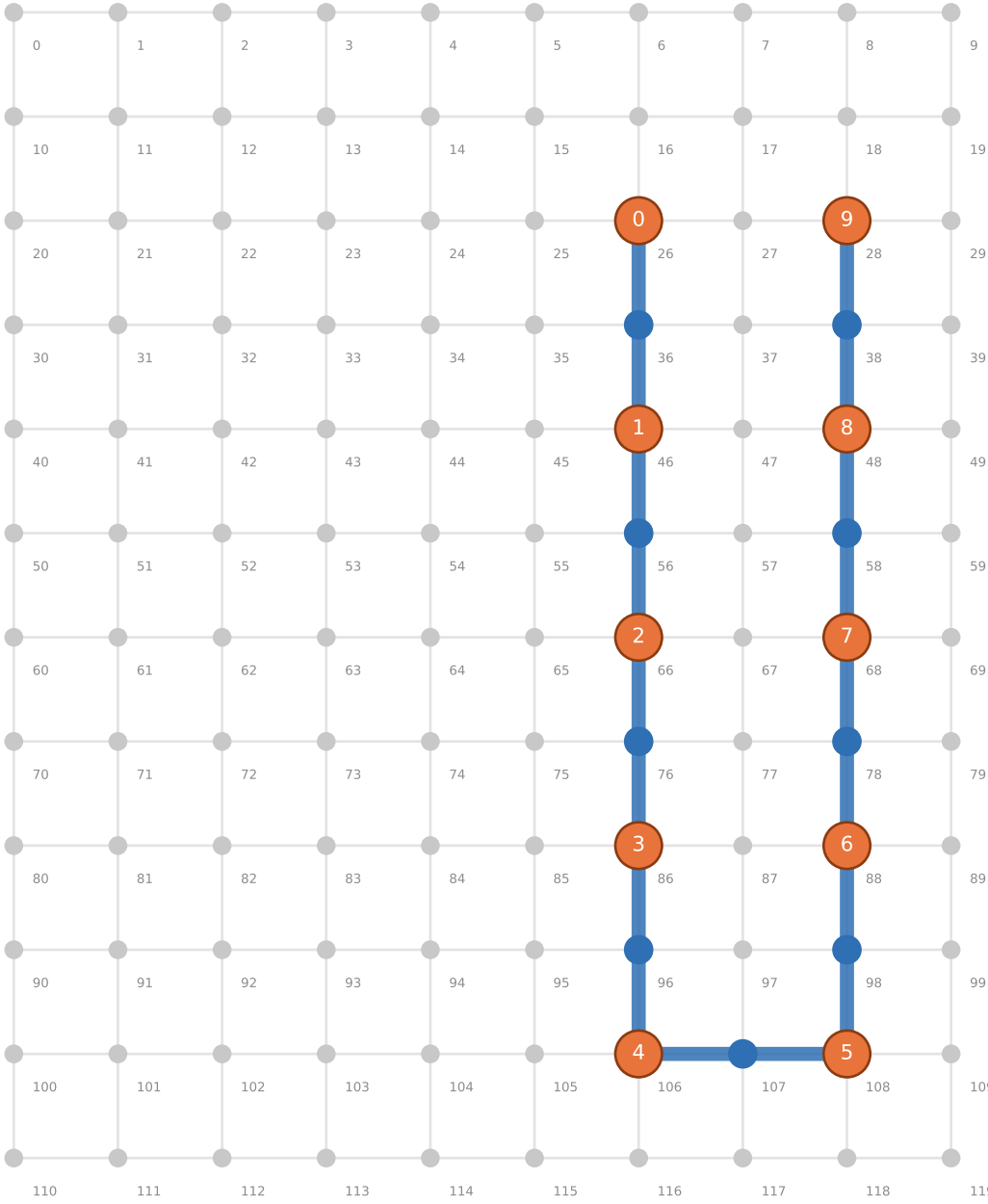
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers



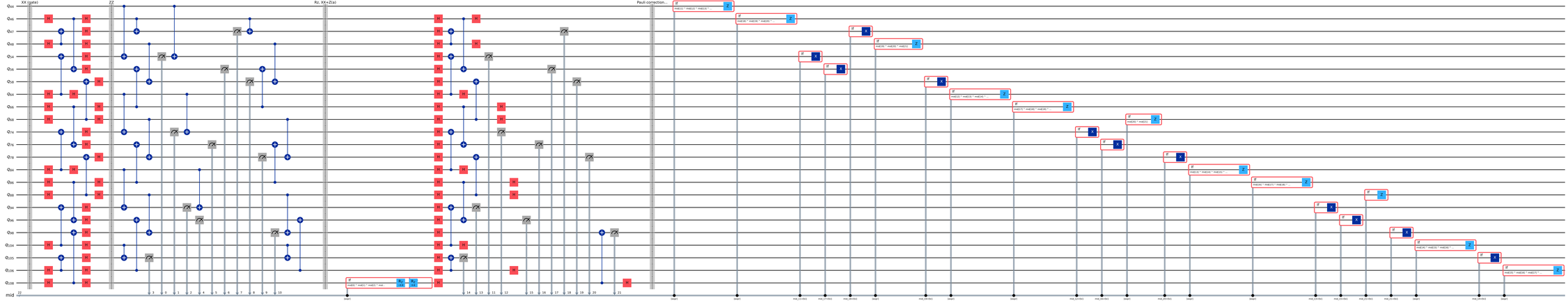
[Z10] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 74, 64, 54, 44, 43, 33, 34] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)



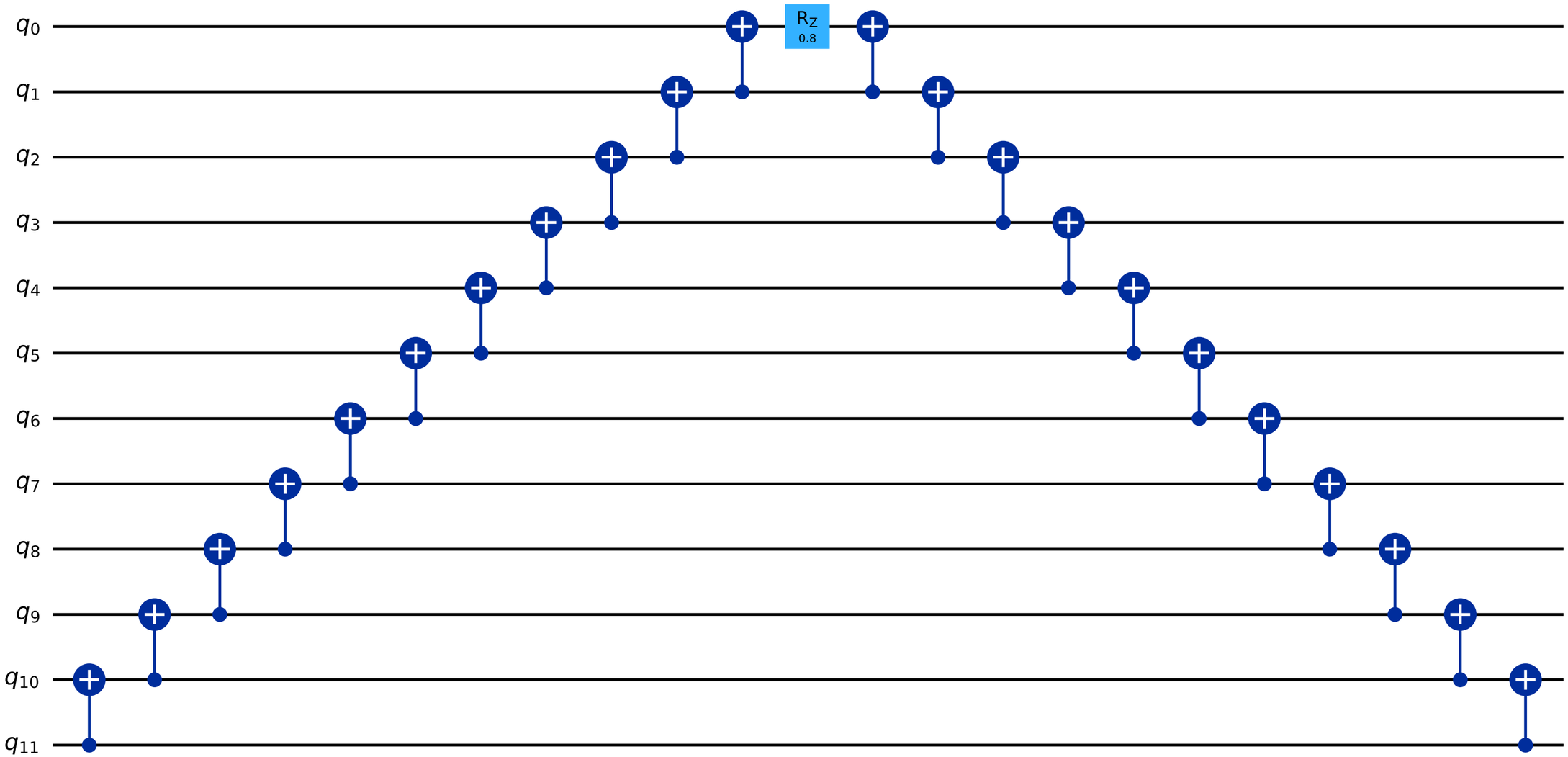
[Z10] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 36



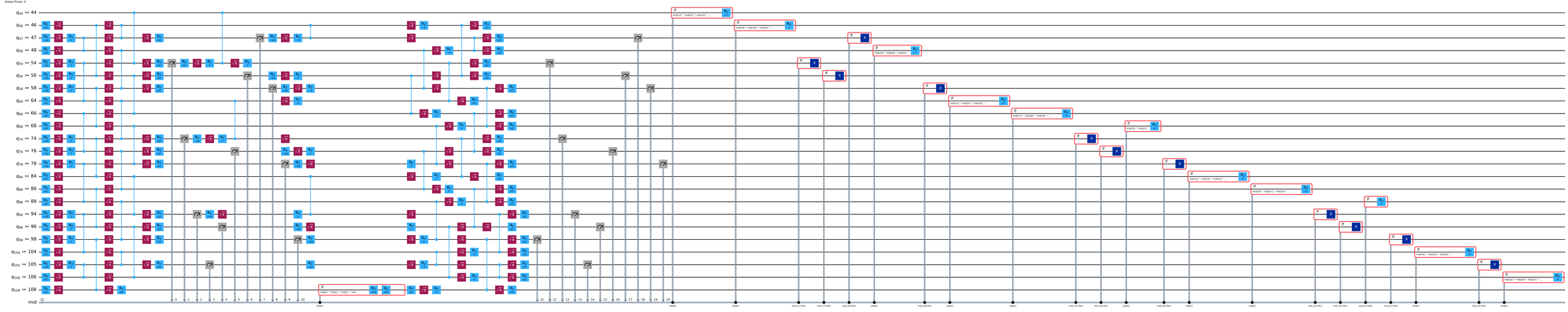
[212] DYNAMIC constant-depth circuit, NISQ compact (8 gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4 P = 20 21 Z2 Z3 24 Z5 26 27 Z8 Z9 Z10 Z11 (12 logical-qubits)
2q depth 4, 22 mid-circuit measurements, logical=physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]



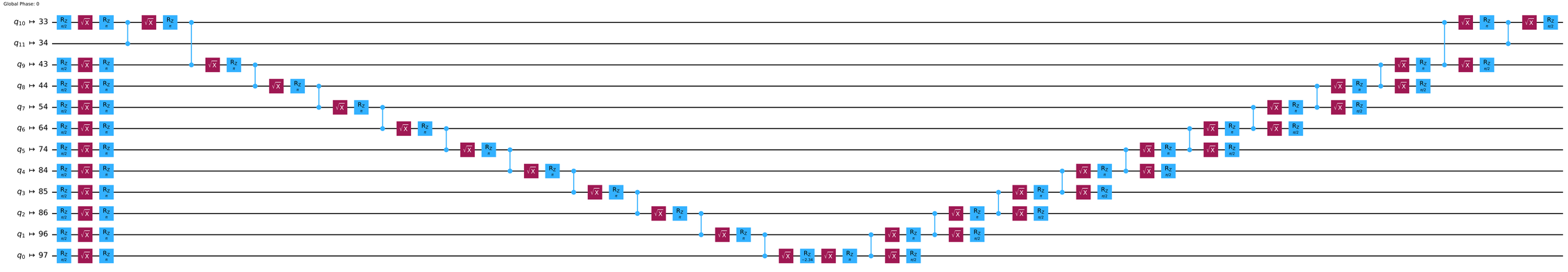
[Z12] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22



[212] DYNAMIC circuit, NISQ compact (9 gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:110 rz:102 cz:44 if_else:23 measure:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 4, total depth 33

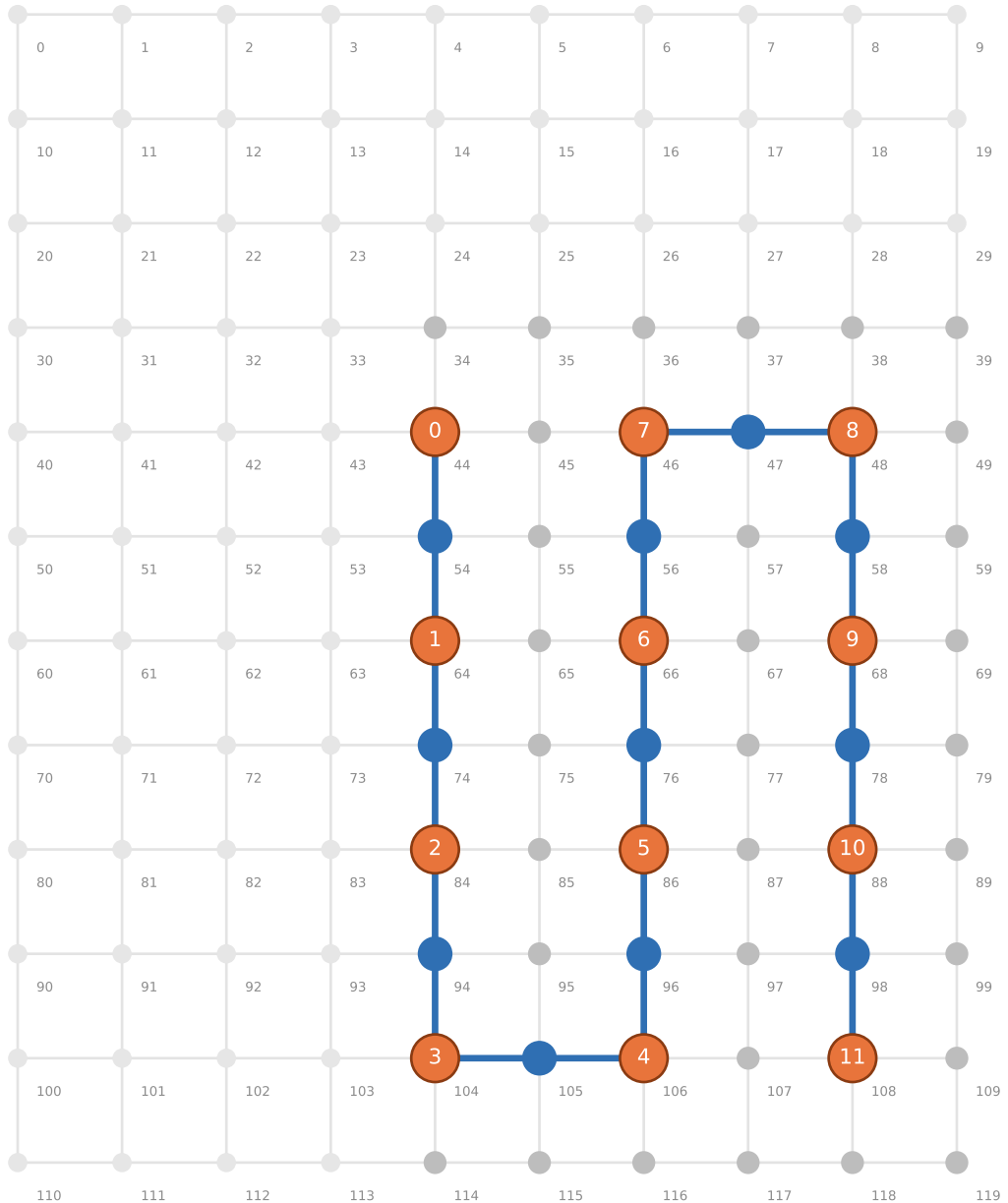


[Z12] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22, total depth 71

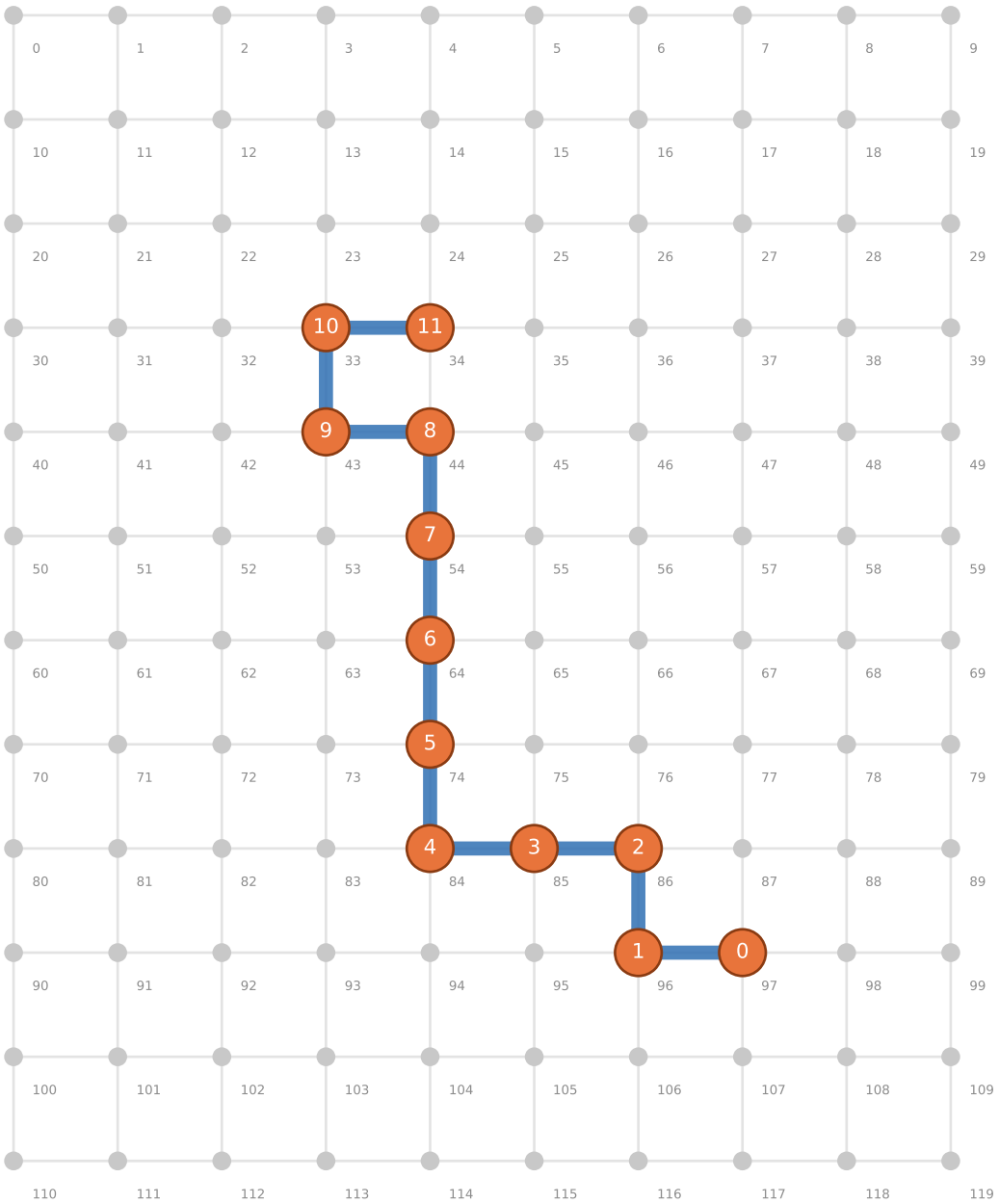


[Z12] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

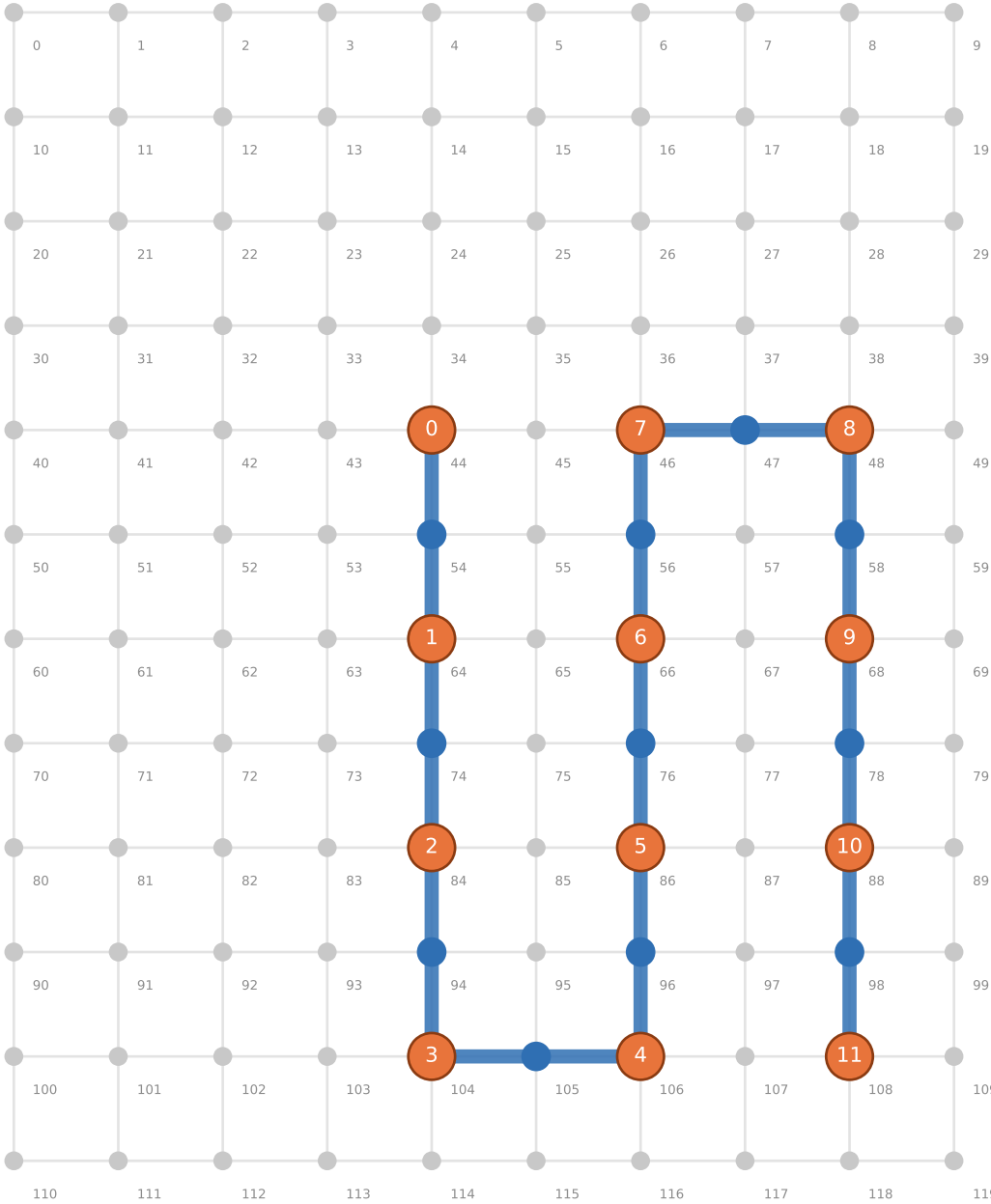
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers



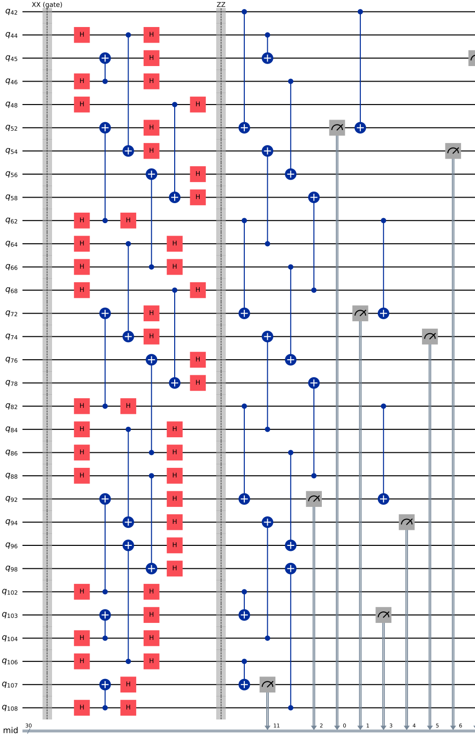
[Z12] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [97, 96, 86, 85, 84, 74, 64, 54, 44, 43, 33, 34] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)

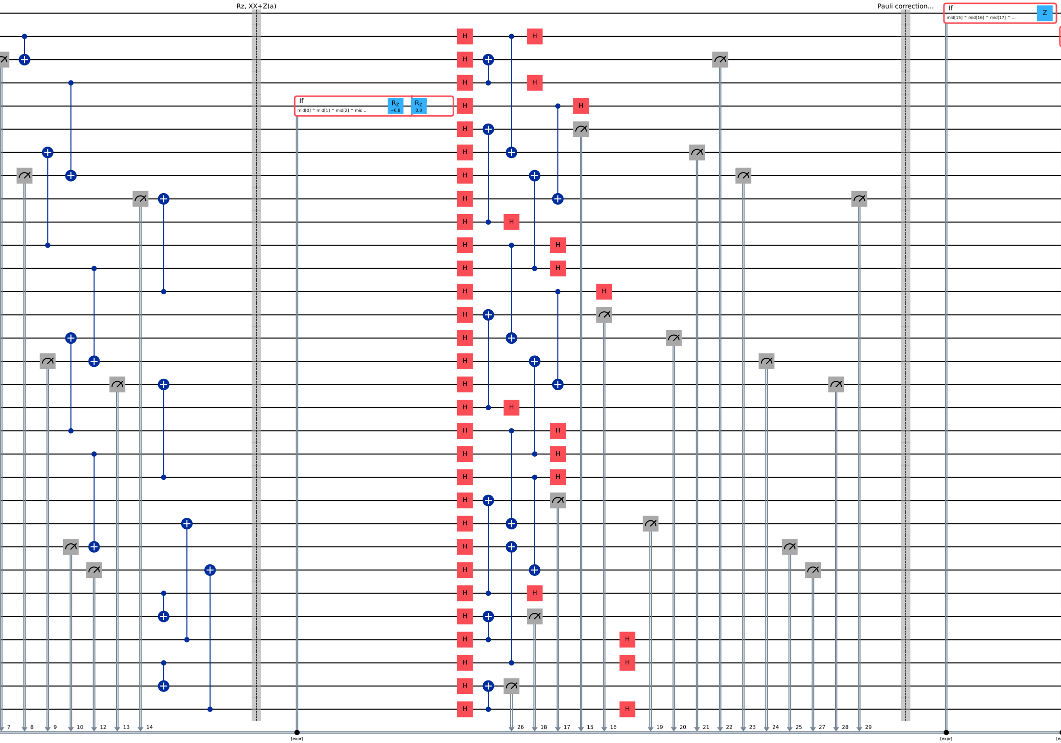


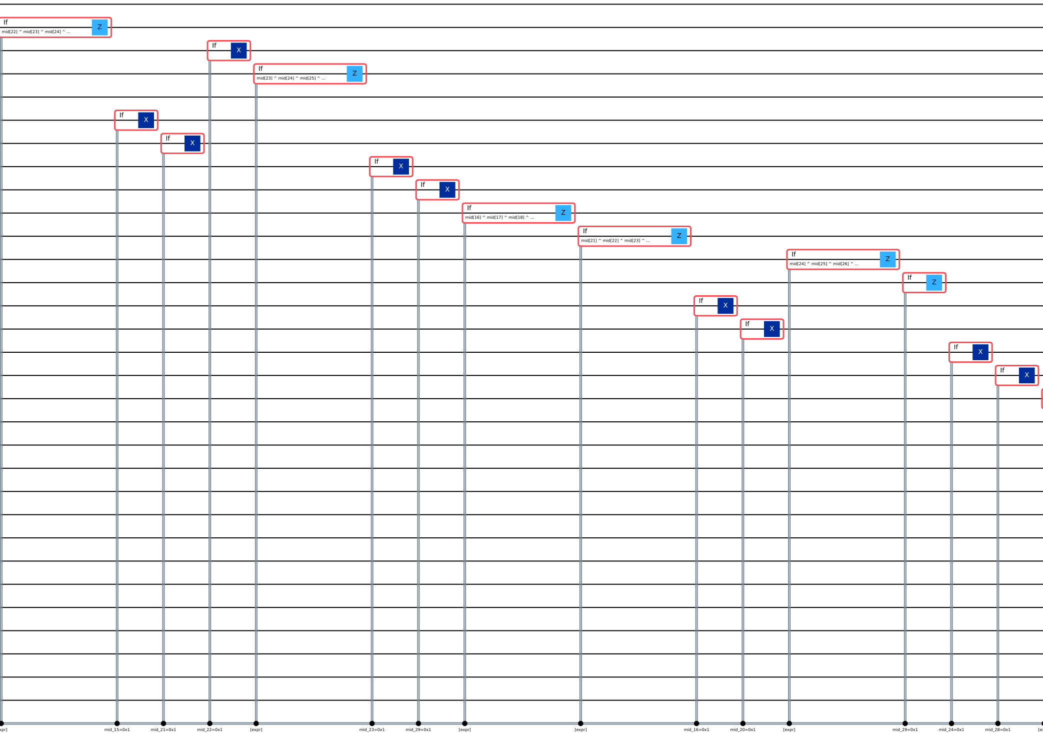
[Z12] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 44

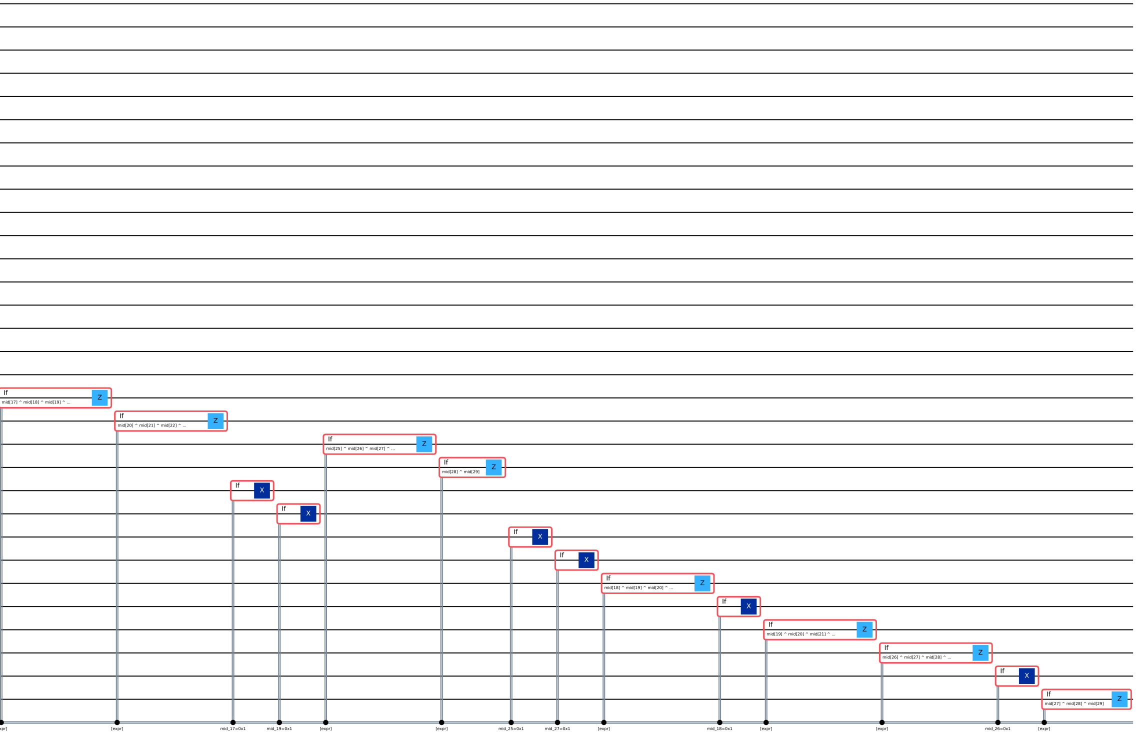


[Z16] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 4, 30 mid-circuit measurements, logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]

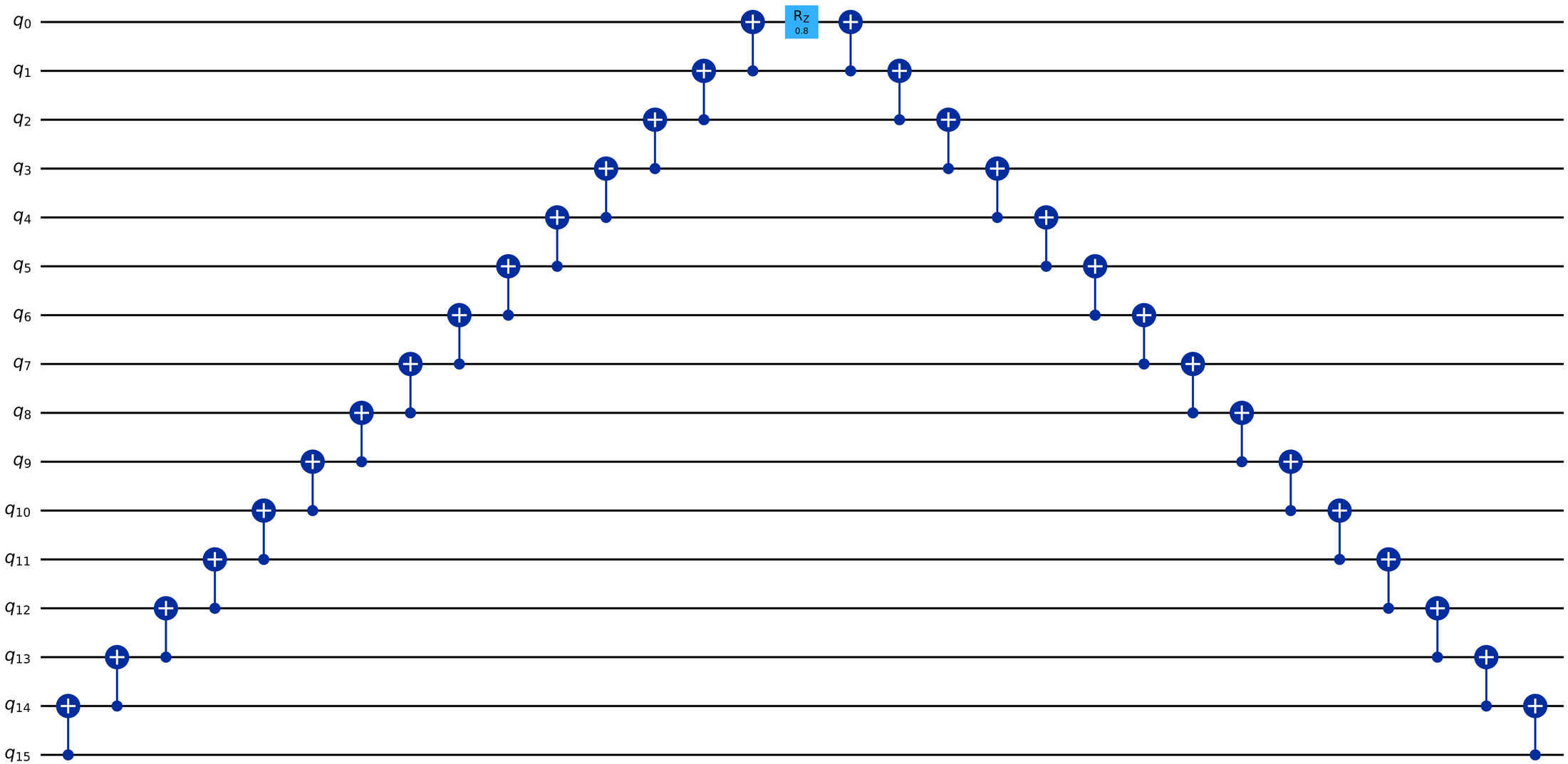

XX (gate)
ZZ
q_{42}
q_{44}
q_{45}
q_{46}
q_{48}
q_{52}
q_{54}
q_{56}
q_{58}
q_{62}
q_{64}
q_{66}
q_{68}
q_{72}
q_{74}
q_{76}
q_{78}
q_{82}
q_{84}
q_{86}
q_{88}
q_{92}
q_{94}
q_{96}
q_{98}
q_{102}
q_{103}
q_{104}
q_{106}
q_{107}
q_{108}
mid
30
H
11
2
0
1
3
4
5
6


Rz, XX+Z(a)
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z −0.8
R_Z 0.8
H
Pauli correction...
If
mid[15] ^ mid[16] ^ mid[17] ^ ...
Z
7
8
9
10
12
13
14
[expr]
26
18
17
15
16
19
20
21
22
23
24
25
27
28
29
[expr]

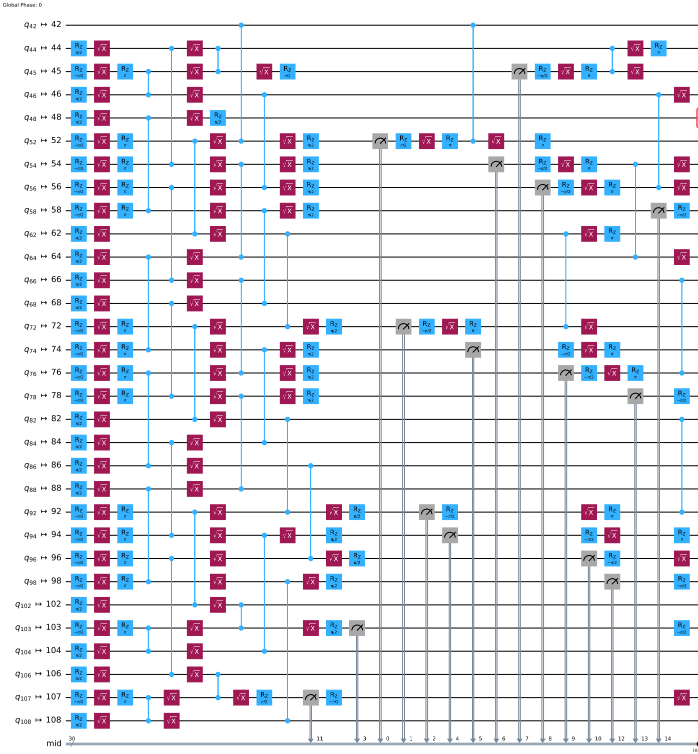

If
mid[22] ^ mid[23] ^ mid[24] ^ ...
Z
If
X
If
mid[23] ^ mid[24] ^ mid[25] ^ ...
Z
If
X
If
X
If
X
If
X
If
mid[16] ^ mid[17] ^ mid[18] ^ ...
Z
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
Z
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
Z
If
Z
If
X
If
X
If
X
If
X
xpr]
mid_15=0x1
mid_21=0x1
mid_22=0x1
[expr]
mid_23=0x1
mid_29=0x1
[expr]
[expr]
mid_16=0x1
mid_20=0x1
[expr]
mid_29=0x1
mid_24=0x1
mid_28=0x1
[e

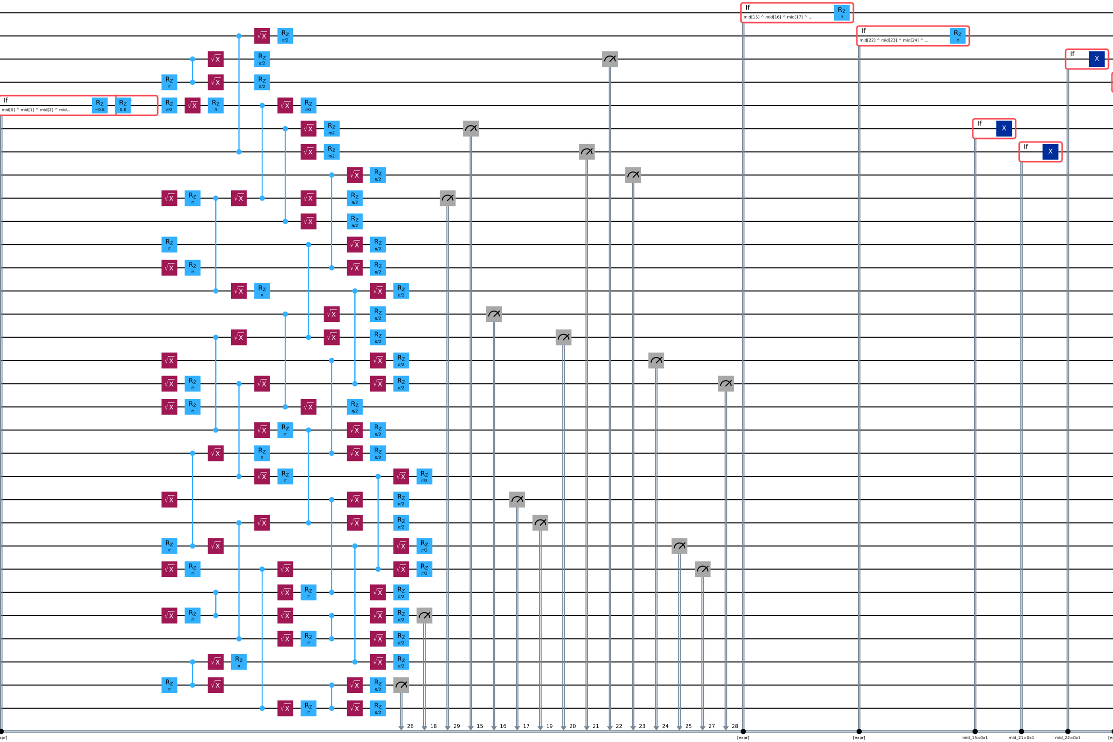

If
mid[17] ^ mid[18] ^ mid[19] ^ ...
Z
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
Z
If
X
If
X
If
mid[25] ^ mid[26] ^ mid[27] ^ ...
Z
If
mid[28] ^ mid[29]
Z
If
X
If
X
If
mid[18] ^ mid[19] ^ mid[20] ^ ...
Z
If
X
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
Z
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
Z
If
X
If
mid[27] ^ mid[28] ^ mid[29]
Z
expr]
[expr]
mid_17=0x1
mid_19=0x1
[expr]
[expr]
mid_25=0x1
mid_27=0x1
[expr]
mid_18=0x1
[expr]
[expr]
mid_26=0x1
[expr]

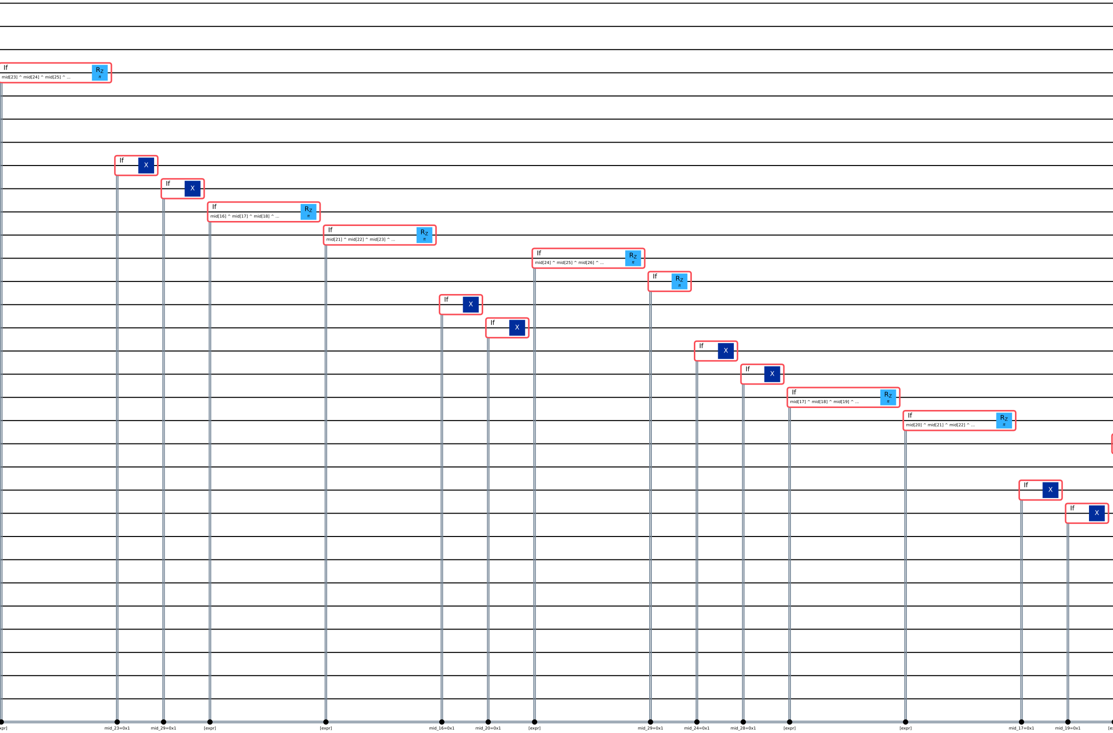
[Z16] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 30

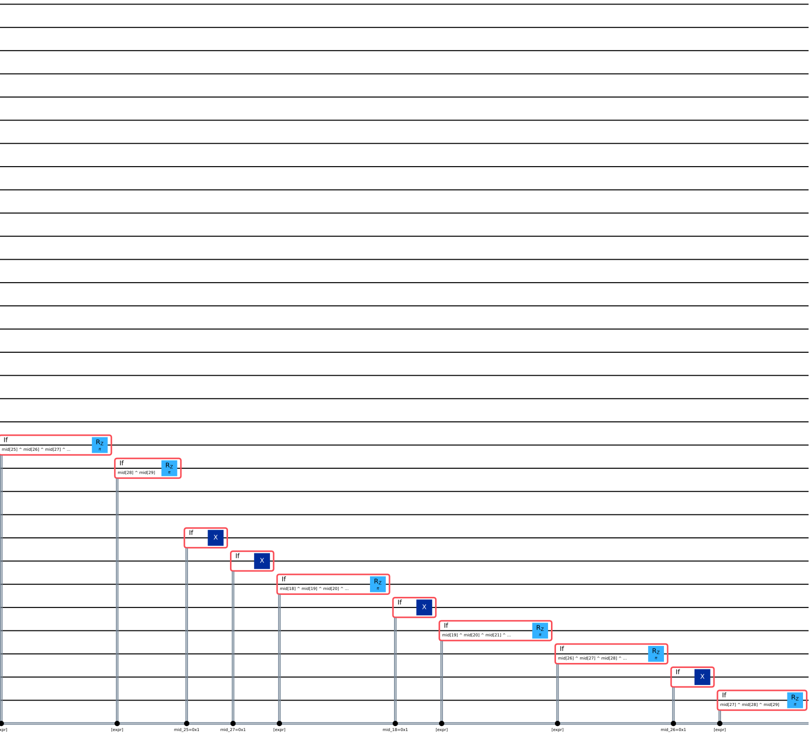


[Z16] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:150 rz:138 cz:60 if_else:31 measure:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 4, total depth 35

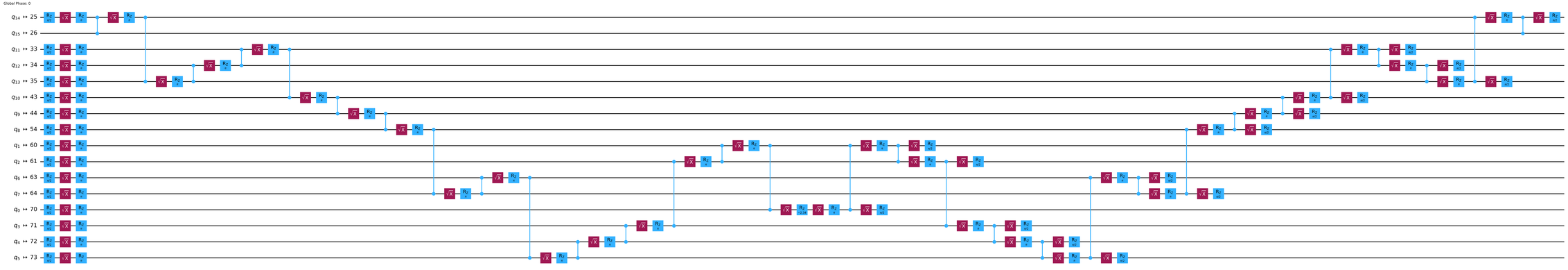

Global Phase: 0
$q_{42} \mapsto 42$
$q_{44} \mapsto 44$
$q_{45} \mapsto 45$
$q_{46} \mapsto 46$
$q_{48} \mapsto 48$
$q_{52} \mapsto 52$
$q_{54} \mapsto 54$
$q_{56} \mapsto 56$
$q_{58} \mapsto 58$
$q_{62} \mapsto 62$
$q_{64} \mapsto 64$
$q_{66} \mapsto 66$
$q_{68} \mapsto 68$
$q_{72} \mapsto 72$
$q_{74} \mapsto 74$
$q_{76} \mapsto 76$
$q_{78} \mapsto 78$
$q_{82} \mapsto 82$
$q_{84} \mapsto 84$
$q_{86} \mapsto 86$
$q_{88} \mapsto 88$
$q_{92} \mapsto 92$
$q_{94} \mapsto 94$
$q_{96} \mapsto 96$
$q_{98} \mapsto 98$
$q_{102} \mapsto 102$
$q_{103} \mapsto 103$
$q_{104} \mapsto 104$
$q_{106} \mapsto 106$
$q_{107} \mapsto 107$
$q_{108} \mapsto 108$
mid
30
R_Z
$\sqrt{X}$
π/2
−π/2
π
11
3
0
1
2
4
5
6
7
8
9
10
12
13
14


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
If
mid[15] ^ mid[16] ^ mid[17] ^ ...
If
mid[22] ^ mid[23] ^ mid[24] ^ ...
If
If
If
26
18
29
15
16
17
19
20
21
22
23
24
25
27
28
xpr]
[expr]
[expr]
mid_15=0x1
mid_21=0x1
mid_22=0x1


If
mid[23] ^ mid[24] ^ mid[25] ^ ...
R_Z
π
If
X
If
X
If
mid[16] ^ mid[17] ^ mid[18] ^ ...
R_Z
π
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
R_Z
π
If
X
If
X
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
R_Z
π
If
R_Z
π
If
X
If
X
If
mid[17] ^ mid[18] ^ mid[19] ^ ...
R_Z
π
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
R_Z
π
If
X
If
X
xpr]
mid_23=0x1
mid_29=0x1
[expr]
[expr]
mid_16=0x1
mid_20=0x1
[expr]
mid_29=0x1
mid_24=0x1
mid_28=0x1
[expr]
[expr]
mid_17=0x1
mid_19=0x1

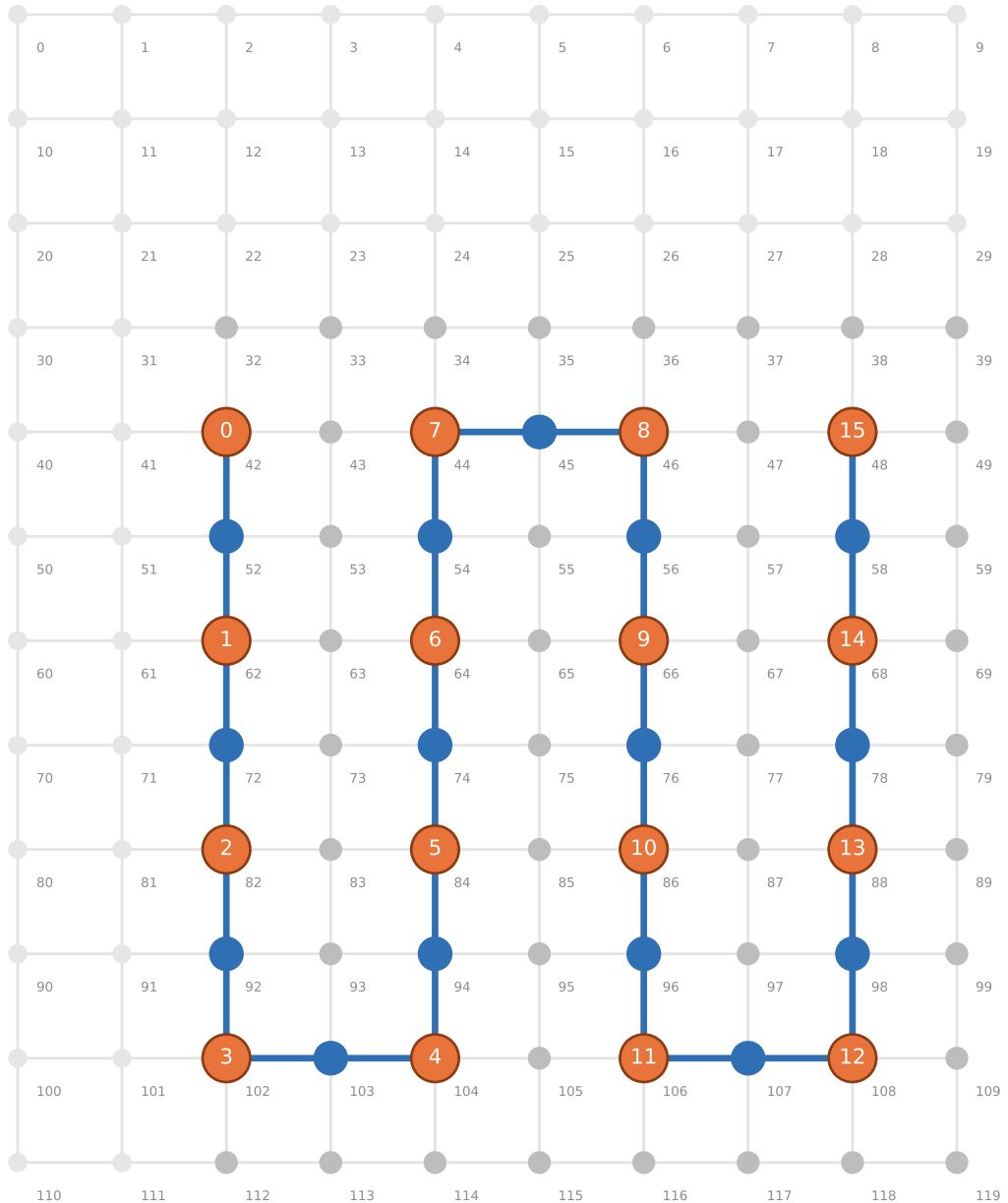

If
mid[25] ^ mid[26] ^ mid[27] ^ ...
RZ
π
If
mid[28] ^ mid[29]
RZ
π
If
X
If
X
If
mid[18] ^ mid[19] ^ mid[20] ^ ...
RZ
π
If
X
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
RZ
π
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
RZ
π
If
X
If
mid[27] ^ mid[28] ^ mid[29]
RZ
π
xpr]
[expr]
mid_25=0x1
mid_27=0x1
[expr]
mid_18=0x1
[expr]
[expr]
mid_26=0x1
[expr]

[Z16] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

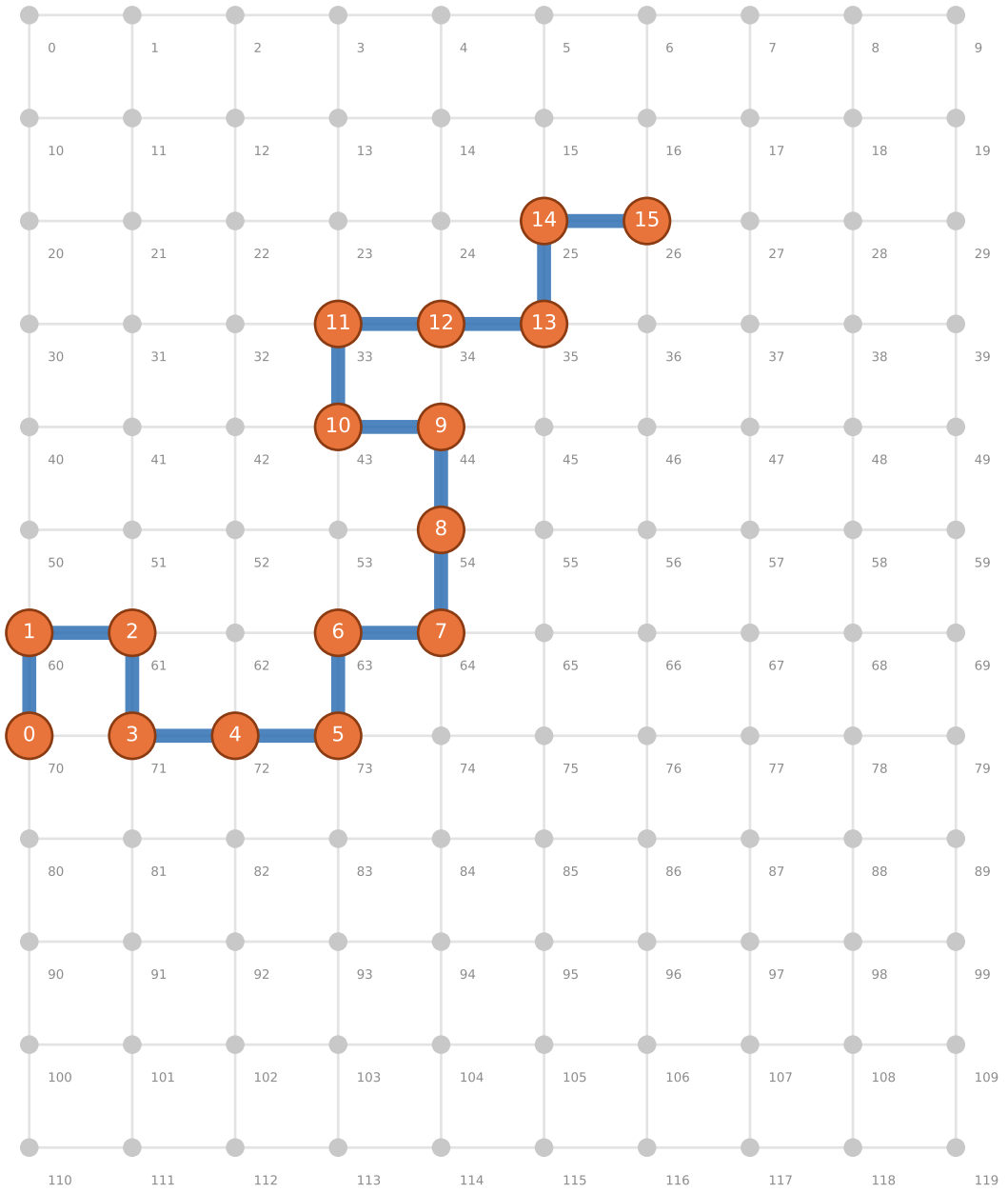


[Z16] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

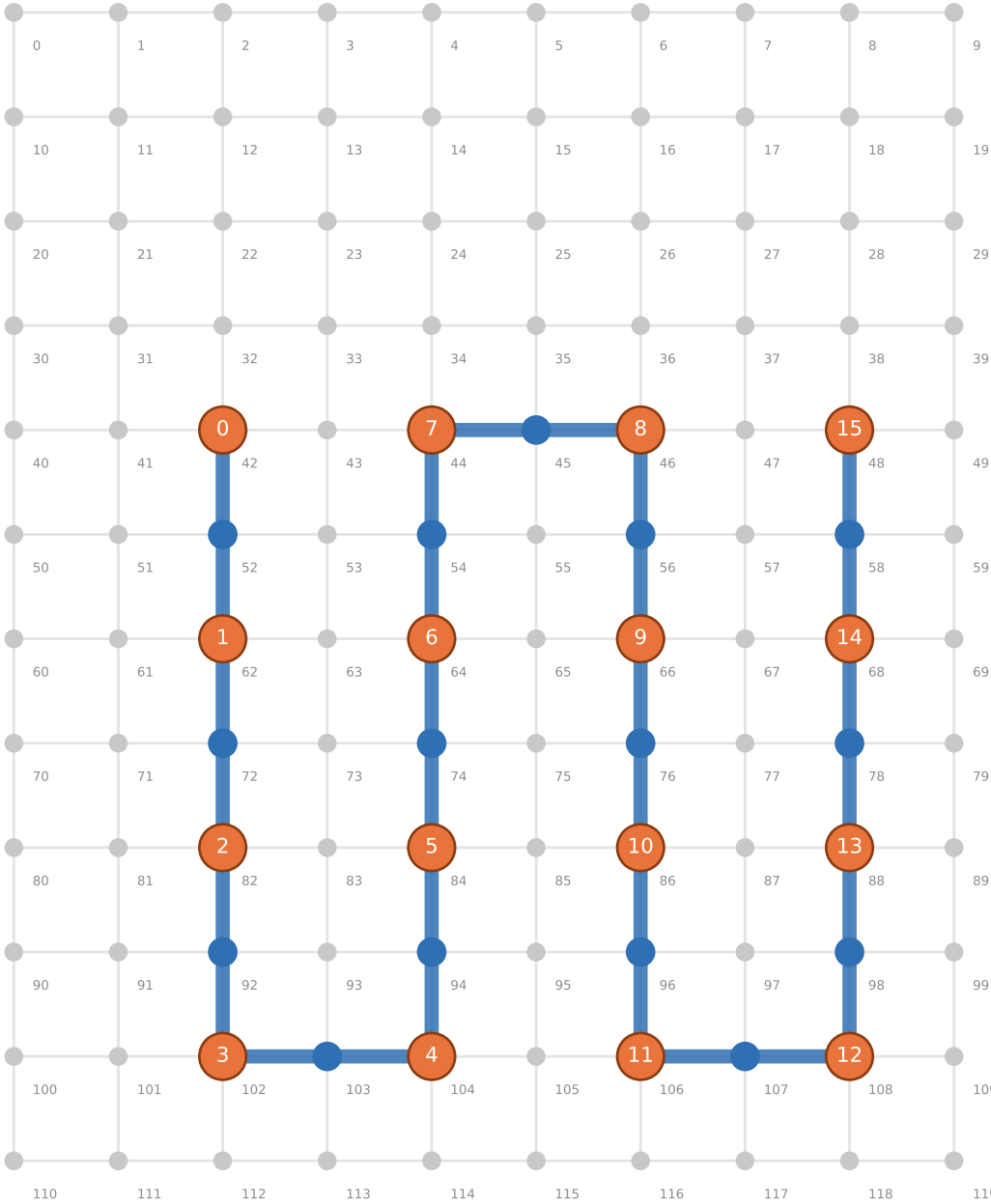
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers



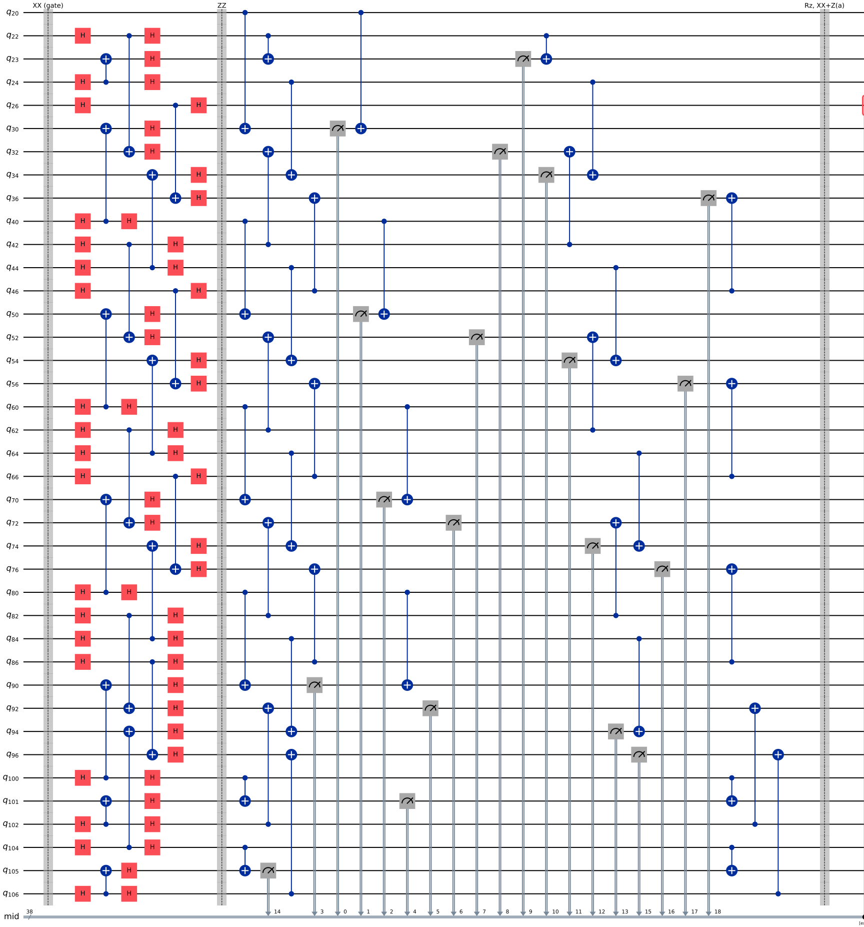
[Z16] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [70, 60, 61, 71, 72, 73, 63, 64, 54, 44, 43, 33, 34, 35, 25, 26] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)

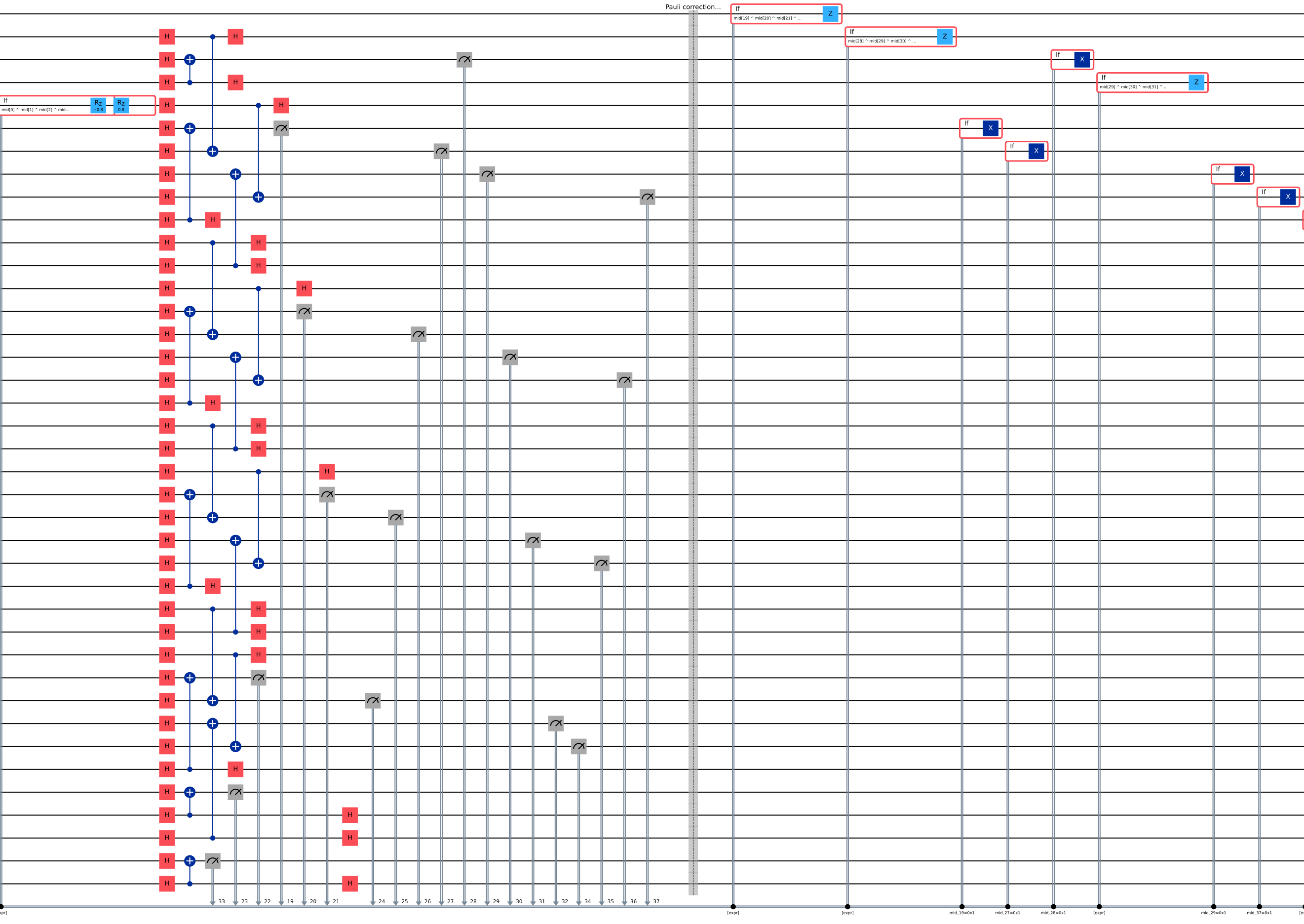


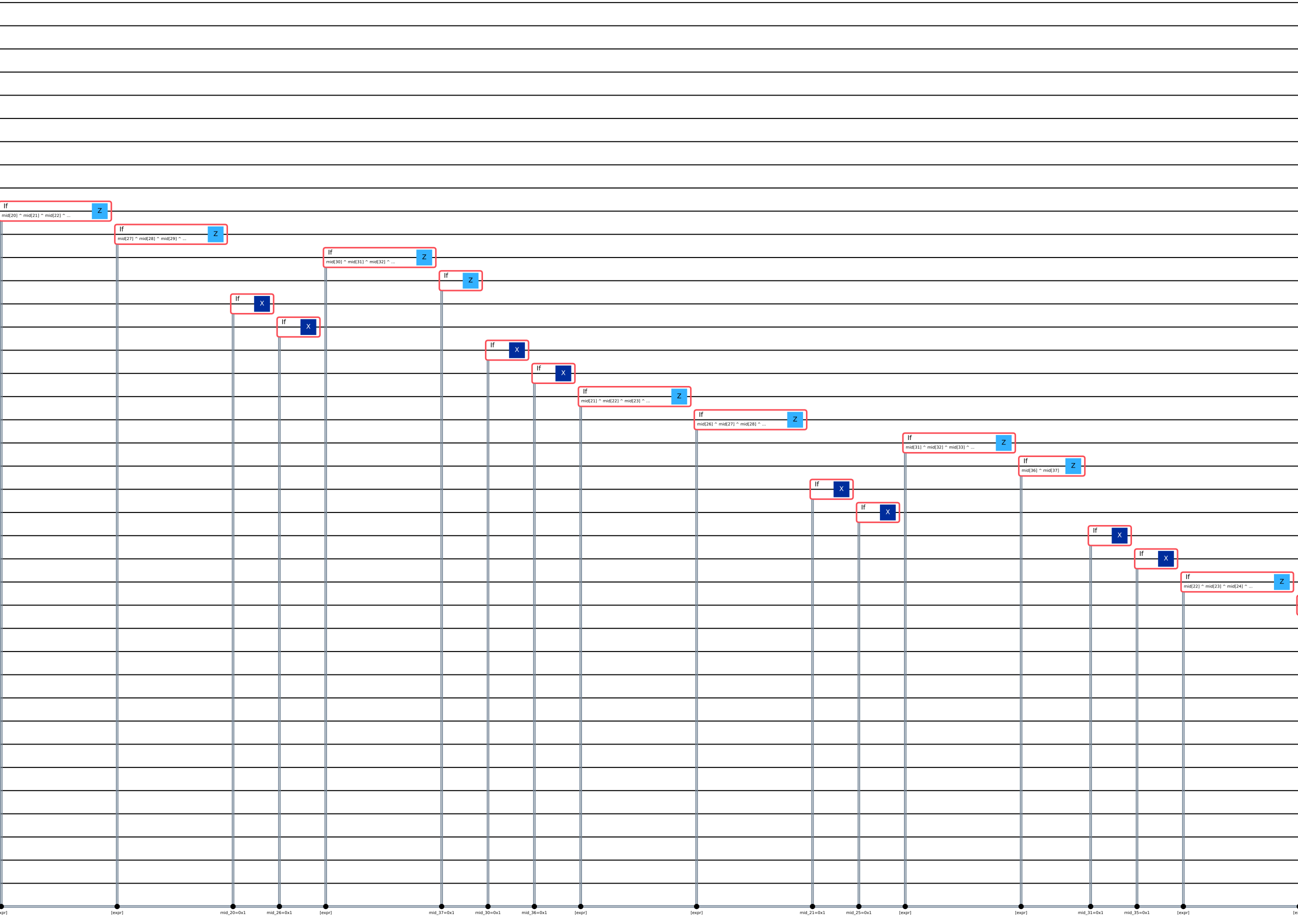
[Z16] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 60

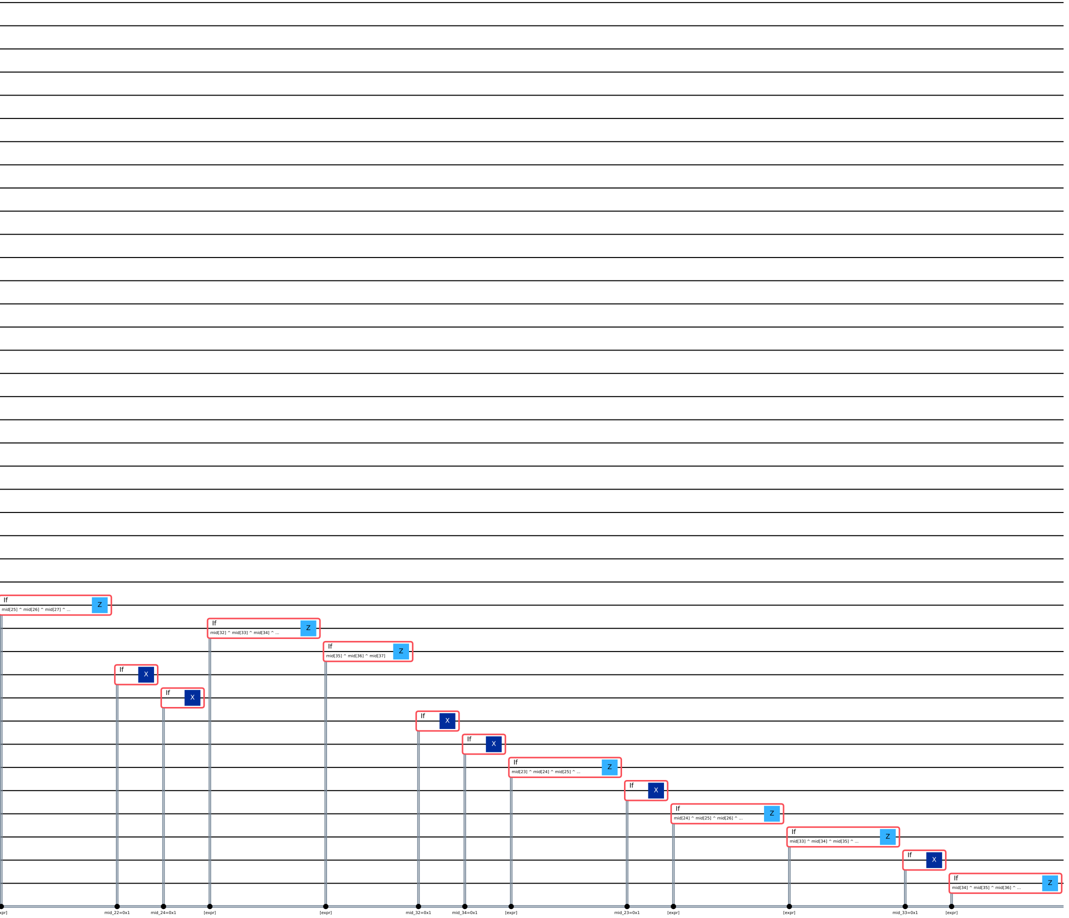


[Z20] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 4, 38 mid-circuit measurements, logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]

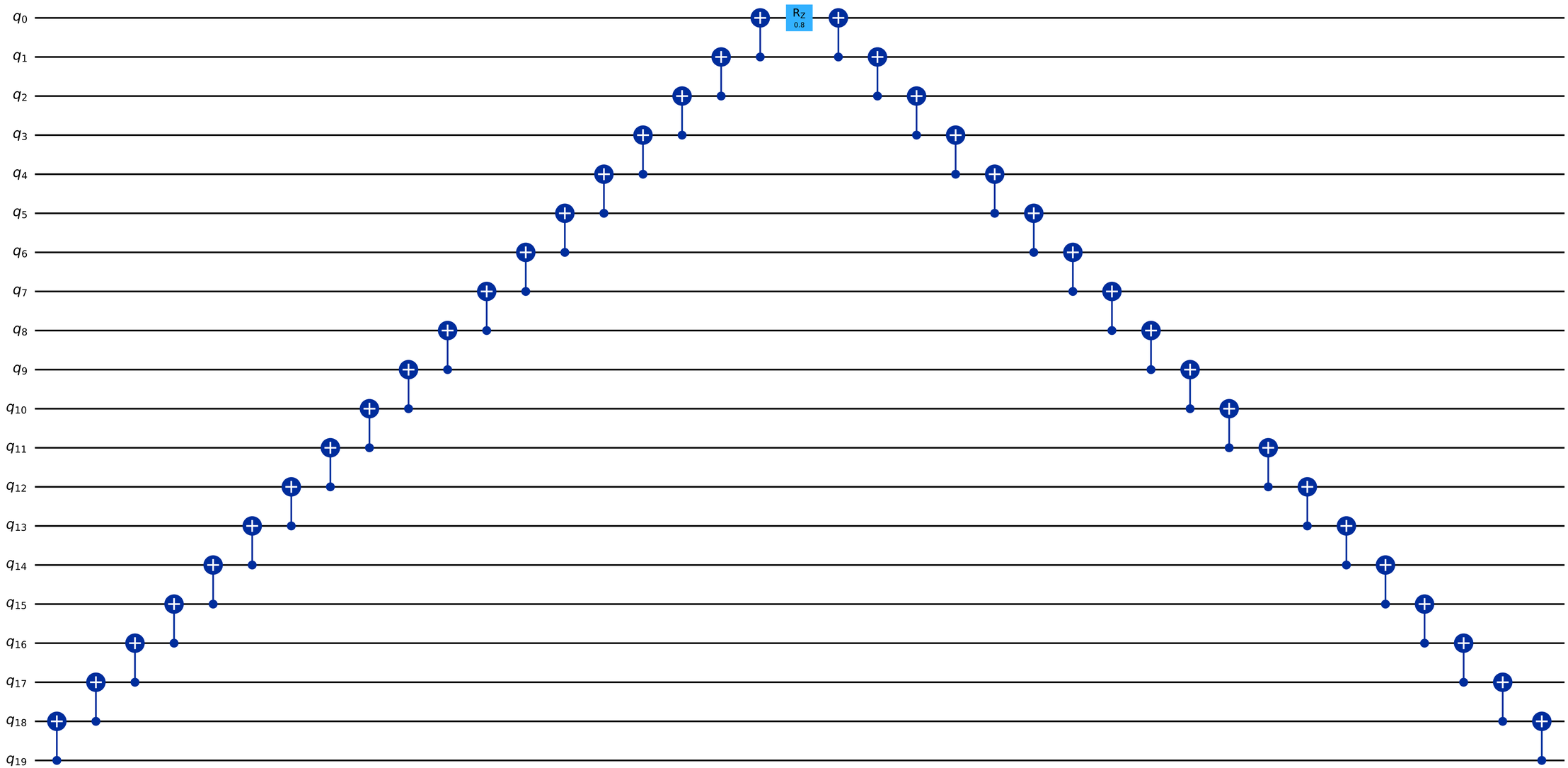

XX (gate)
ZZ
Rz, XX+Z(a)
q20
q22
q23
q24
q26
q30
q32
q34
q36
q40
q42
q44
q46
q50
q52
q54
q56
q60
q62
q64
q66
q70
q72
q74
q76
q80
q82
q84
q86
q90
q92
q94
q96
q100
q101
q102
q104
q105
q106
mid
38
H
14
3
0
1
2
4
5
6
7
8
9
10
11
12
13
15
16
17
18


Pauli correction...
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
Z
If
mid[28] ^ mid[29] ^ mid[30] ^ ...
Z
If
X
If
mid[29] ^ mid[30] ^ mid[31] ^ ...
Z
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z
−0.8
R_Z
0.8
If
X
If
X
If
X
If
X
H
33
23
22
19
20
21
24
25
26
27
28
29
30
31
32
34
35
36
37
[expr]
[expr]
mid_19=0x1
mid_27=0x1
mid_28=0x1
[expr]
mid_29=0x1
mid_37=0x1

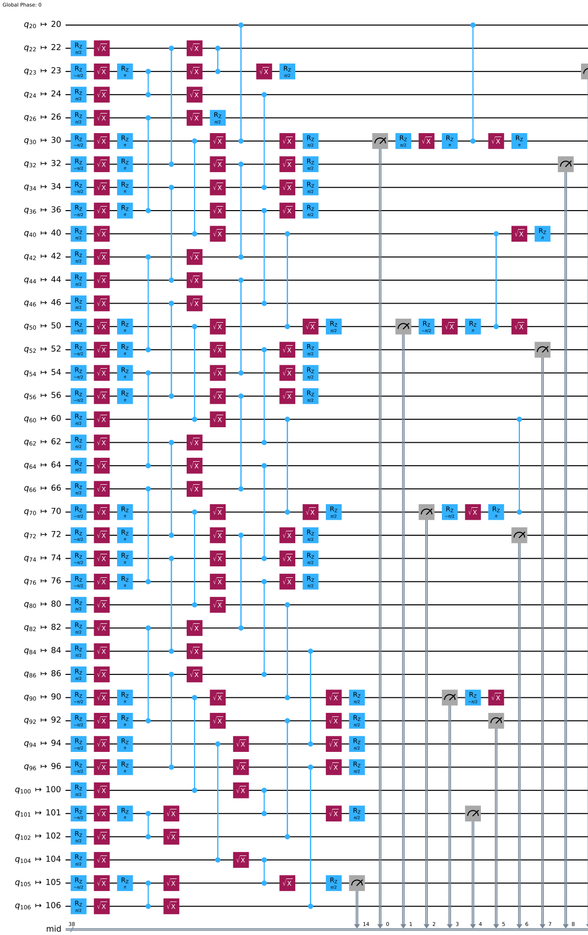

If
mid[20] ^ mid[21] ^ mid[22] ^ ...
Z
If
mid[27] ^ mid[28] ^ mid[29] ^ ...
Z
If
X
If
X
If
mid[30] ^ mid[31] ^ mid[32] ^ ...
Z
If
Z
If
X
If
X
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
Z
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
Z
If
X
If
X
If
mid[31] ^ mid[32] ^ mid[33] ^ ...
Z
If
mid[36] ^ mid[37]
Z
If
X
If
X
If
mid[22] ^ mid[23] ^ mid[24] ^ ...
Z
[expr]
[expr]
mid_20=0x1
mid_26=0x1
[expr]
mid_37=0x1
mid_30=0x1
mid_36=0x1
[expr]
[expr]
mid_21=0x1
mid_25=0x1
[expr]
[expr]
mid_31=0x1
mid_35=0x1
[expr]

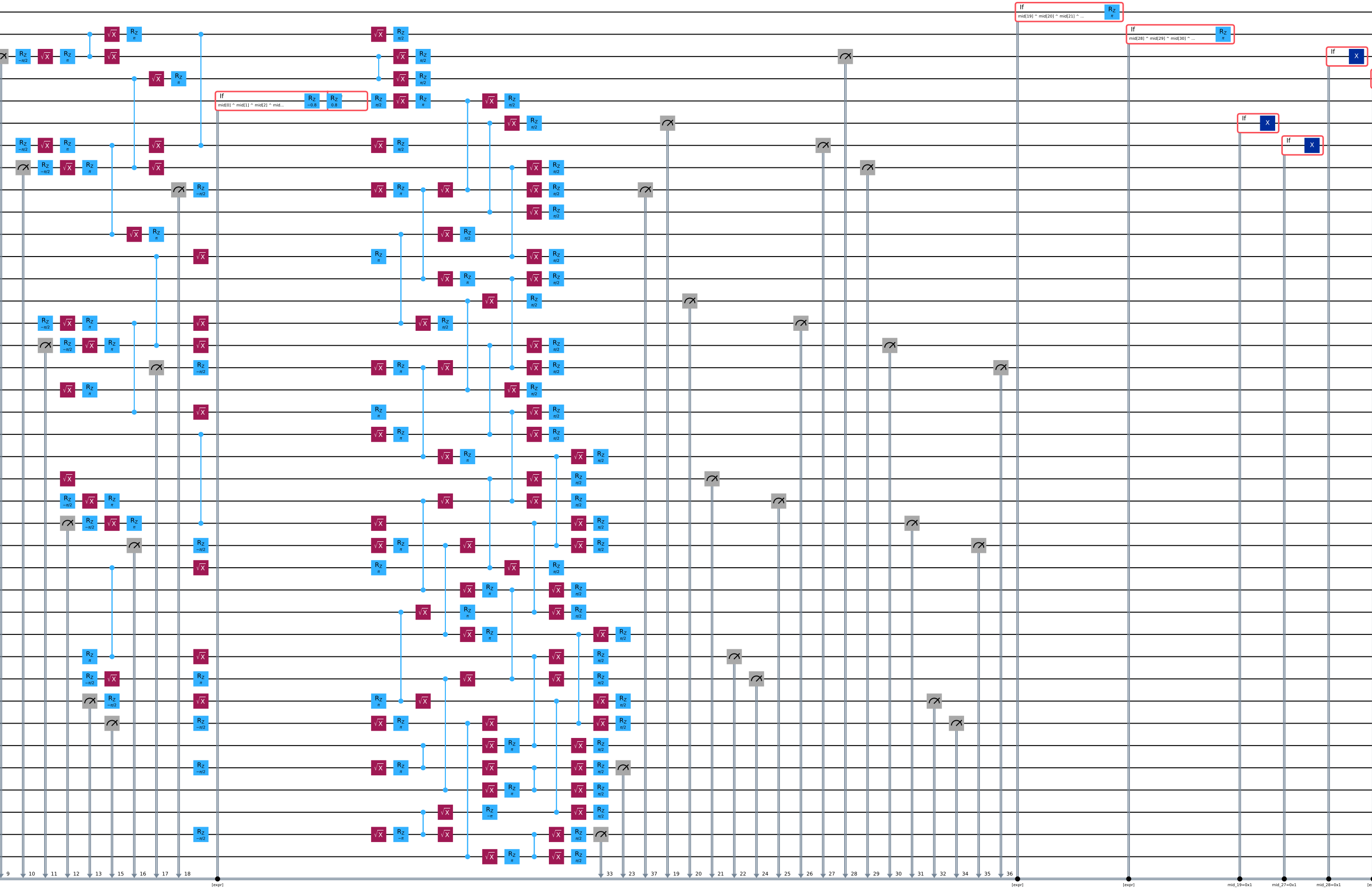

If
mid[25] ^ mid[26] ^ mid[27] ^ ...
Z
If
X
If
X
If
mid[32] ^ mid[33] ^ mid[34] ^ ...
Z
If
mid[35] ^ mid[36] ^ mid[37]
Z
If
X
If
X
If
mid[23] ^ mid[24] ^ mid[25] ^ ...
Z
If
X
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
Z
If
mid[33] ^ mid[34] ^ mid[35] ^ ...
Z
If
X
If
mid[34] ^ mid[35] ^ mid[36] ^ ...
Z
xpr]
mid_22=0x1
mid_24=0x1
[expr]
[expr]
mid_32=0x1
mid_34=0x1
[expr]
mid_23=0x1
[expr]
[expr]
mid_33=0x1
[expr]

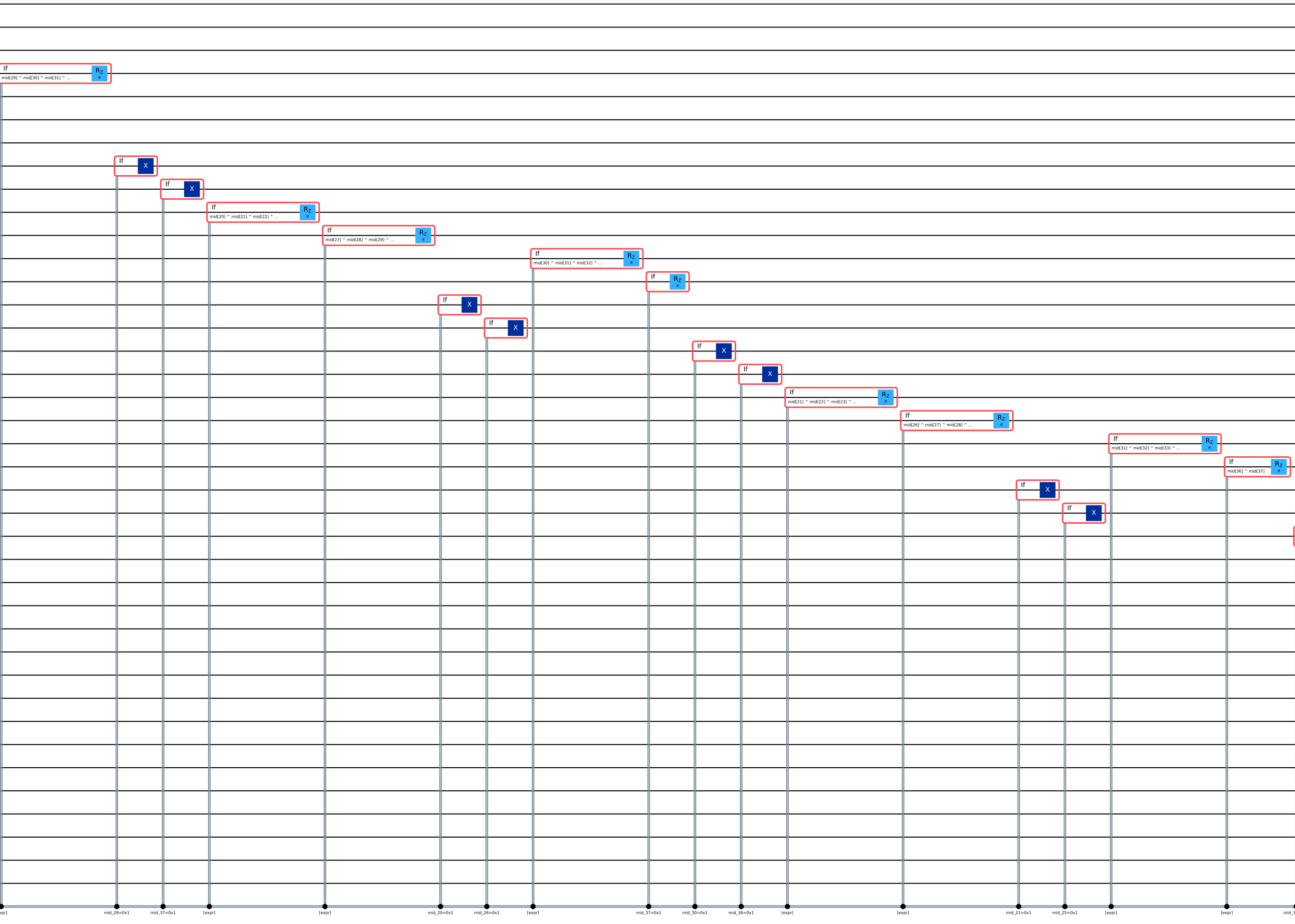
[Z20] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

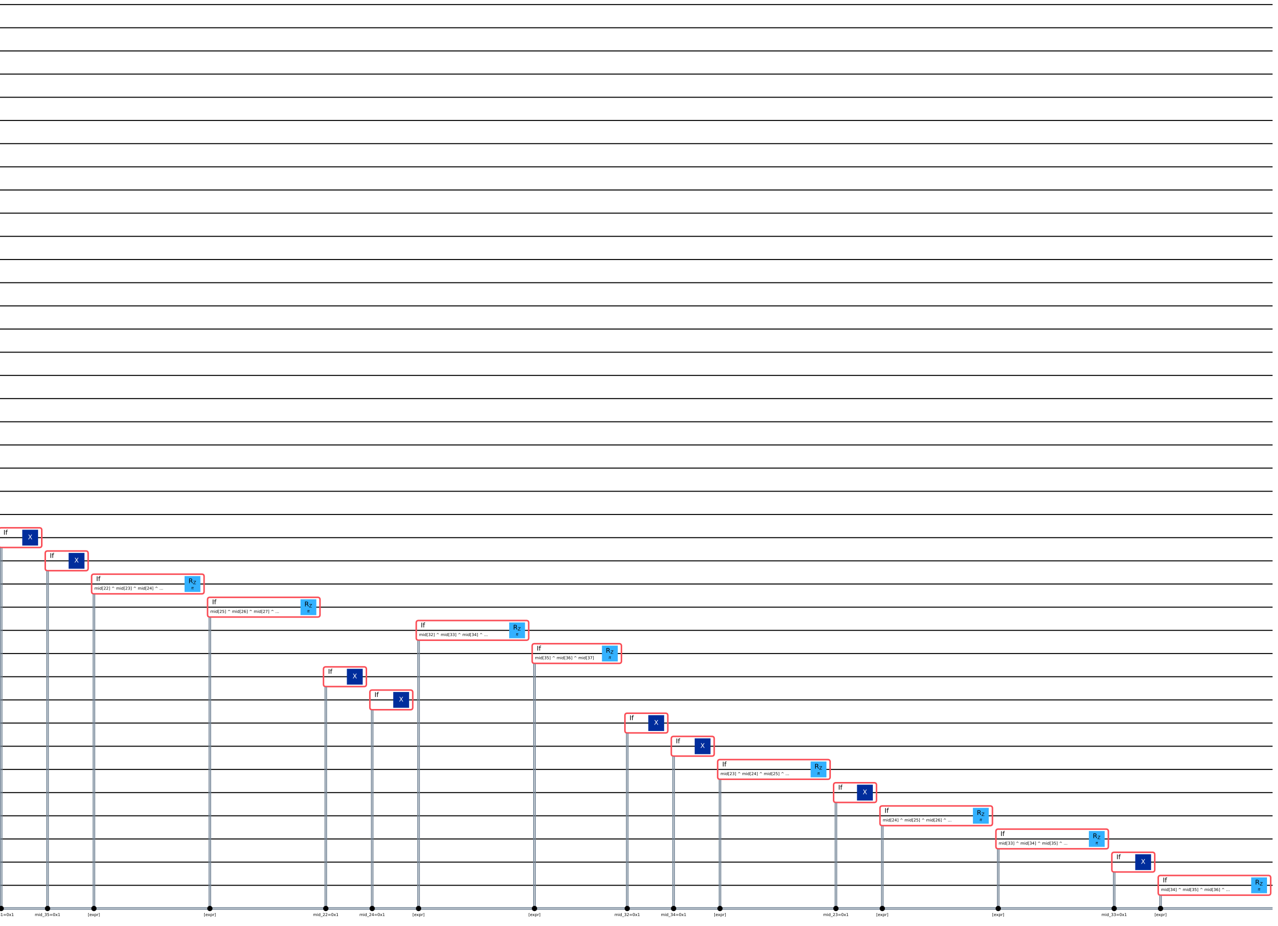


[Z20] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:190 rz:174 cz:76 if_else:39 measure:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 4, total depth 39

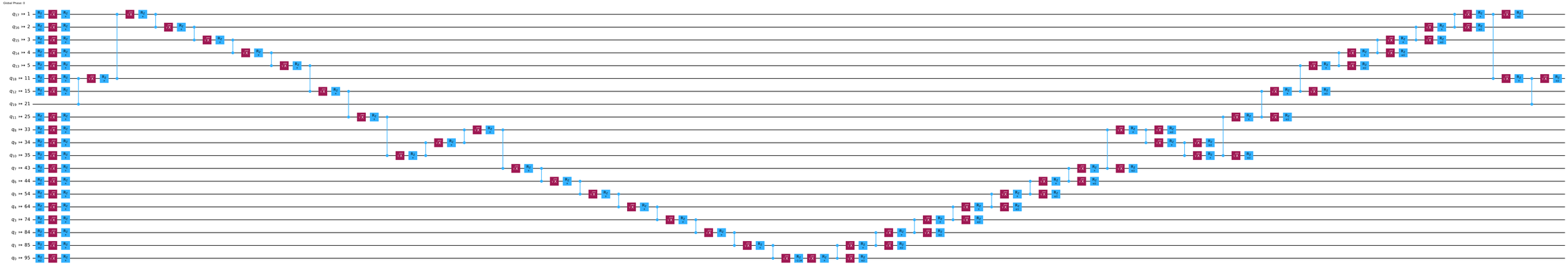

Global Phase: 0
q_{20} ↦ 20
q_{22} ↦ 22
q_{23} ↦ 23
q_{24} ↦ 24
q_{26} ↦ 26
q_{30} ↦ 30
q_{32} ↦ 32
q_{34} ↦ 34
q_{36} ↦ 36
q_{40} ↦ 40
q_{42} ↦ 42
q_{44} ↦ 44
q_{46} ↦ 46
q_{50} ↦ 50
q_{52} ↦ 52
q_{54} ↦ 54
q_{56} ↦ 56
q_{60} ↦ 60
q_{62} ↦ 62
q_{64} ↦ 64
q_{66} ↦ 66
q_{70} ↦ 70
q_{72} ↦ 72
q_{74} ↦ 74
q_{76} ↦ 76
q_{80} ↦ 80
q_{82} ↦ 82
q_{84} ↦ 84
q_{86} ↦ 86
q_{90} ↦ 90
q_{92} ↦ 92
q_{94} ↦ 94
q_{96} ↦ 96
q_{100} ↦ 100
q_{101} ↦ 101
q_{102} ↦ 102
q_{104} ↦ 104
q_{105} ↦ 105
q_{106} ↦ 106
mid
38
R_Z
π/2
−π/2
π
√X
14
0
1
2
3
4
5
6
7
8


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z −0.8
R_Z 0.8
If
mid[19] ^ mid[20] ^ mid[21] ^ ...
R_Z π
If
mid[28] ^ mid[29] ^ mid[30] ^ ...
R_Z π
If
X
If
X
If
X
√X
R_Z π
R_Z π/2
R_Z −π/2
R_Z −π
9
10
11
12
13
14
15
16
17
18
[expr]
33
23
37
19
20
21
22
24
25
26
27
28
29
30
31
32
34
35
36
[expr]
[expr]
mid_19=0x1
mid_27=0x1
mid_28=0x1


If
mid[29] ^ mid[30] ^ mid[31] ^ ...
Rz π
If X
If X
If
mid[20] ^ mid[21] ^ mid[22] ^ ...
Rz π
If
mid[27] ^ mid[28] ^ mid[29] ^ ...
Rz π
If X
If X
If
mid[30] ^ mid[31] ^ mid[32] ^ ...
Rz π
If Rz π
If X
If X
If
mid[21] ^ mid[22] ^ mid[23] ^ ...
Rz π
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
Rz π
If X
If X
If
mid[31] ^ mid[32] ^ mid[33] ^ ...
Rz π
If
mid[36] ^ mid[37]
Rz π
[expr]
mid_29=0x1
mid_37=0x1
[expr]
[expr]
mid_20=0x1
mid_26=0x1
[expr]
mid_37=0x1
mid_30=0x1
mid_36=0x1
[expr]
[expr]
mid_21=0x1
mid_25=0x1
[expr]
[expr]

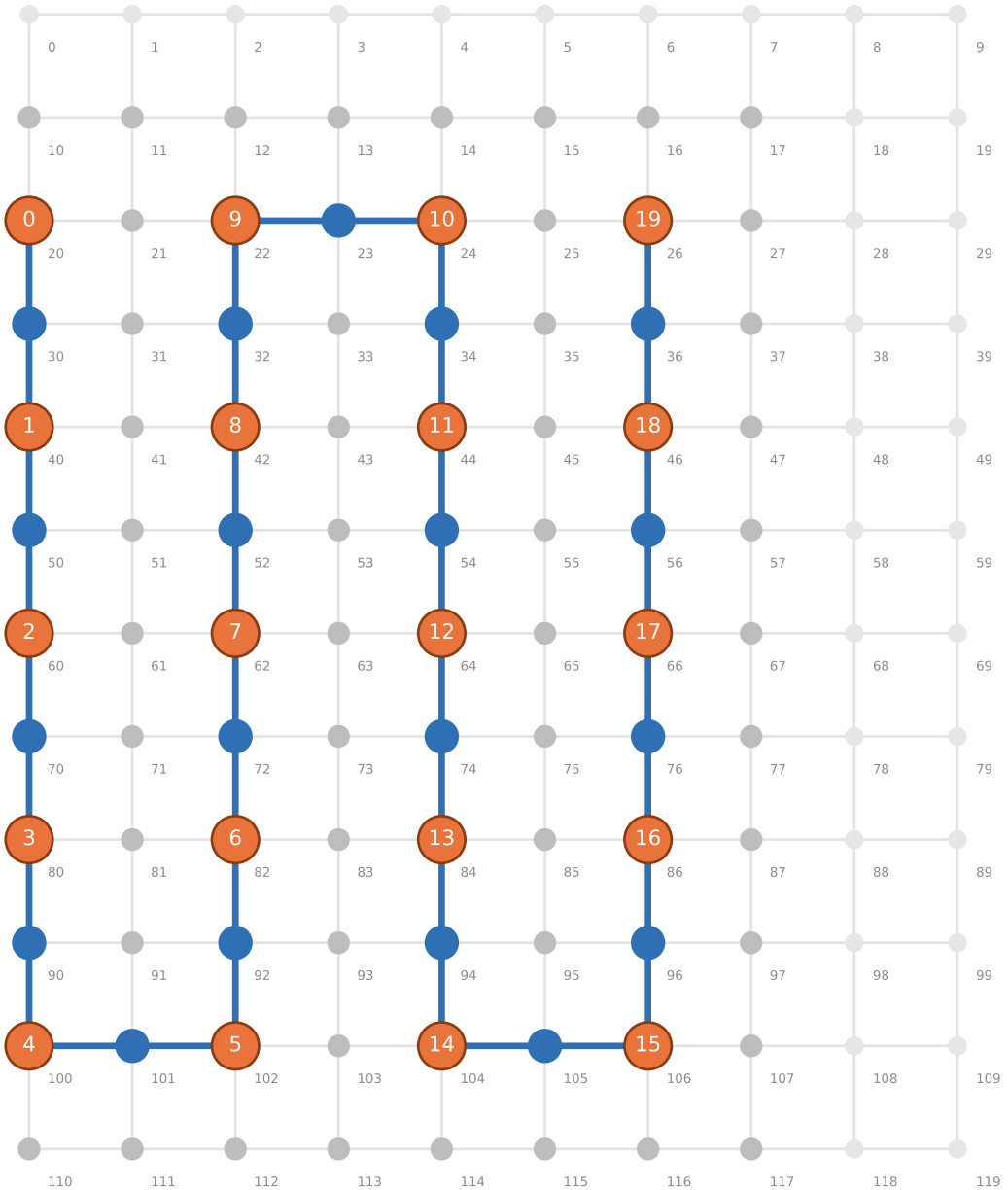

If
X
If
X
If
mid[22] ^ mid[23] ^ mid[24] ^ ...
R_Z
π
If
mid[25] ^ mid[26] ^ mid[27] ^ ...
R_Z
π
If
mid[32] ^ mid[33] ^ mid[34] ^ ...
R_Z
π
If
mid[35] ^ mid[36] ^ mid[37]
R_Z
π
If
X
If
X
If
X
If
X
If
mid[23] ^ mid[24] ^ mid[25] ^ ...
R_Z
π
If
X
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
R_Z
π
If
mid[33] ^ mid[34] ^ mid[35] ^ ...
R_Z
π
If
X
If
mid[34] ^ mid[35] ^ mid[36] ^ ...
R_Z
π
1=0x1
mid_35=0x1
[expr]
[expr]
mid_22=0x1
mid_24=0x1
[expr]
[expr]
mid_32=0x1
mid_34=0x1
[expr]
mid_23=0x1
[expr]
[expr]
mid_33=0x1
[expr]

[Z20] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119

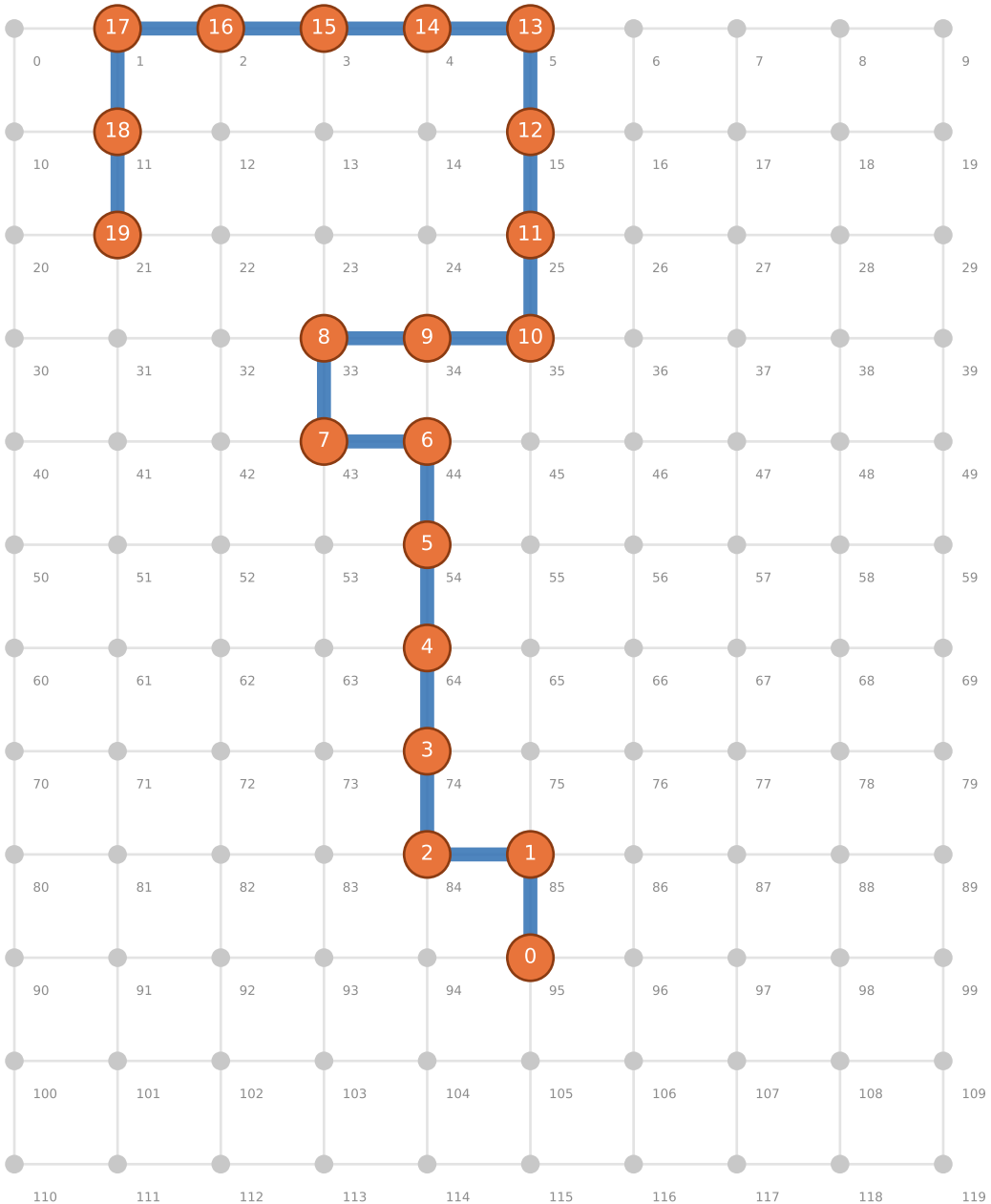


[Z20] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

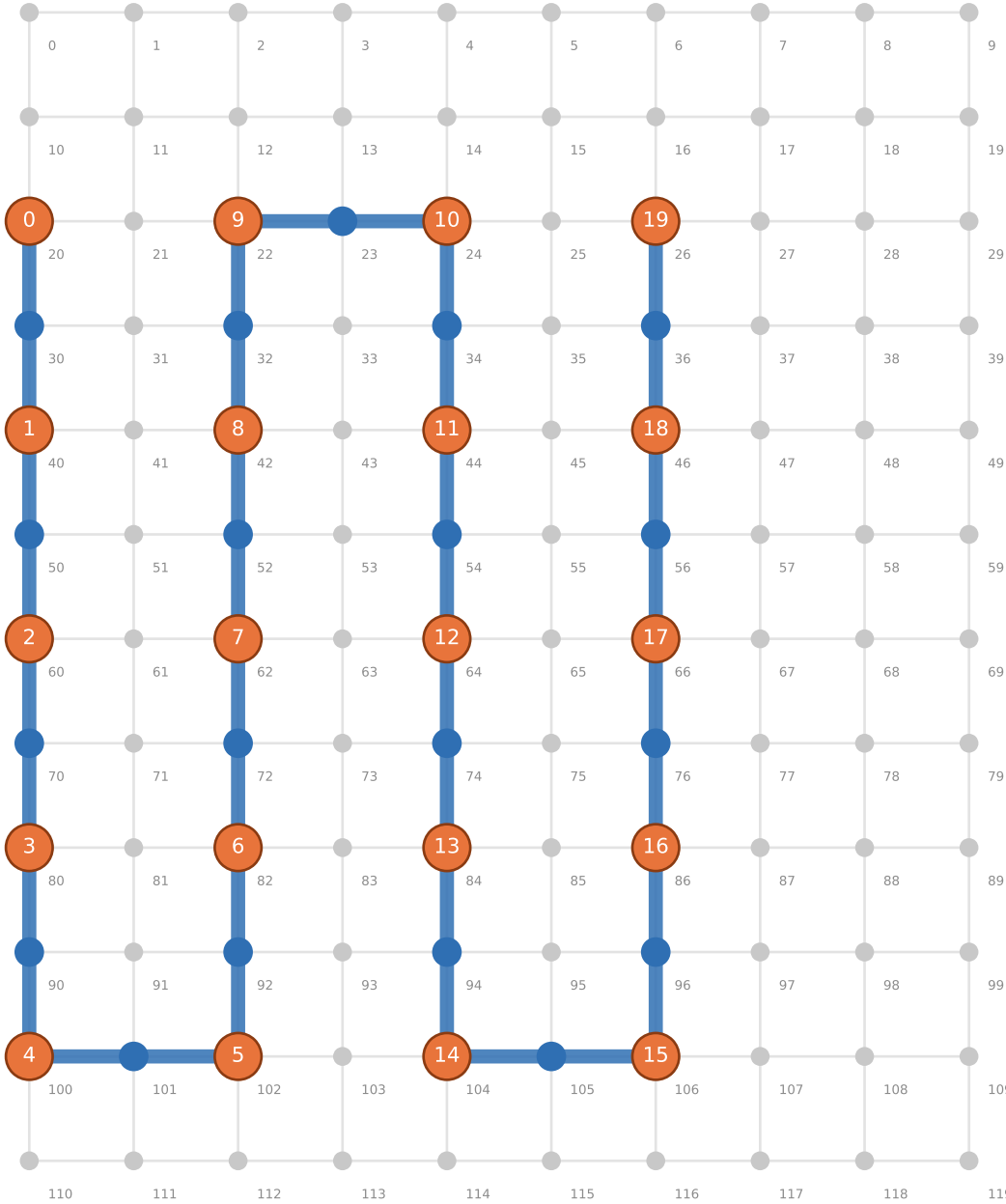
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers



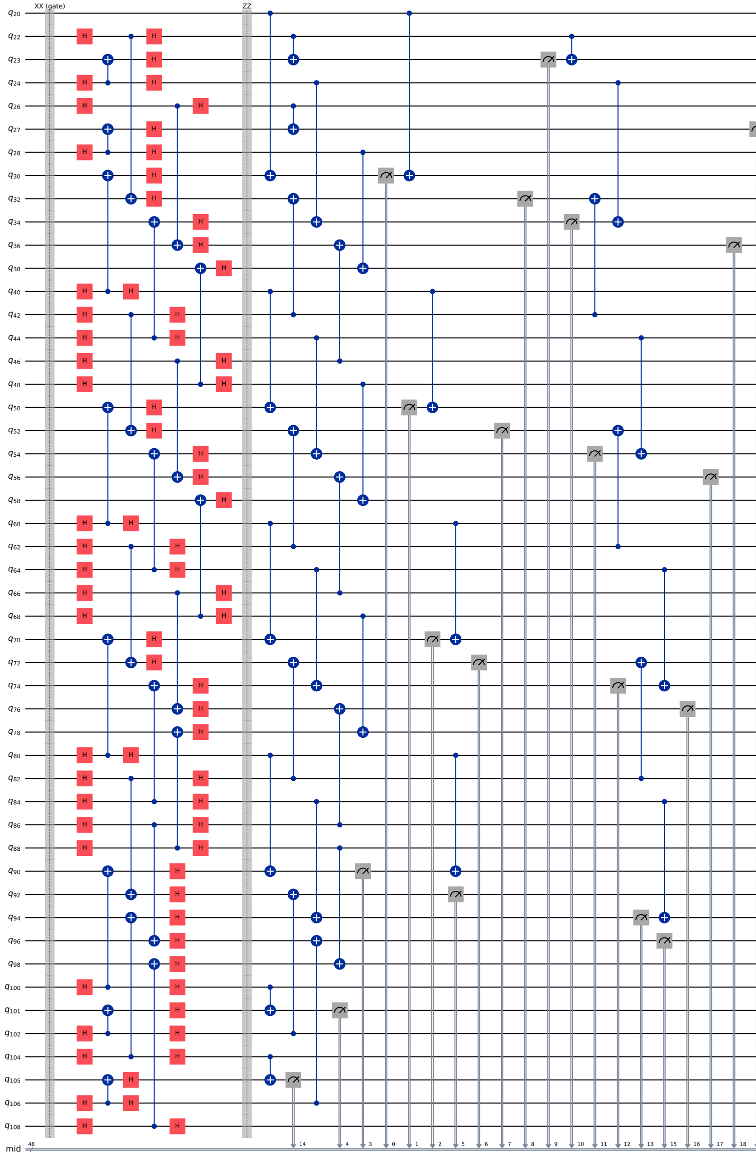
[Z20] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 74, 64, 54, 44, 43, 33, 34, 35, 25, 15, 5, 4, 3, 2, 1, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)

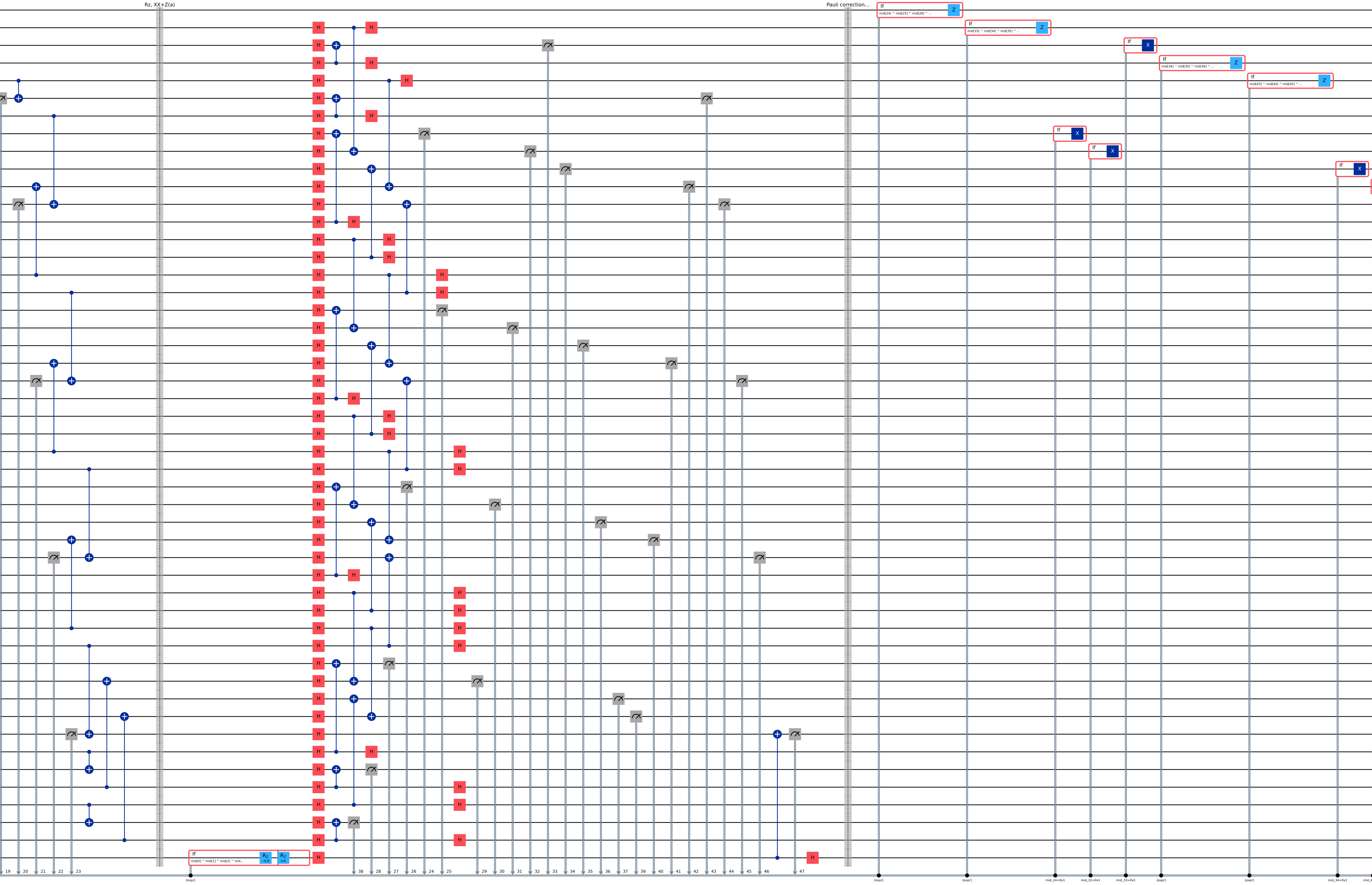


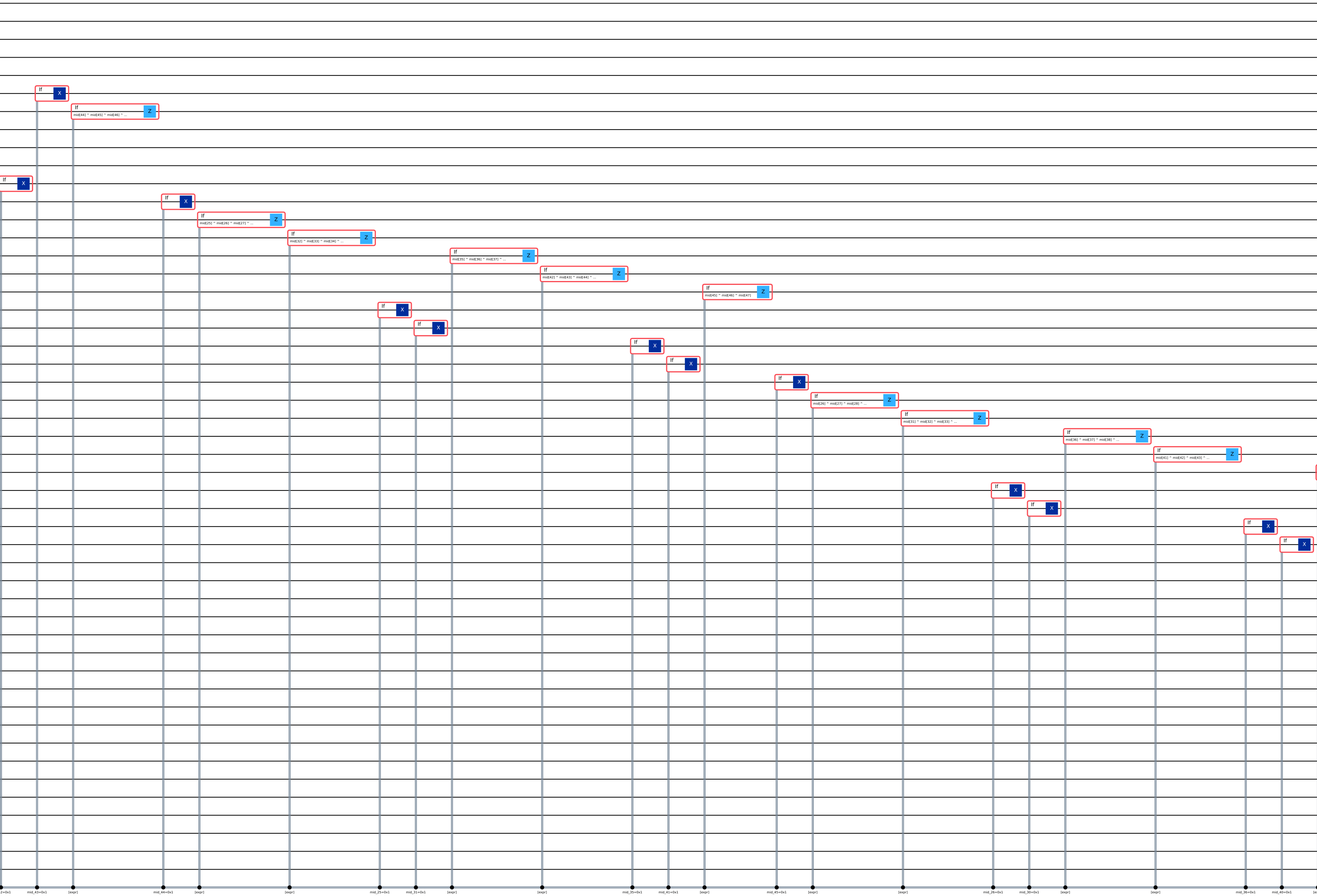
[Z20] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 76

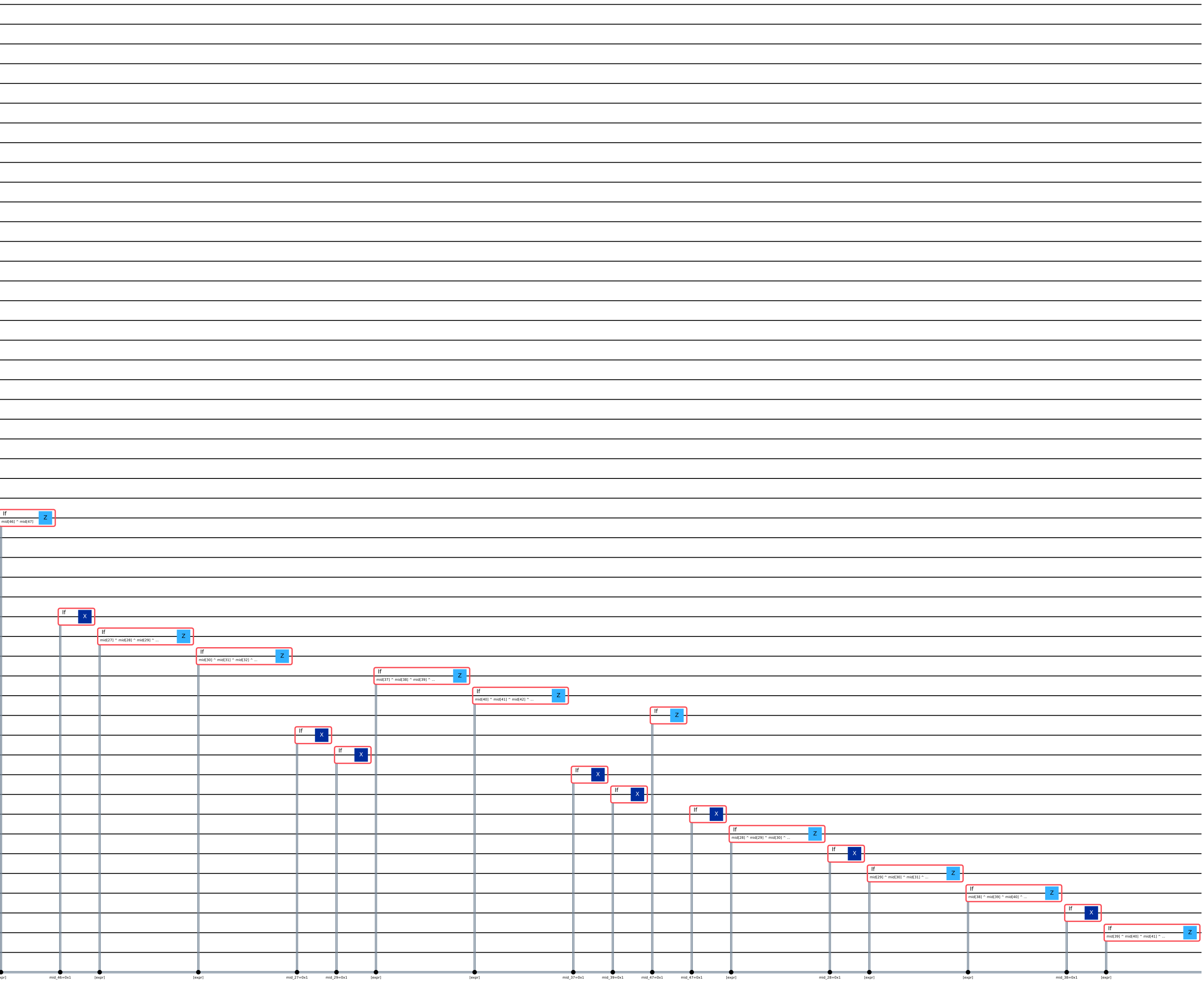


[Z25] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 4, 48 mid-circuit measurements, logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]

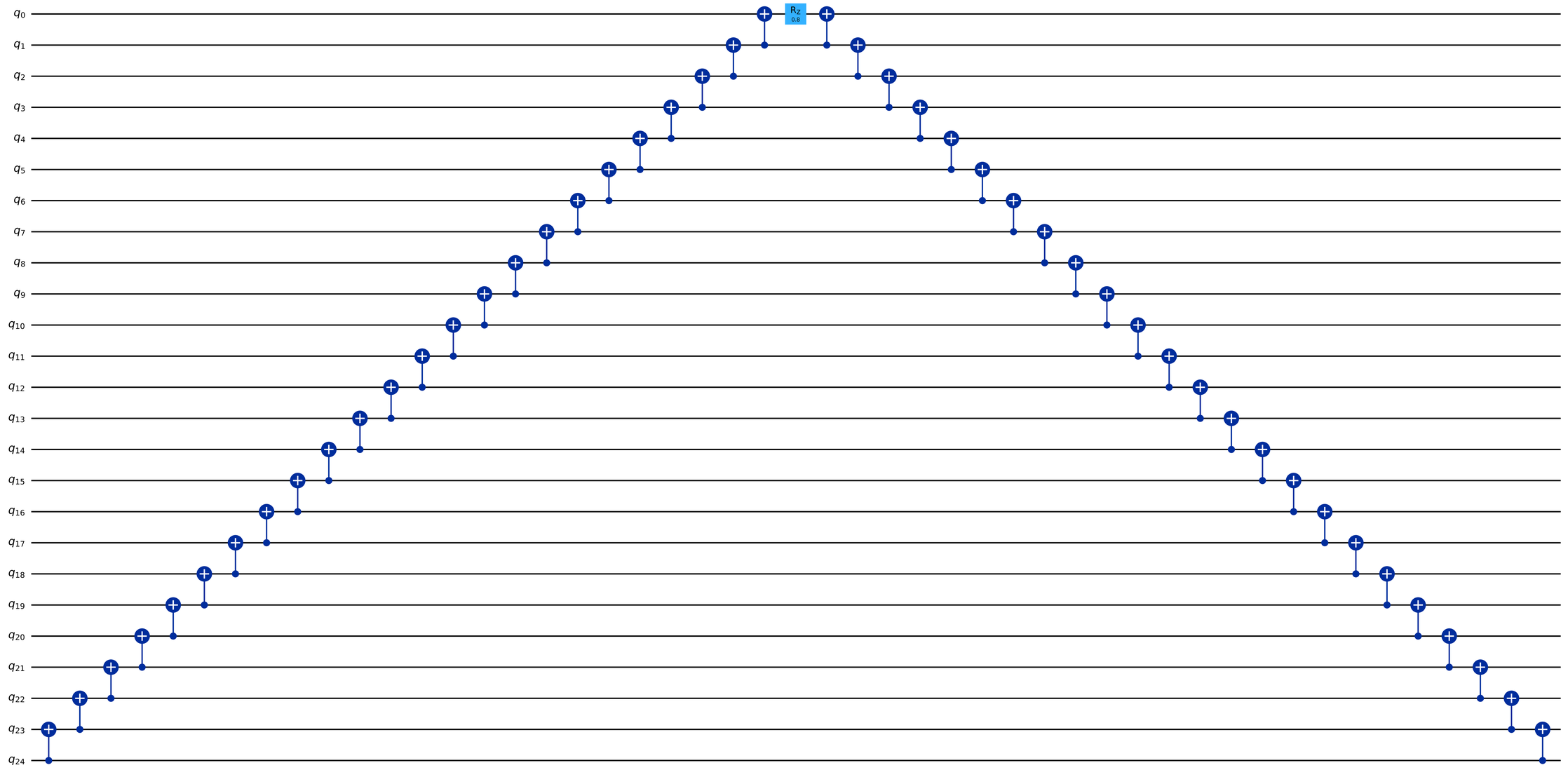

XX (gate)
ZZ
q_{20}
q_{22}
q_{23}
q_{24}
q_{26}
q_{27}
q_{28}
q_{30}
q_{32}
q_{34}
q_{36}
q_{38}
q_{40}
q_{42}
q_{44}
q_{46}
q_{48}
q_{50}
q_{52}
q_{54}
q_{56}
q_{58}
q_{60}
q_{62}
q_{64}
q_{66}
q_{68}
q_{70}
q_{72}
q_{74}
q_{76}
q_{78}
q_{80}
q_{82}
q_{84}
q_{86}
q_{88}
q_{90}
q_{92}
q_{94}
q_{96}
q_{98}
q_{100}
q_{101}
q_{102}
q_{104}
q_{105}
q_{106}
q_{108}
mid
48
H
14
4
3
0
1
2
5
6
7
8
9
10
11
12
13
15
16
17
18


Rz, XX+Z(a)
Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Rz −0.8
Rz 0.8
If
mid[24] ^ mid[25] ^ mid[26] ^ ...
Z
If
mid[33] ^ mid[34] ^ mid[35] ^ ...
Z
If
X
If
mid[34] ^ mid[35] ^ mid[36] ^ ...
Z
If
mid[43] ^ mid[44] ^ mid[45] ^ ...
Z
If
X
If
X
If
X
H
[expr]
mid_24=0x1
mid_32=0x1
mid_33=0x1
mid_34=0x1
19 20 21 22 23 38 28 27 26 24 25 29 30 31 32 33 34 35 36 37 39 40 41 42 43 44 45 46 47

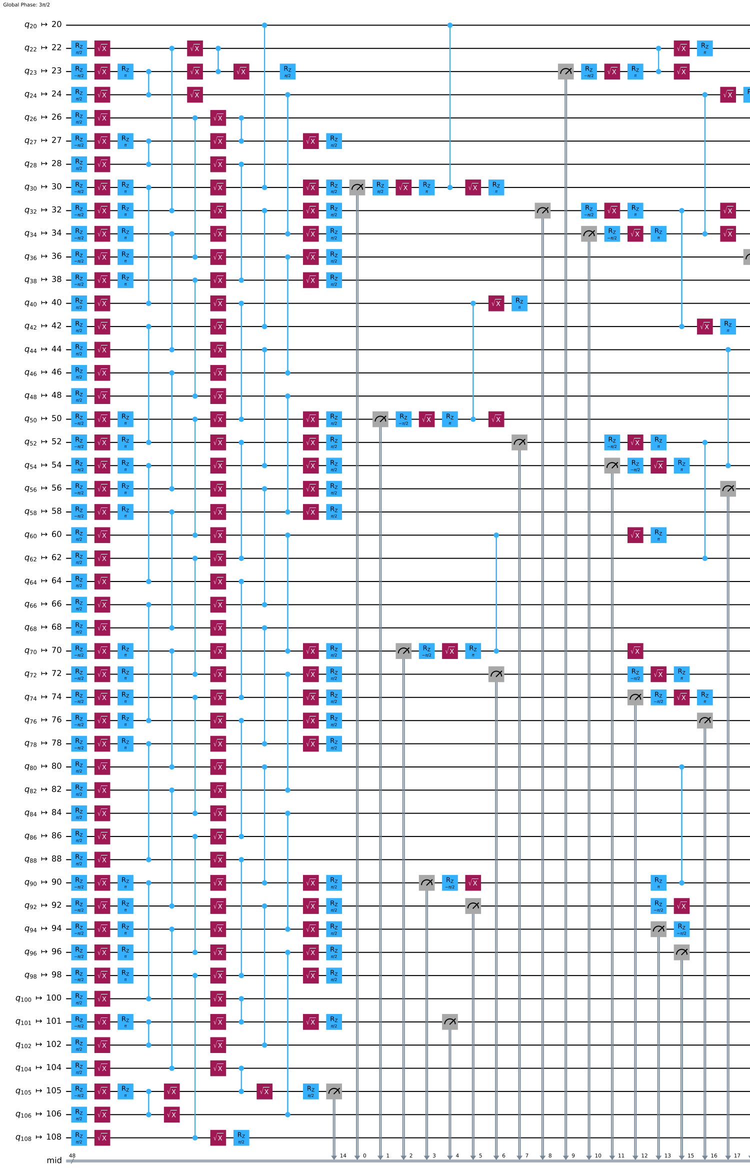

If
X
If
mid[44] ^ mid[45] ^ mid[46] ^ ...
Z
If
X
If
X
If
mid[25] ^ mid[26] ^ mid[27] ^ ...
Z
If
mid[32] ^ mid[33] ^ mid[34] ^ ...
Z
If
mid[35] ^ mid[36] ^ mid[37] ^ ...
Z
If
mid[42] ^ mid[43] ^ mid[44] ^ ...
Z
If
mid[45] ^ mid[46] ^ mid[47]
Z
If
X
If
X
If
X
If
X
If
X
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
Z
If
mid[31] ^ mid[32] ^ mid[33] ^ ...
Z
If
mid[36] ^ mid[37] ^ mid[38] ^ ...
Z
If
mid[41] ^ mid[42] ^ mid[43] ^ ...
Z
If
X
If
X
If
X
If
X
2=0x1
mid_43=0x1
[expr]
mid_44=0x1
[expr]
[expr]
mid_25=0x1
mid_31=0x1
[expr]
[expr]
mid_35=0x1
mid_41=0x1
[expr]
mid_45=0x1
[expr]
[expr]
mid_26=0x1
mid_30=0x1
[expr]
[expr]
mid_36=0x1
mid_40=0x1

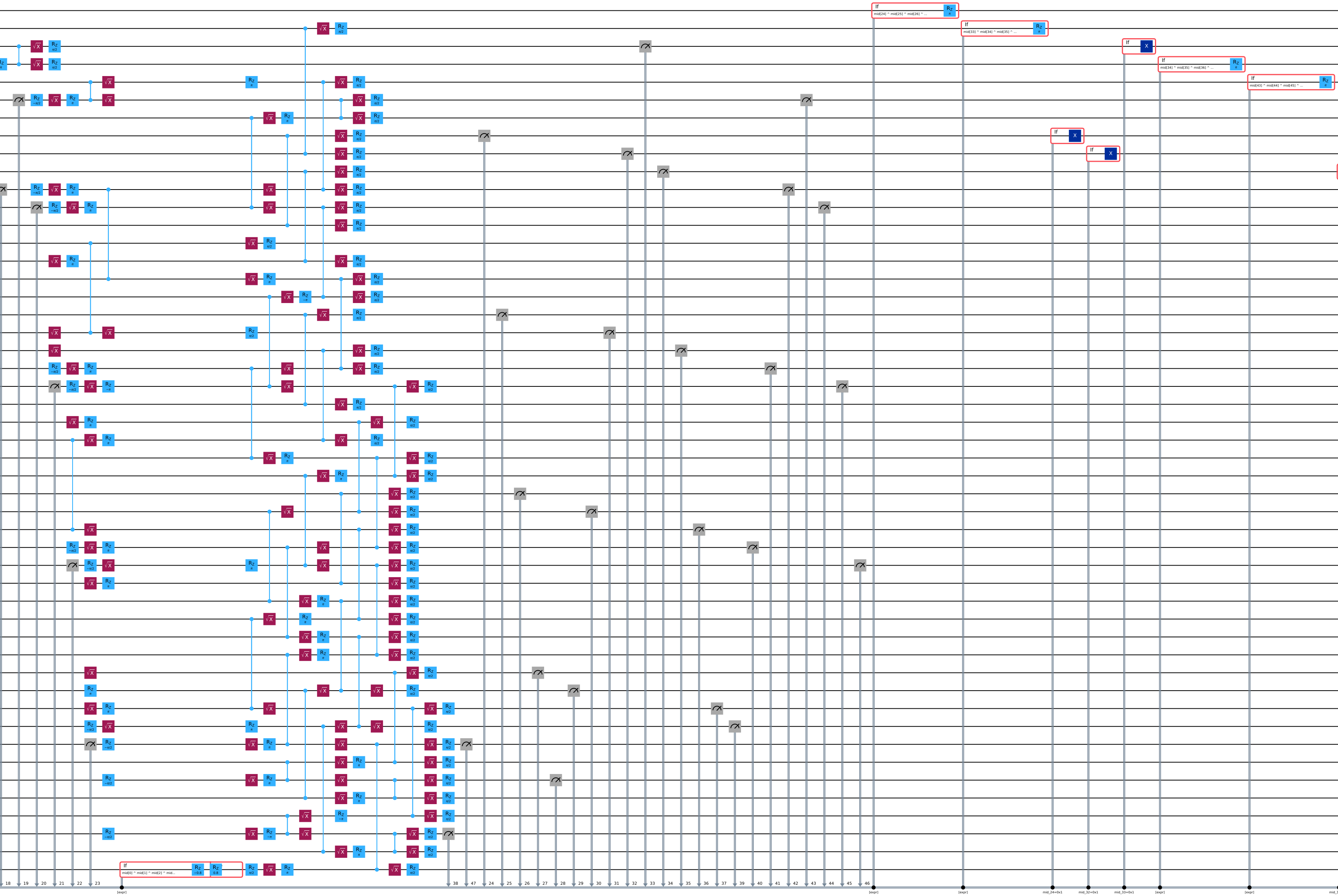

If
Z
mid[46] ^ mid[47]
If
X
If
Z
mid[27] ^ mid[28] ^ mid[29] ^ ...
If
Z
mid[30] ^ mid[31] ^ mid[32] ^ ...
If
X
If
X
If
Z
mid[37] ^ mid[38] ^ mid[39] ^ ...
If
Z
mid[40] ^ mid[41] ^ mid[42] ^ ...
If
X
If
X
If
Z
If
X
If
Z
mid[28] ^ mid[29] ^ mid[30] ^ ...
If
X
If
Z
mid[29] ^ mid[30] ^ mid[31] ^ ...
If
Z
mid[38] ^ mid[39] ^ mid[40] ^ ...
If
X
If
Z
mid[39] ^ mid[40] ^ mid[41] ^ ...
xpr]
mid_46=0x1
[expr]
[expr]
mid_27=0x1
mid_29=0x1
[expr]
[expr]
mid_37=0x1
mid_39=0x1
mid_47=0x1
mid_47=0x1
[expr]
mid_28=0x1
[expr]
[expr]
mid_38=0x1
[expr]

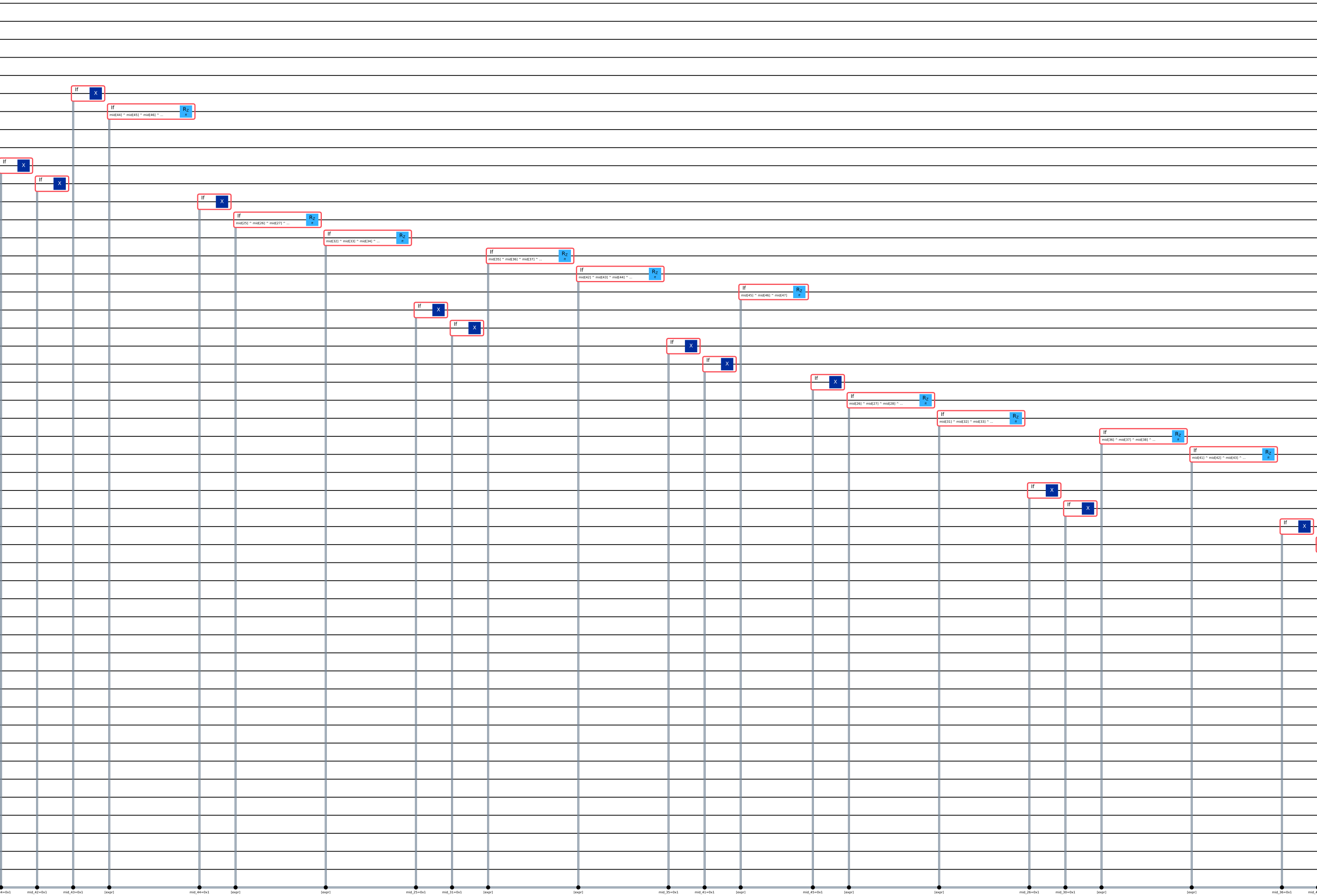
[Z25] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48

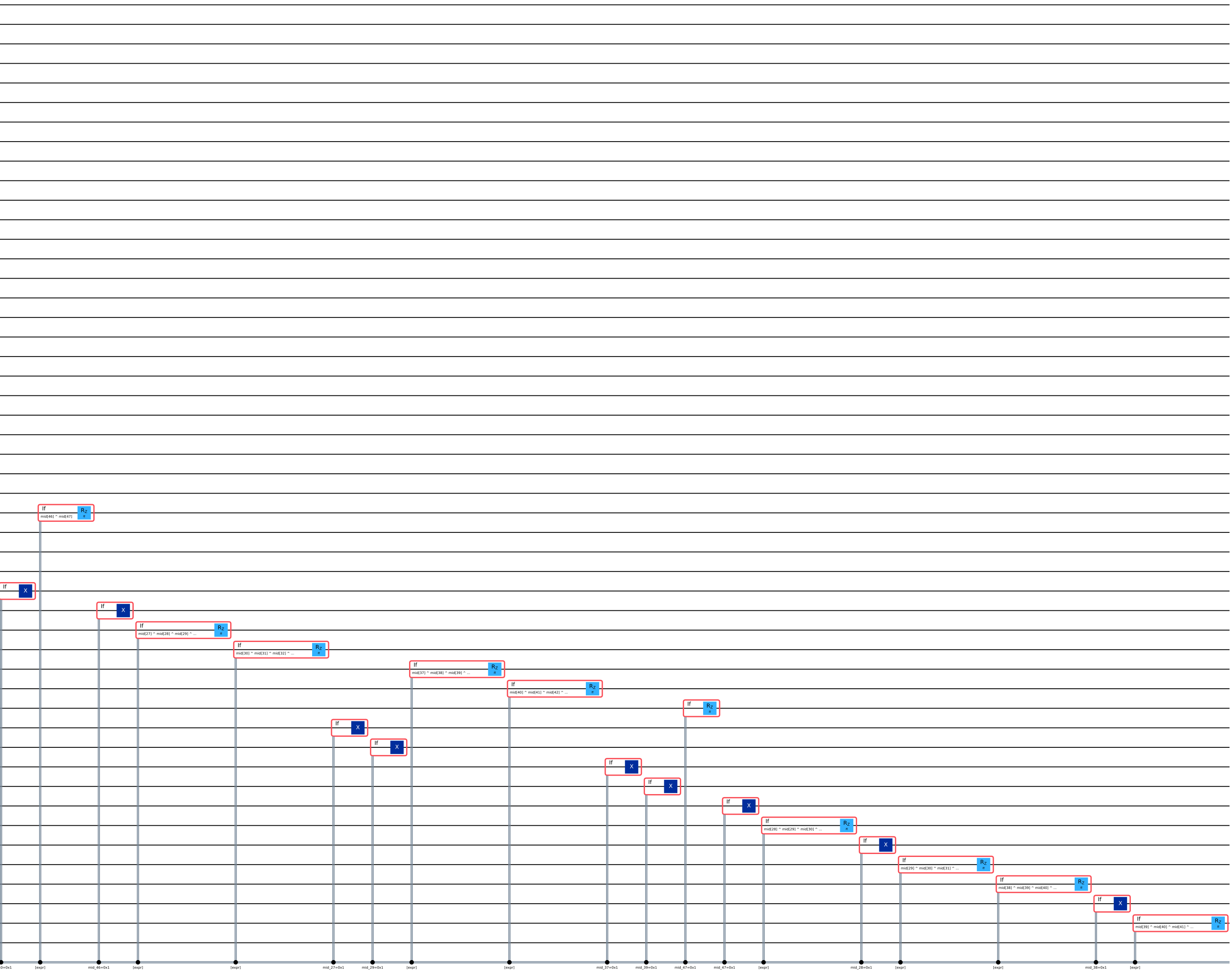


[Z25] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:240 rz:219 cz:96 if_else:49 measure:48)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 4, total depth 47


Global Phase: 3π/2
$q_{20} \mapsto 20$
$q_{22} \mapsto 22$
$q_{23} \mapsto 23$
$q_{24} \mapsto 24$
$q_{26} \mapsto 26$
$q_{27} \mapsto 27$
$q_{28} \mapsto 28$
$q_{30} \mapsto 30$
$q_{32} \mapsto 32$
$q_{34} \mapsto 34$
$q_{36} \mapsto 36$
$q_{38} \mapsto 38$
$q_{40} \mapsto 40$
$q_{42} \mapsto 42$
$q_{44} \mapsto 44$
$q_{46} \mapsto 46$
$q_{48} \mapsto 48$
$q_{50} \mapsto 50$
$q_{52} \mapsto 52$
$q_{54} \mapsto 54$
$q_{56} \mapsto 56$
$q_{58} \mapsto 58$
$q_{60} \mapsto 60$
$q_{62} \mapsto 62$
$q_{64} \mapsto 64$
$q_{66} \mapsto 66$
$q_{68} \mapsto 68$
$q_{70} \mapsto 70$
$q_{72} \mapsto 72$
$q_{74} \mapsto 74$
$q_{76} \mapsto 76$
$q_{78} \mapsto 78$
$q_{80} \mapsto 80$
$q_{82} \mapsto 82$
$q_{84} \mapsto 84$
$q_{86} \mapsto 86$
$q_{88} \mapsto 88$
$q_{90} \mapsto 90$
$q_{92} \mapsto 92$
$q_{94} \mapsto 94$
$q_{96} \mapsto 96$
$q_{98} \mapsto 98$
$q_{100} \mapsto 100$
$q_{101} \mapsto 101$
$q_{102} \mapsto 102$
$q_{104} \mapsto 104$
$q_{105} \mapsto 105$
$q_{106} \mapsto 106$
$q_{108} \mapsto 108$
mid
48
R_Z
π/2
−π/2
π
$\sqrt{X}$
14
0
1
2
3
4
5
6
7
8
9
10
11
12
13
15
16
17

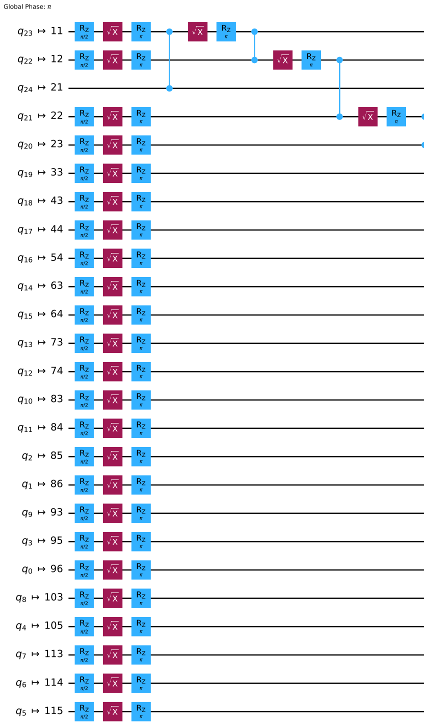

If
mid[24] ^ mid[25] ^ mid[26] ^ ...
If
mid[33] ^ mid[34] ^ mid[35] ^ ...
If
X
If
mid[34] ^ mid[35] ^ mid[36] ^ ...
If
mid[43] ^ mid[44] ^ mid[45] ^ ...
If
X
If
X
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z
−0.8
R_Z
0.8
R_Z
π/2
−π/2
π
−π
√X
18
19
20
21
22
23
38
47
24
25
26
27
28
29
30
31
32
33
34
35
36
37
39
40
41
42
43
44
45
46
[expr]
[expr]
[expr]
mid_24=0x1
mid_32=0x1
mid_33=0x1
[expr]
[expr]

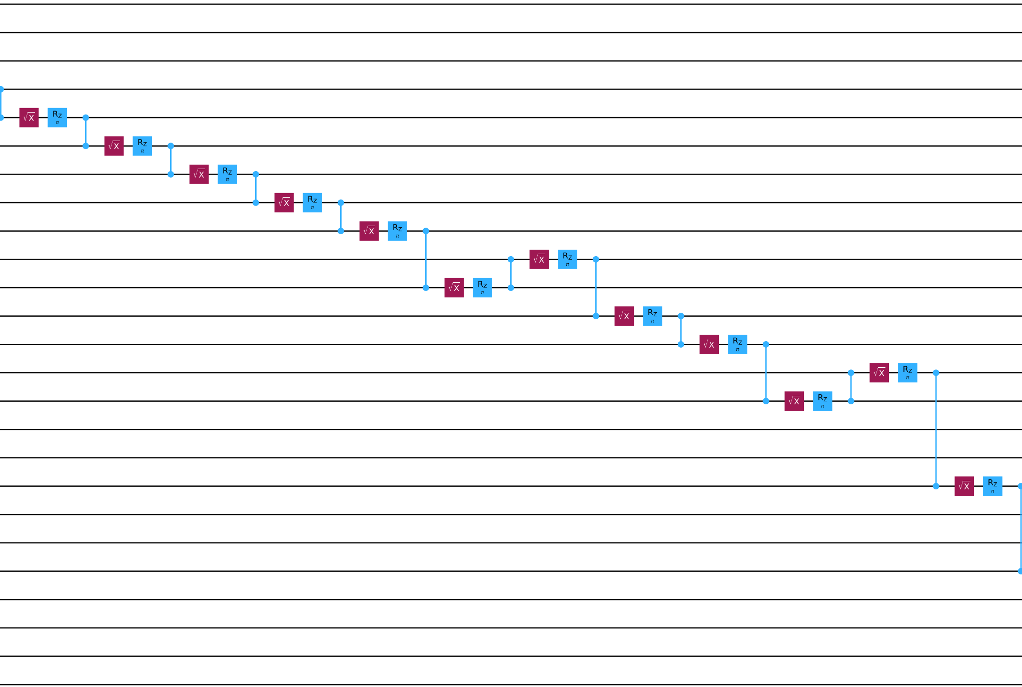

If
X
If
X
If
X
If
mid[44] ^ mid[45] ^ mid[46] ^ ...
R_Z
π
If
X
If
mid[25] ^ mid[26] ^ mid[27] ^ ...
R_Z
π
If
mid[32] ^ mid[33] ^ mid[34] ^ ...
R_Z
π
If
X
If
X
If
mid[35] ^ mid[36] ^ mid[37] ^ ...
R_Z
π
If
mid[42] ^ mid[43] ^ mid[44] ^ ...
R_Z
π
If
X
If
X
If
mid[45] ^ mid[46] ^ mid[47]
R_Z
π
If
X
If
mid[26] ^ mid[27] ^ mid[28] ^ ...
R_Z
π
If
mid[31] ^ mid[32] ^ mid[33] ^ ...
R_Z
π
If
X
If
X
If
mid[36] ^ mid[37] ^ mid[38] ^ ...
R_Z
π
If
mid[41] ^ mid[42] ^ mid[43] ^ ...
R_Z
π
If
X
4=0x1
mid_42=0x1
mid_43=0x1
[expr]
mid_44=0x1
[expr]
[expr]
mid_25=0x1
mid_31=0x1
[expr]
[expr]
mid_35=0x1
mid_41=0x1
[expr]
mid_45=0x1
[expr]
[expr]
mid_26=0x1
mid_30=0x1
[expr]
[expr]
mid_36=0x1
mid_4

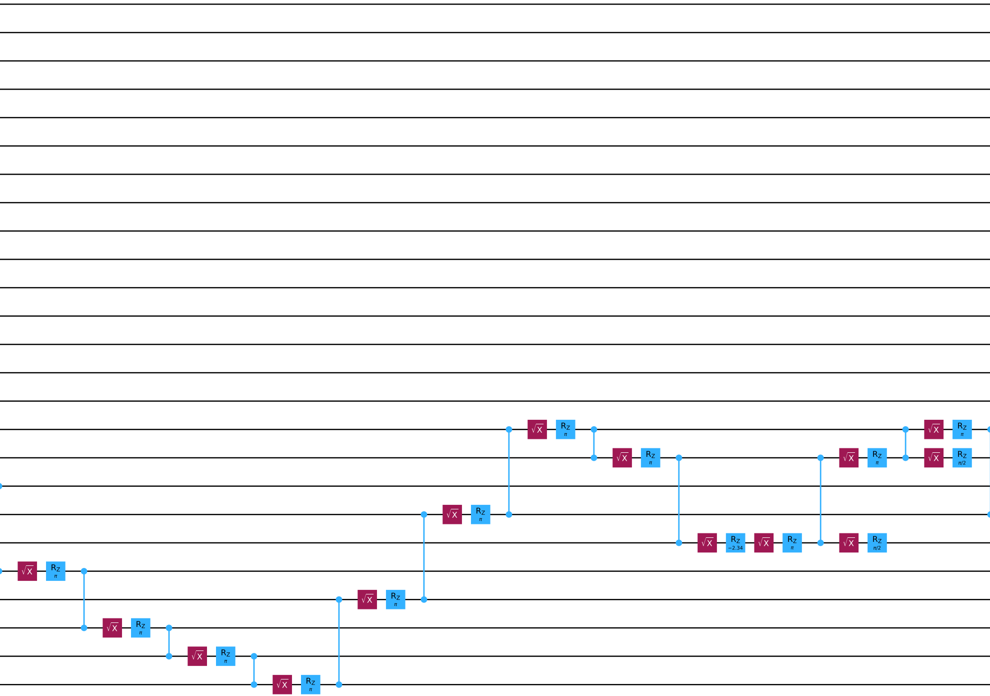


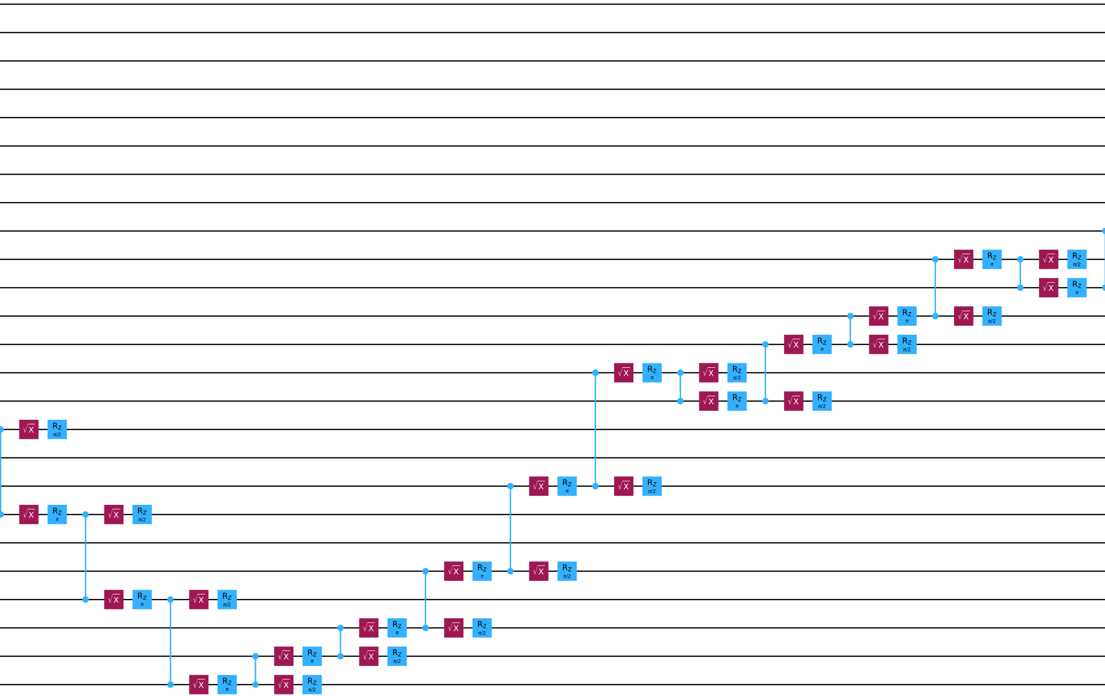
If
X
If
mid[46] ^ mid[47]
R_Z
π
If
X
If
mid[27] ^ mid[28] ^ mid[29] ^ ...
R_Z
π
If
mid[30] ^ mid[31] ^ mid[32] ^ ...
R_Z
π
If
X
If
X
If
mid[37] ^ mid[38] ^ mid[39] ^ ...
R_Z
π
If
mid[40] ^ mid[41] ^ mid[42] ^ ...
R_Z
π
If
X
If
X
If
R_Z
π
If
X
If
mid[28] ^ mid[29] ^ mid[30] ^ ...
R_Z
π
If
X
If
mid[29] ^ mid[30] ^ mid[31] ^ ...
R_Z
π
If
mid[38] ^ mid[39] ^ mid[40] ^ ...
R_Z
π
If
X
If
mid[39] ^ mid[40] ^ mid[41] ^ ...
R_Z
π
0=0x1
[expr]
mid_46=0x1
[expr]
[expr]
mid_27=0x1
mid_29=0x1
[expr]
[expr]
mid_37=0x1
mid_39=0x1
mid_47=0x1
mid_47=0x1
[expr]
mid_28=0x1
[expr]
[expr]
mid_38=0x1
[expr]

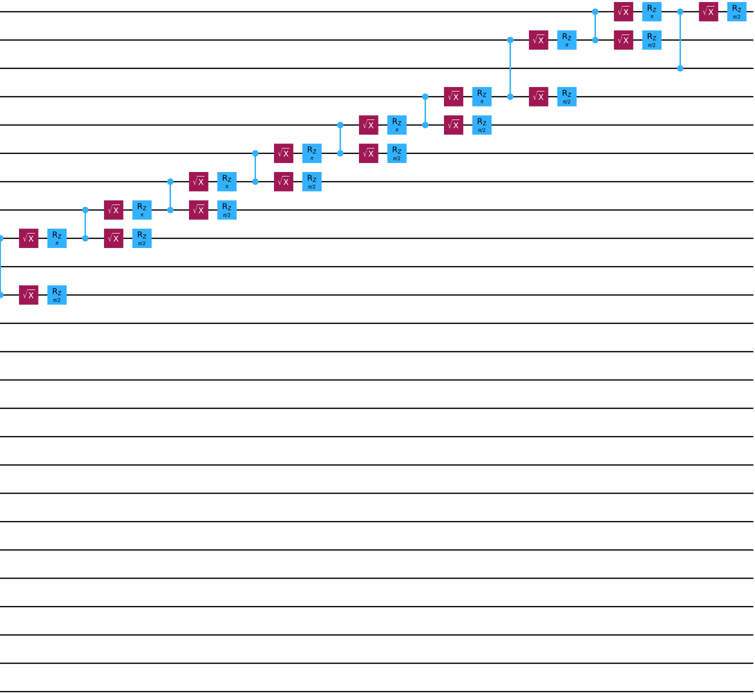
[Z25] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:120 sx:96 cz:48)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48, total depth 149


Global Phase: π
q_{23} ↦ 11
q_{22} ↦ 12
q_{24} ↦ 21
q_{21} ↦ 22
q_{20} ↦ 23
q_{19} ↦ 33
q_{18} ↦ 43
q_{17} ↦ 44
q_{16} ↦ 54
q_{14} ↦ 63
q_{15} ↦ 64
q_{13} ↦ 73
q_{12} ↦ 74
q_{10} ↦ 83
q_{11} ↦ 84
q_2 ↦ 85
q_1 ↦ 86
q_9 ↦ 93
q_3 ↦ 95
q_0 ↦ 96
q_8 ↦ 103
q_4 ↦ 105
q_7 ↦ 113
q_6 ↦ 114
q_5 ↦ 115
R_Z π/2
√X
R_Z π


√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π
√X
RZ π

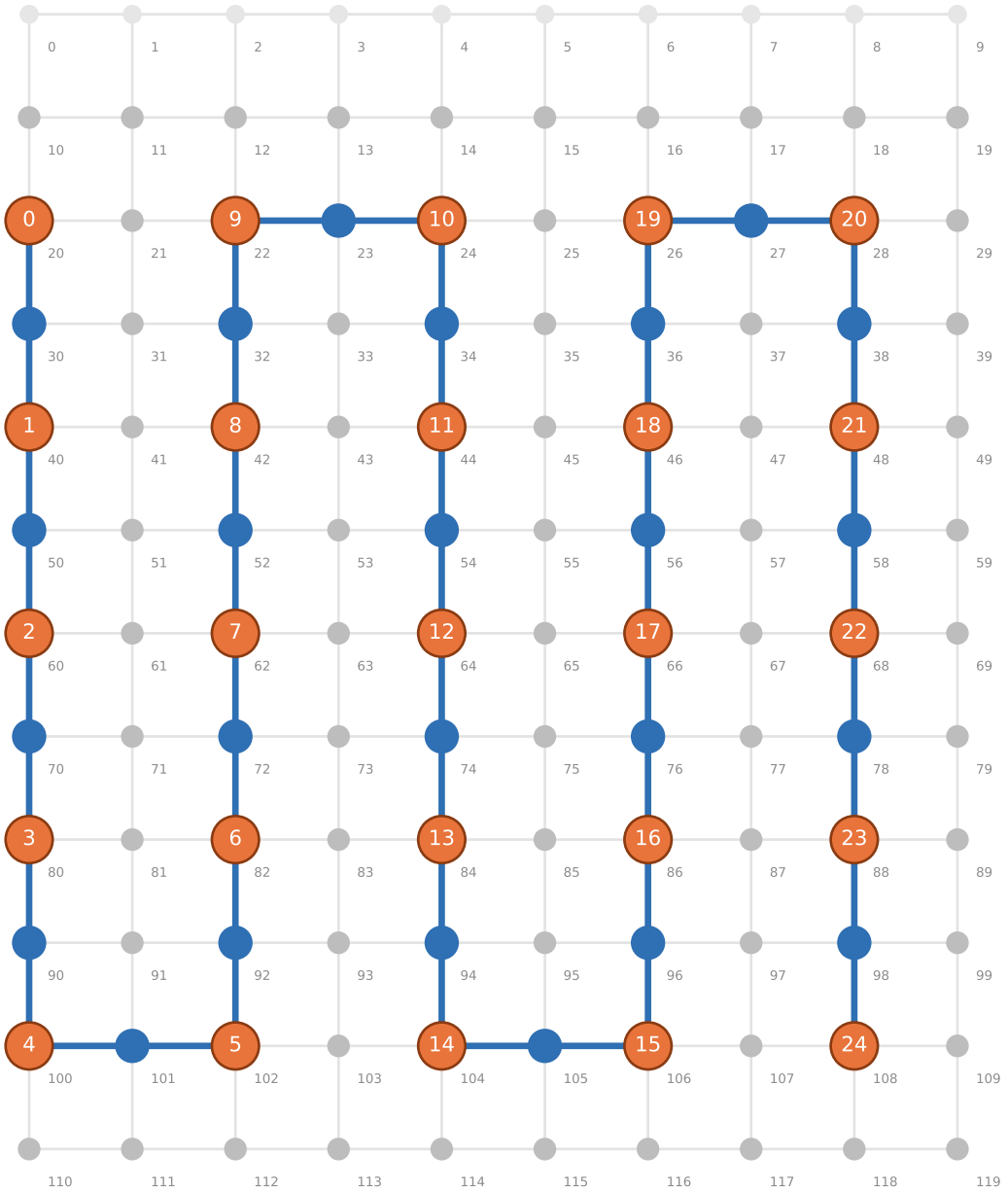




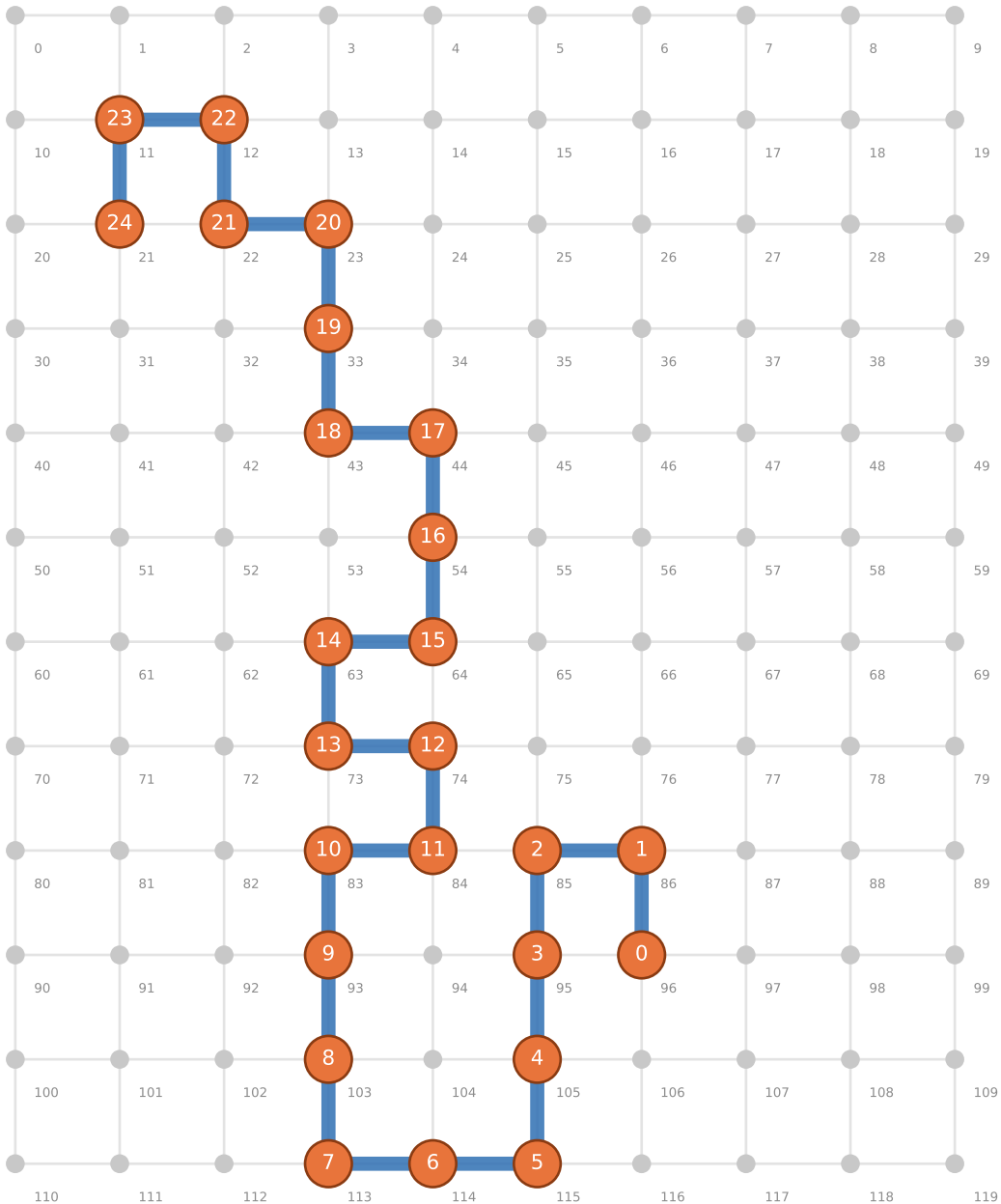


[Z25] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

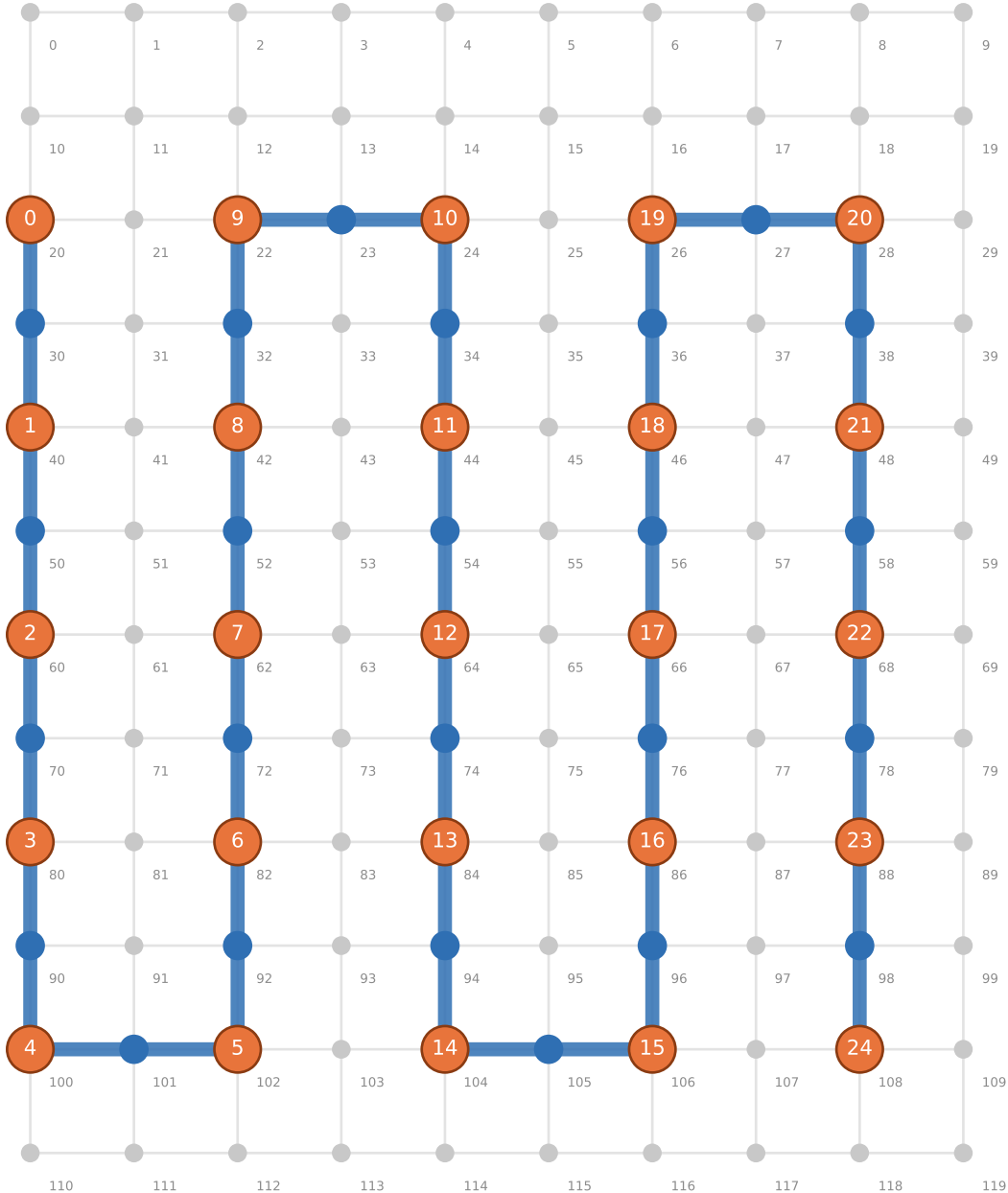
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers



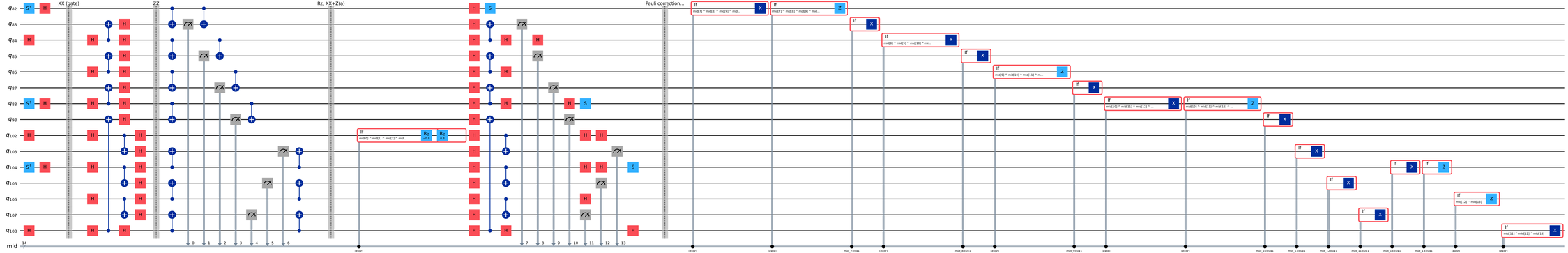
[Z25] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [96, 86, 85, 95, 105, 115, 114, 113, 103, 93, 83, 84, 74, 73, 63, 64, 54, 44, 43, 33, 23, 22, 12, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)



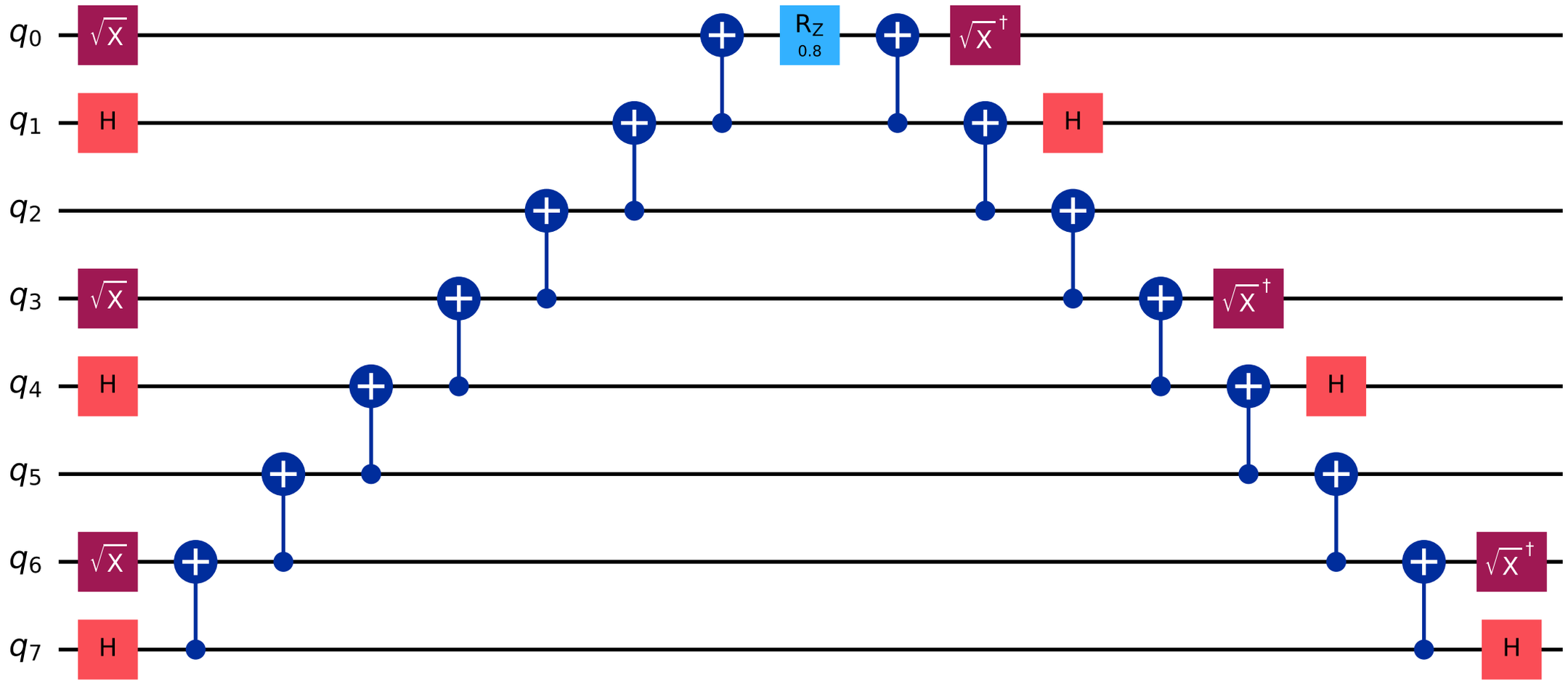
[Z25] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 96



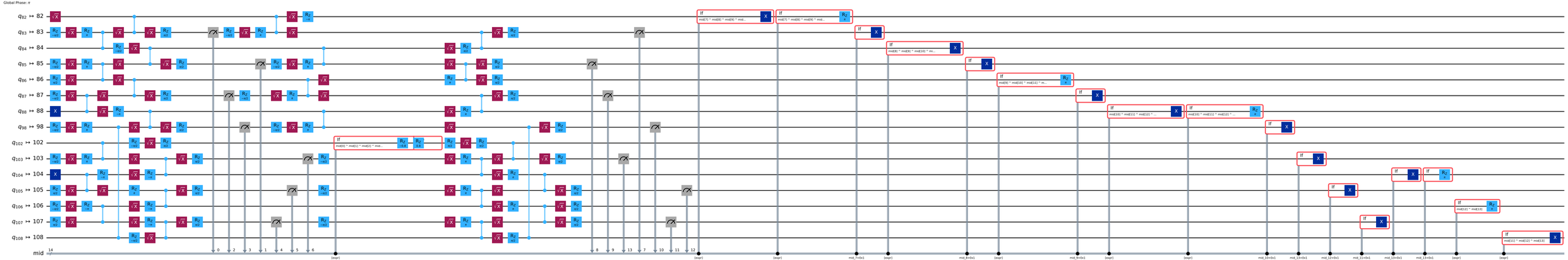
[XYZXYZXY] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 4, 14 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]



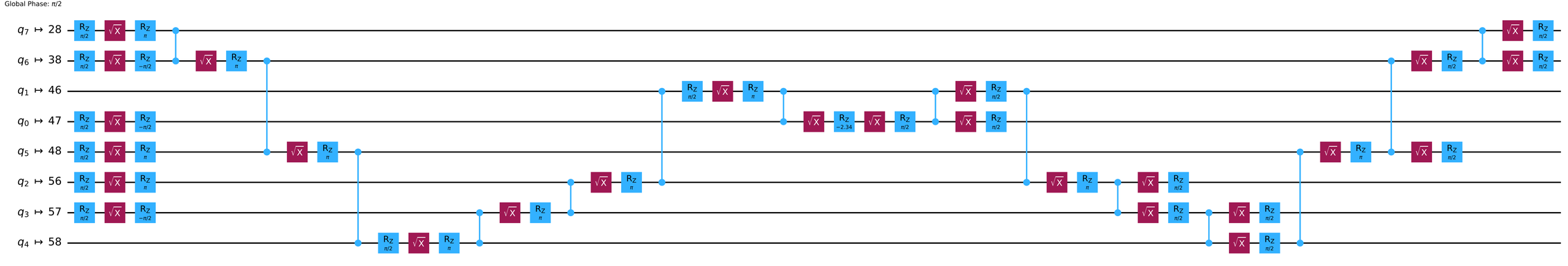
[XYZXYZXY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14

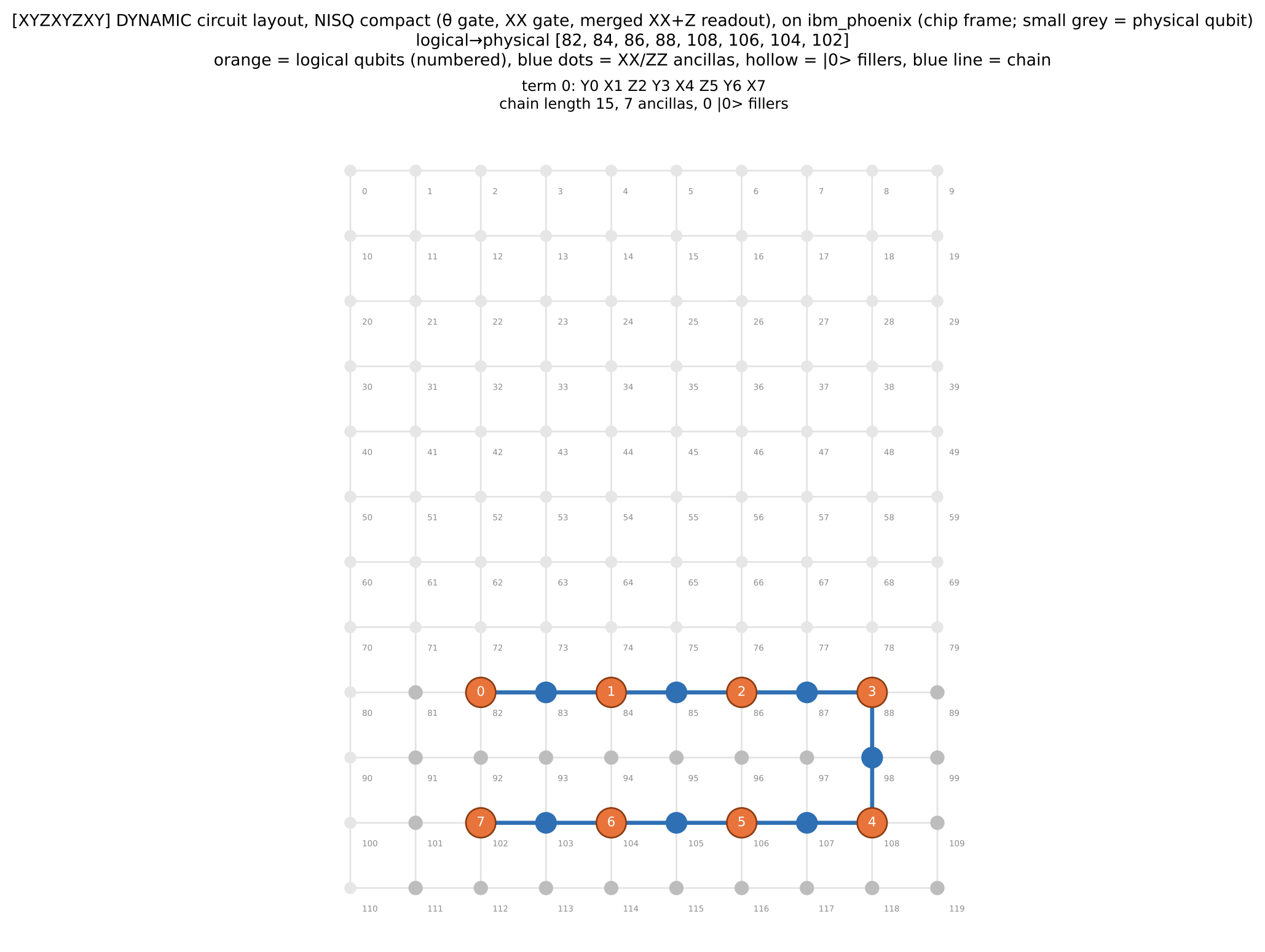


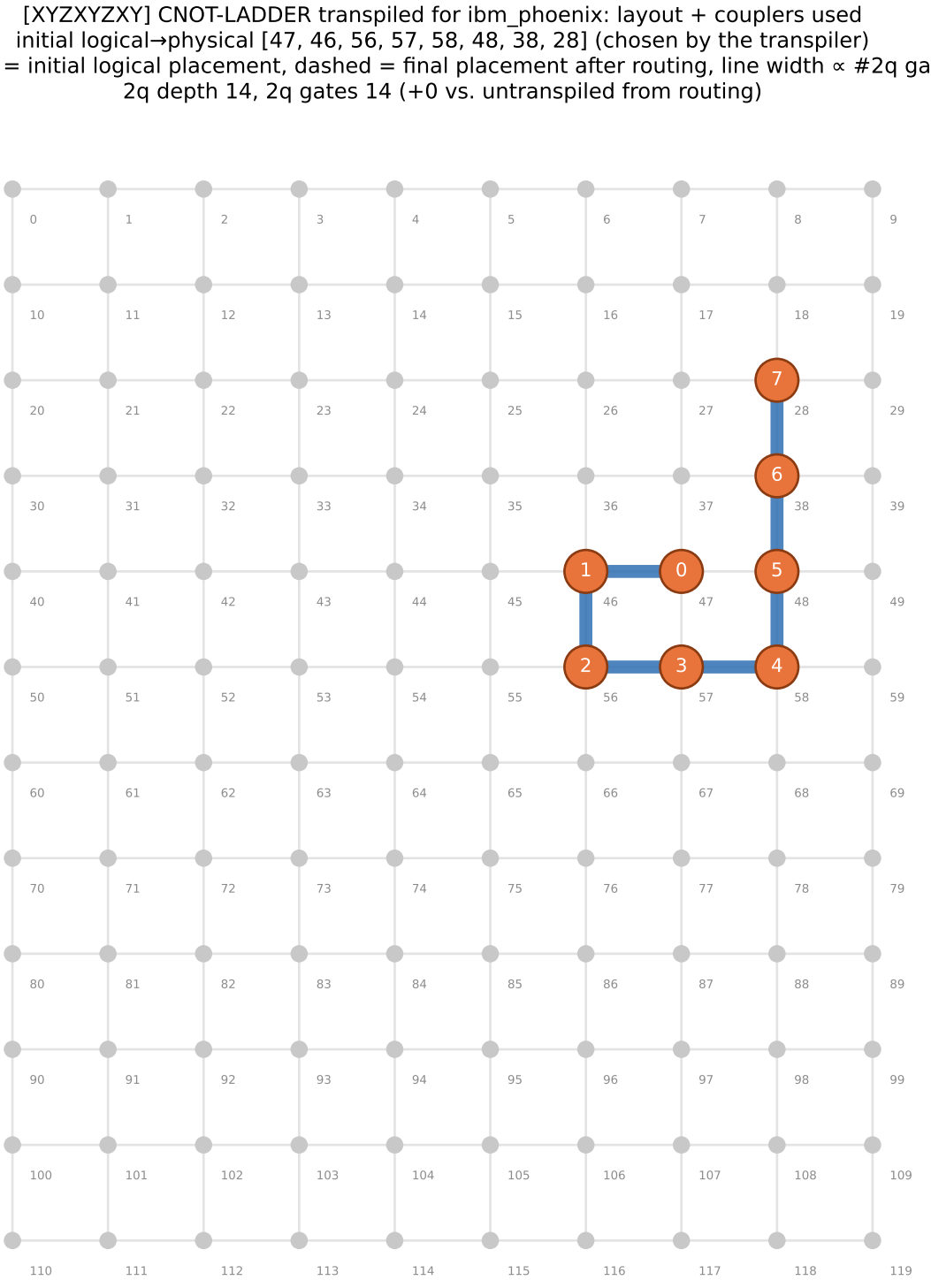
[XYZXYZXY] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: rz:63 sx:62 cz:28 if_else:18 measure:14 x:2)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 4, total depth 31



[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:34 sx:26 cz:14)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 49




[XYZXYZXY] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5
6
7


[XYZXYZXY] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [47, 46, 56, 57, 58, 48, 38, 28] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
7
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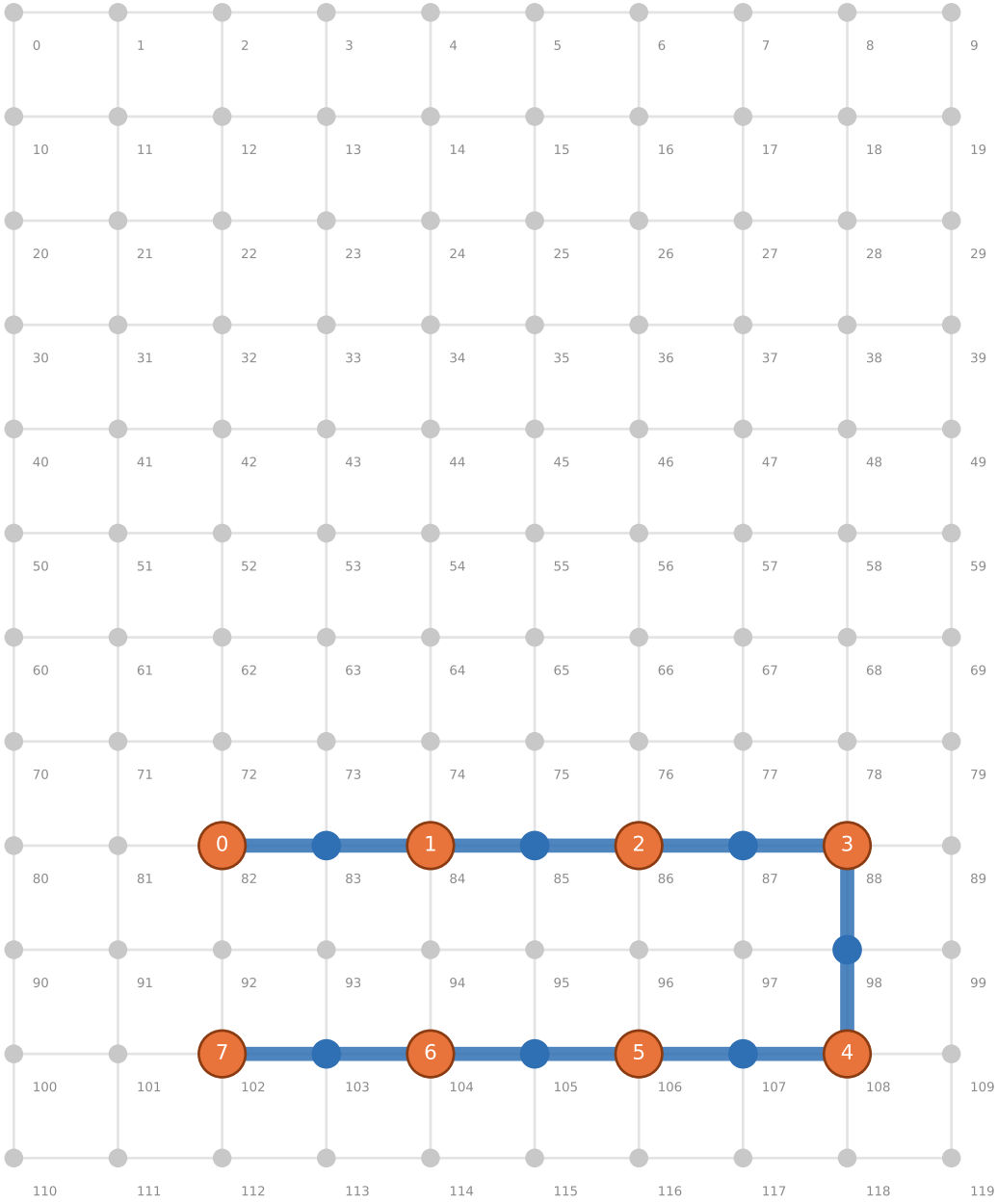
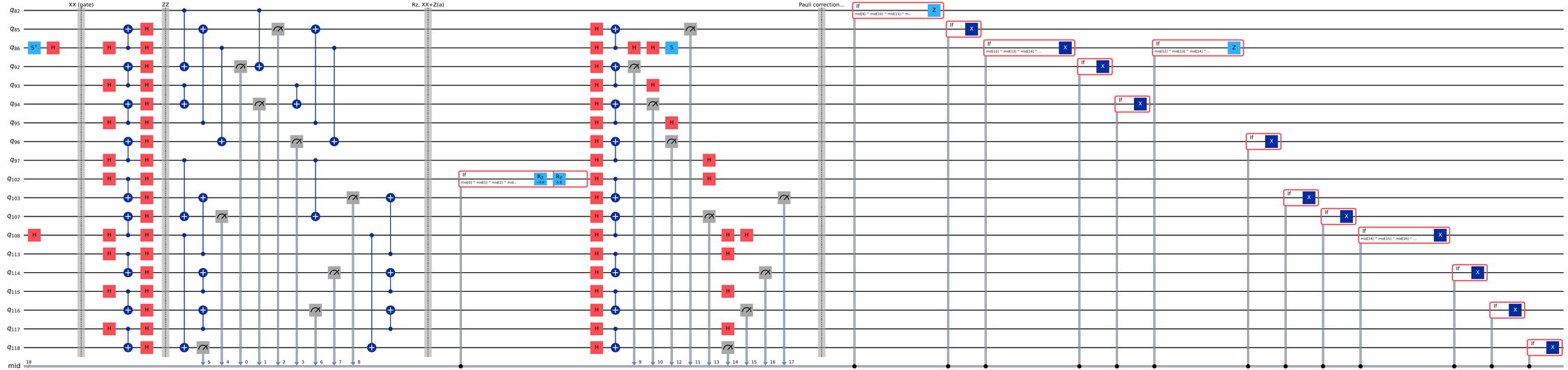
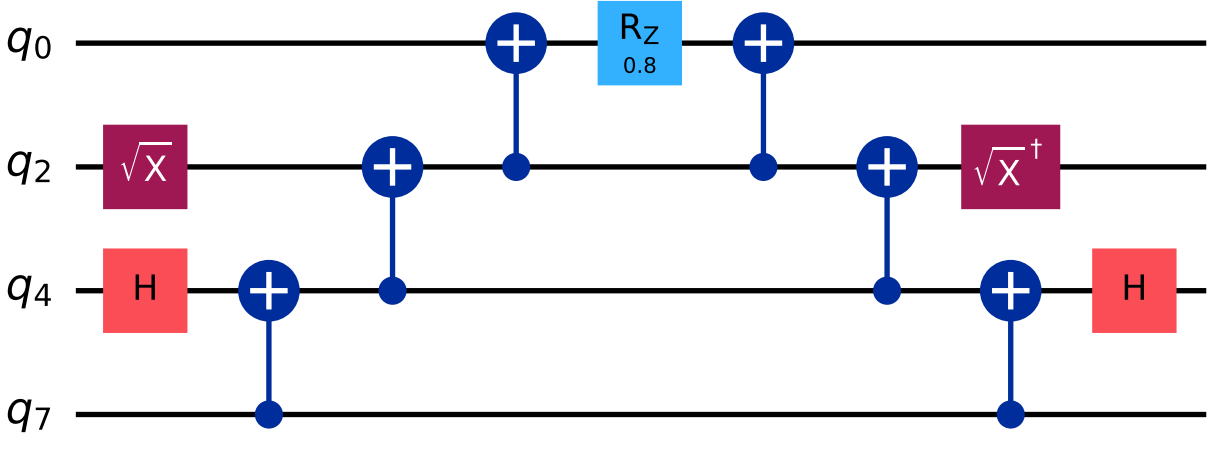
[XYZXYZXY] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 28



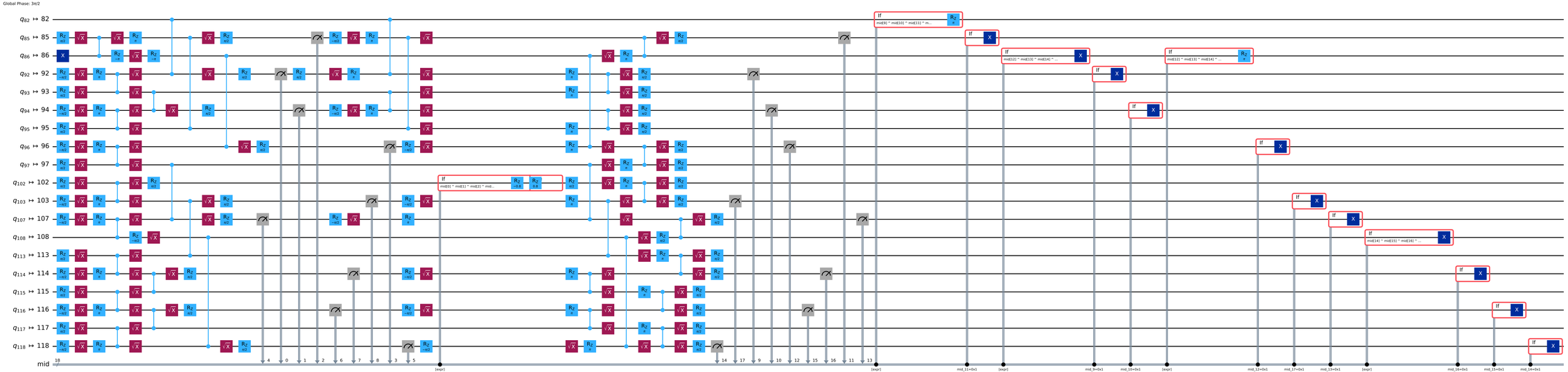
[ZIIXIYIZ_gapped] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 4, 18 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]



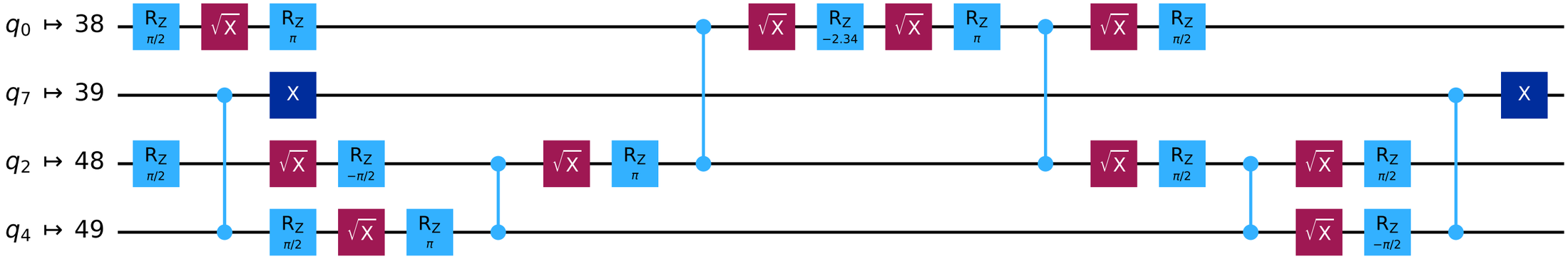
[ZIIXIYIZ_gapped] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6



[ZIIXIYIZ_gapped] DYNAMIC circuit, NISQ compact (8 gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:86 rz:83 cz:36 measure:18 if_else:14 x:1)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 4, total depth 26

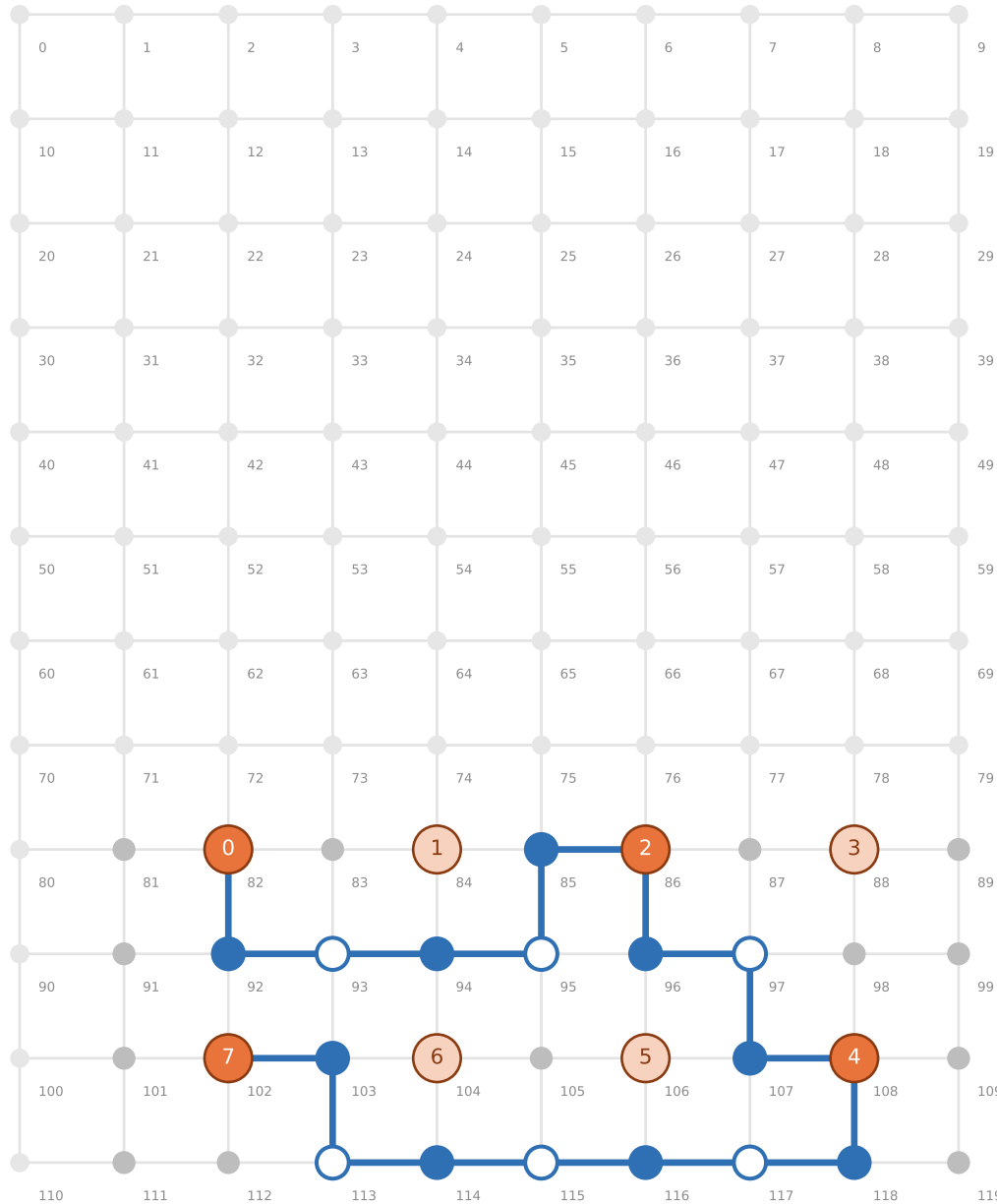


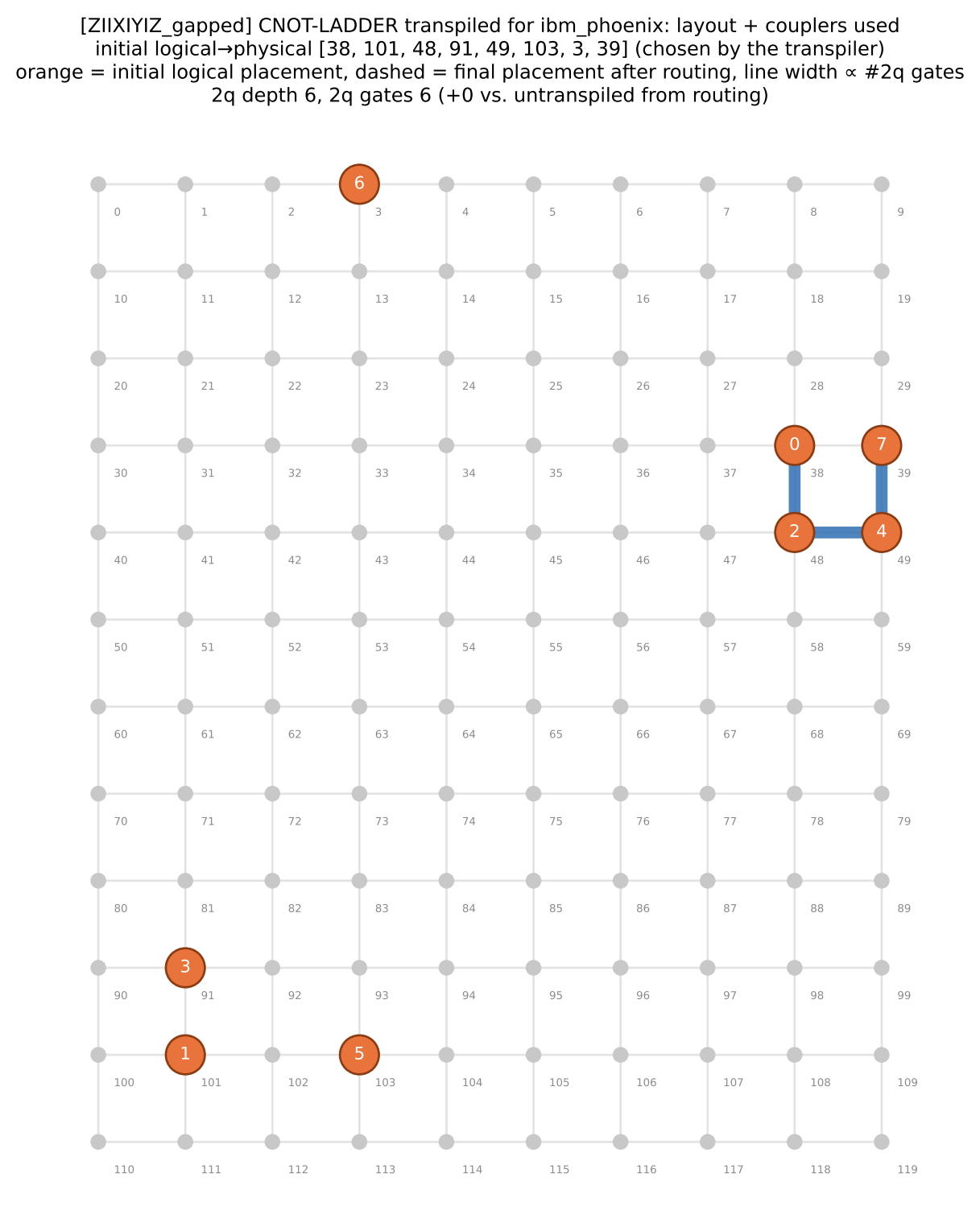
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:13 sx:10 cz:6 x:2)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7 (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 20



[ZIIXIYIZ_gapped] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers




[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 101, 48, 91, 49, 103, 3, 39] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
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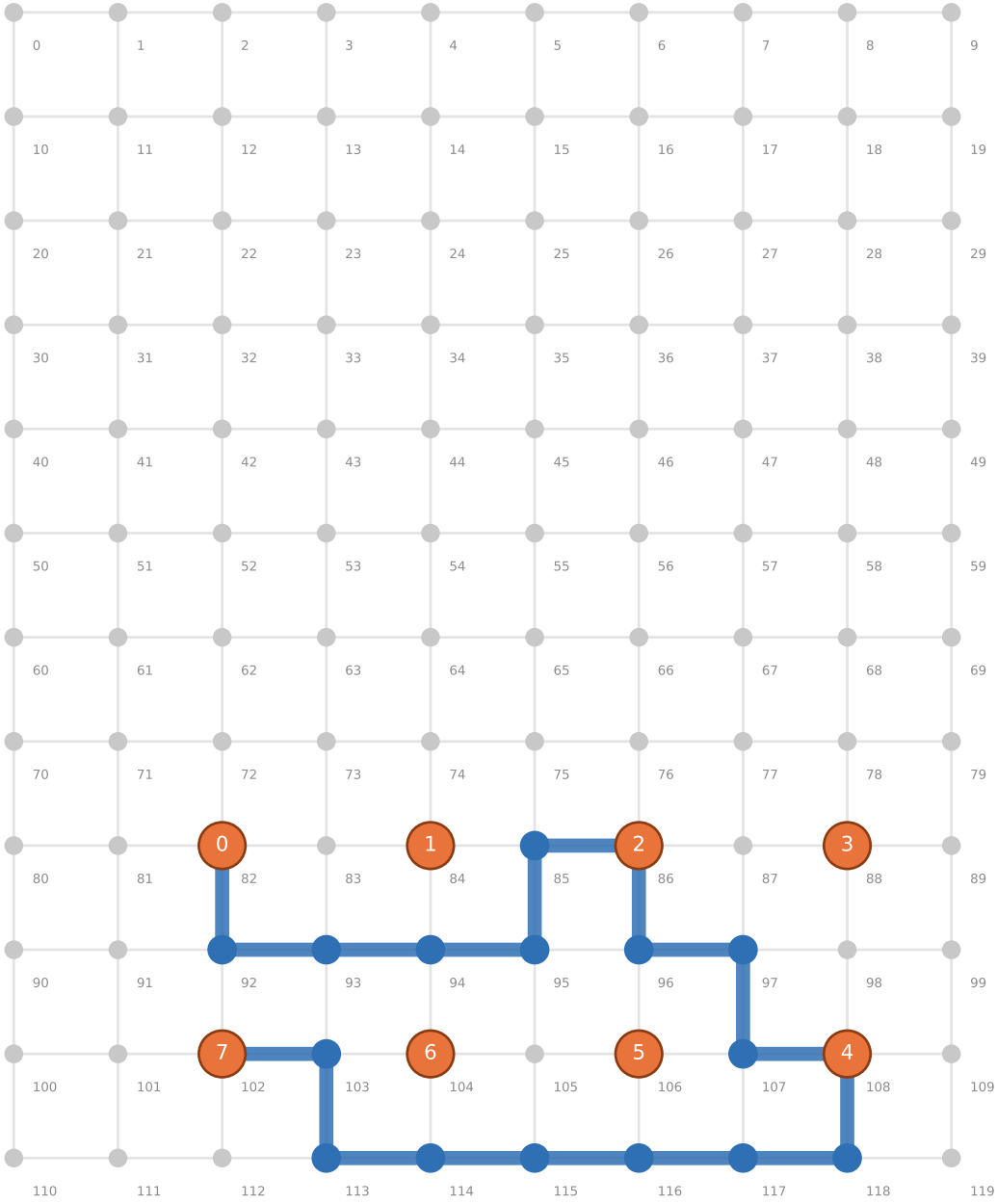
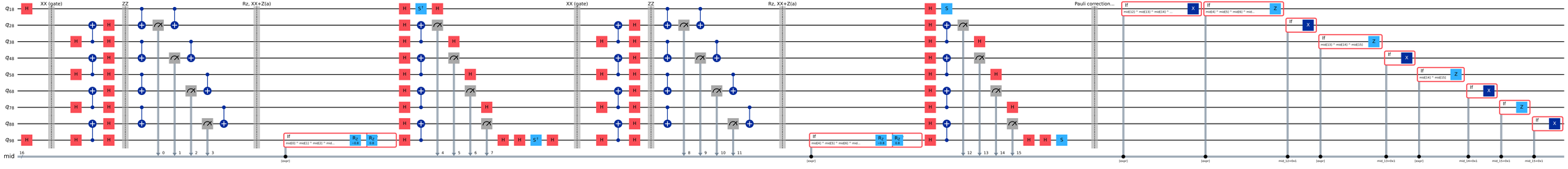
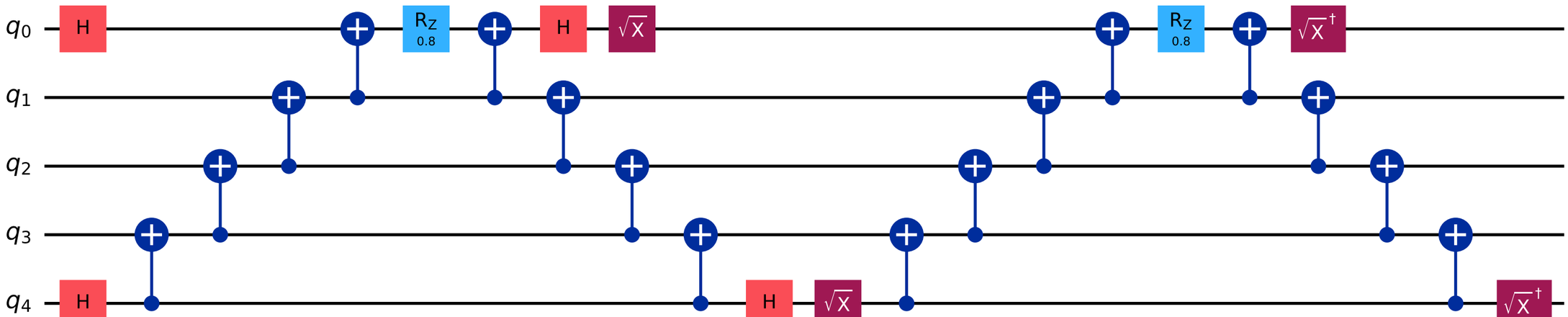
[ZIIXIYIZ_gapped] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 4, 2q gates 36



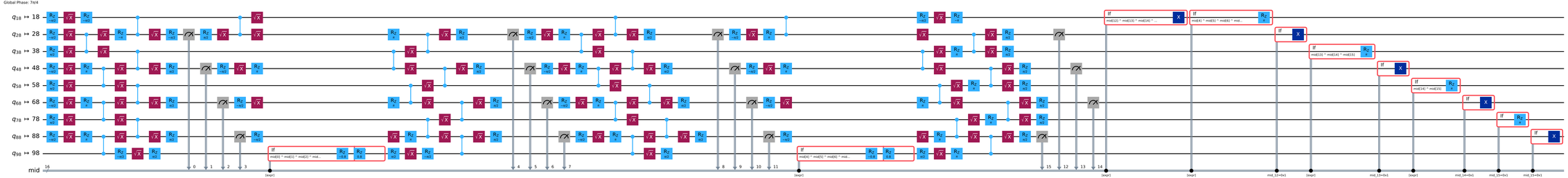
[XZZZX+YZZZY] DYNAMIC constant-depth circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 - Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 8, 16 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



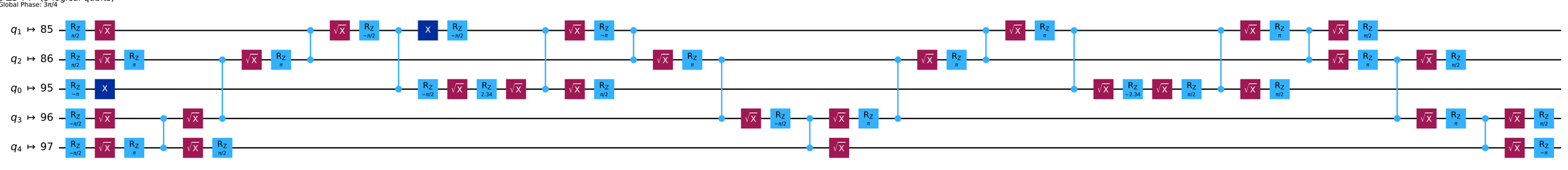
[XZZZX+YZZZY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16



[XZZZX+YZZZY] DYNAMIC circuit, NISQ compact (θ gate, XX gate, merged XX+Z readout), TRANSPILED for ibm_phoenix (ISA: sx:73 rz:68 cz:32 measure:16 if_else:11)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 - Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 8, total depth 43

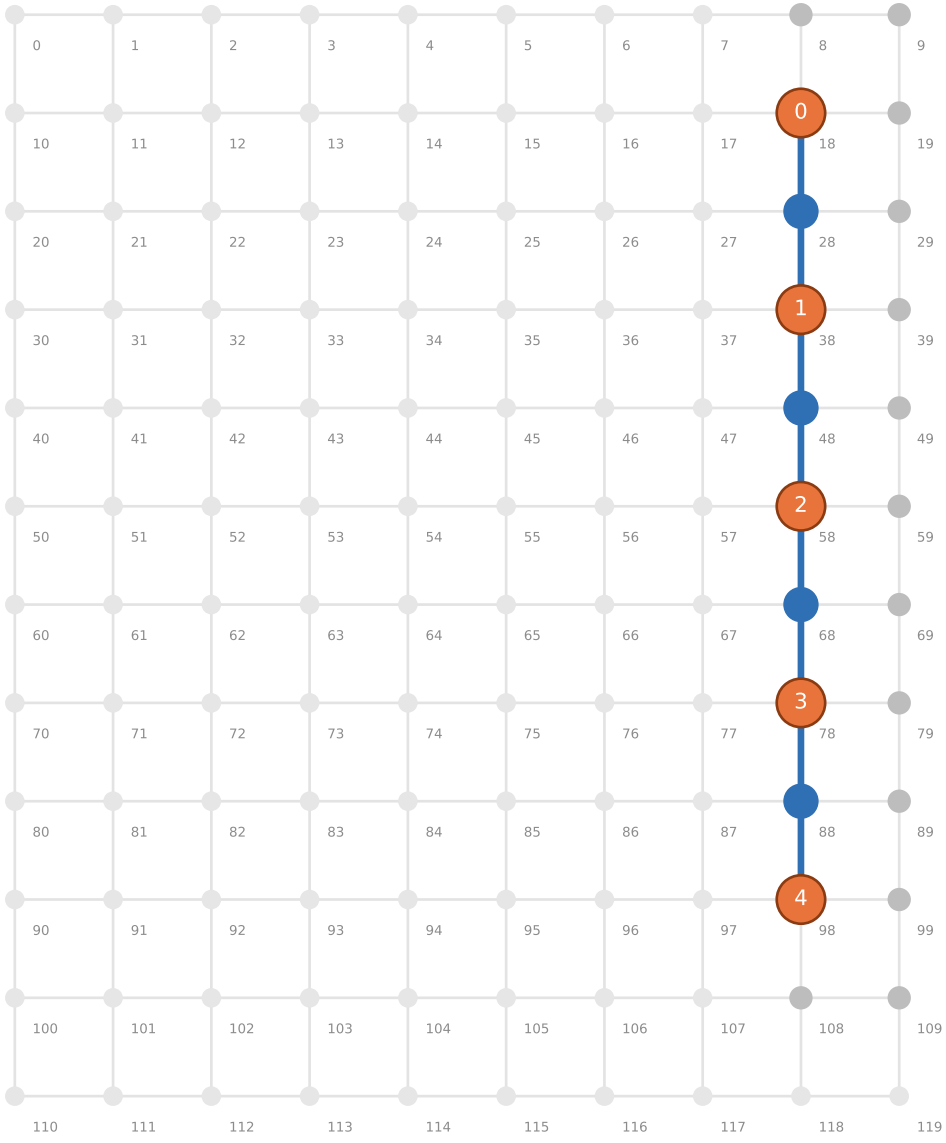


[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:30 sx:28 cz:15 x:2)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical qubits)
2q depth 15, total depth 51

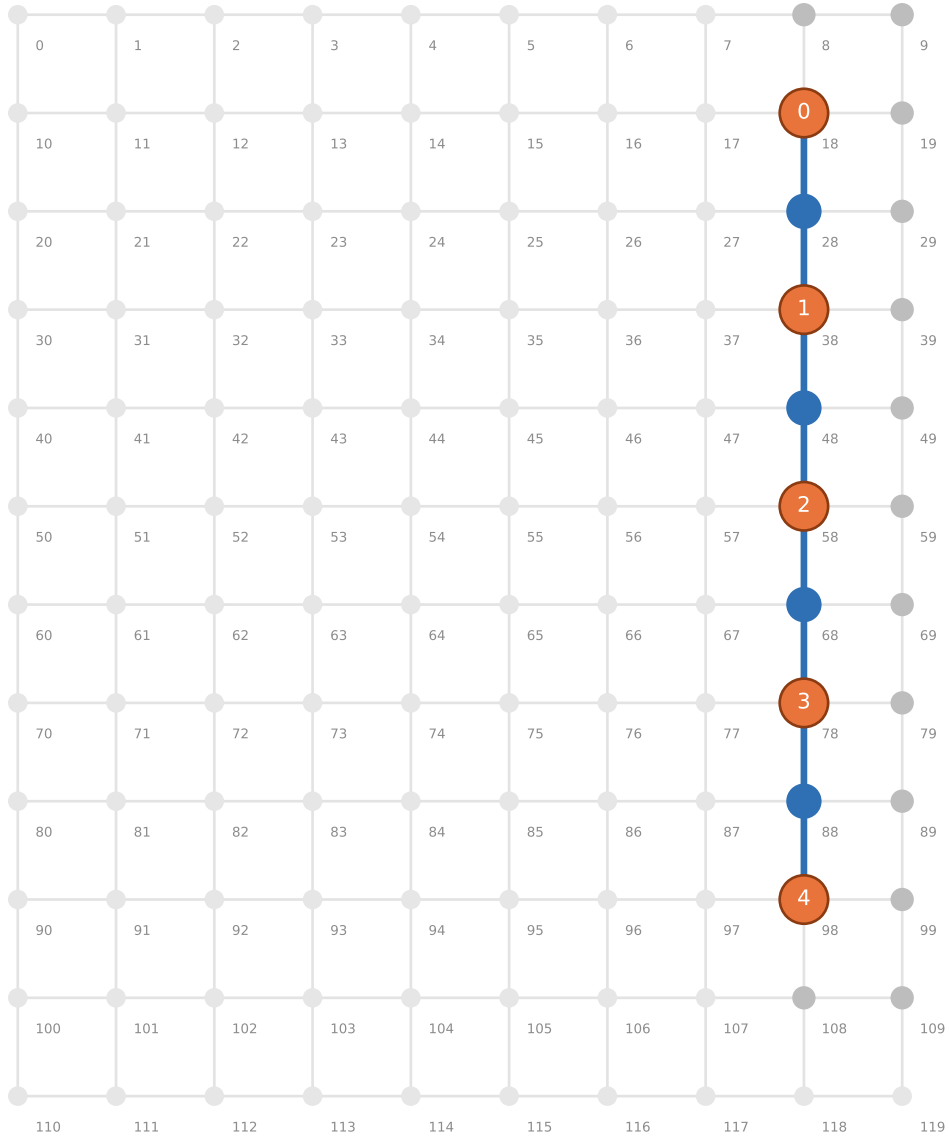


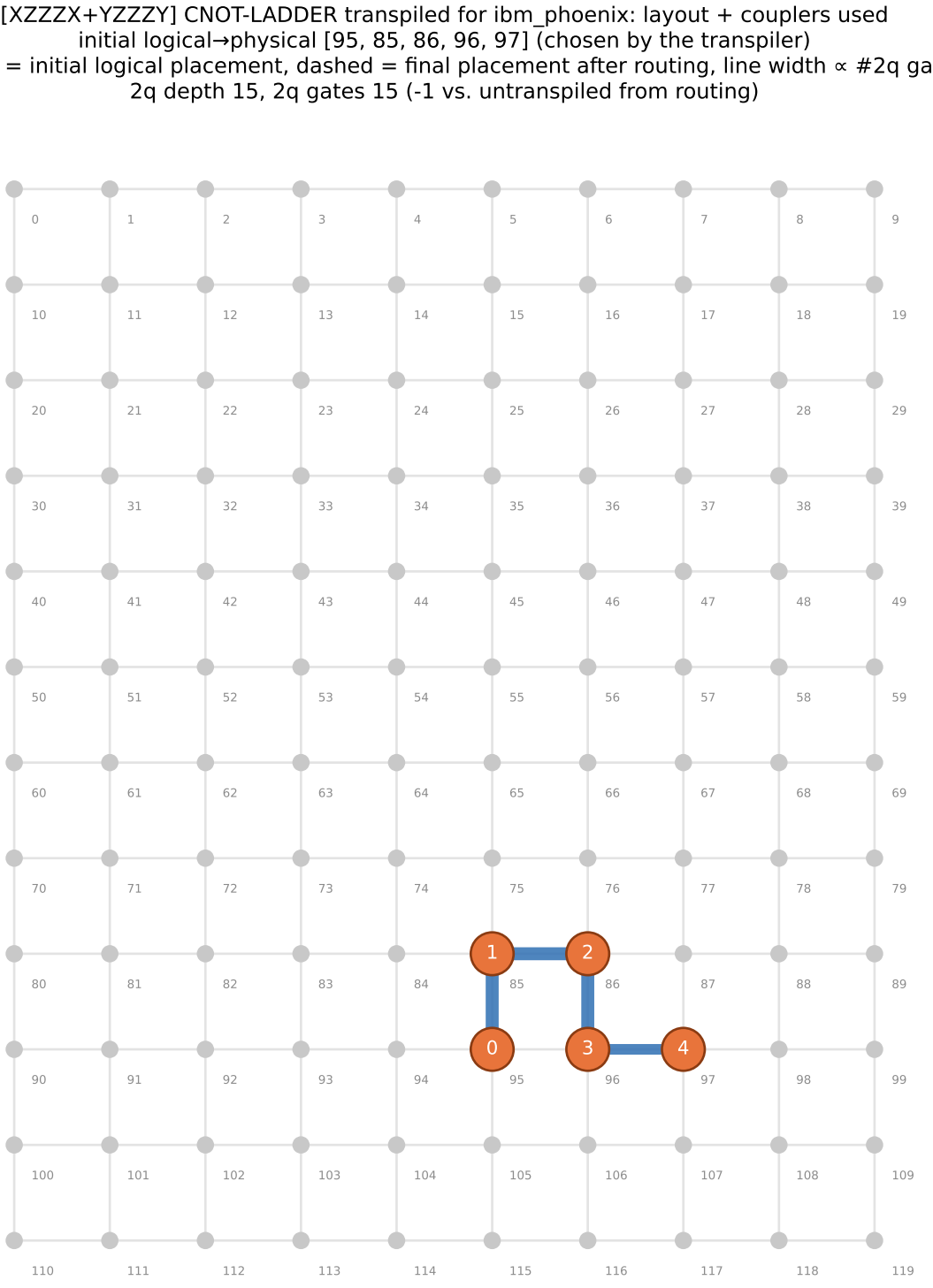
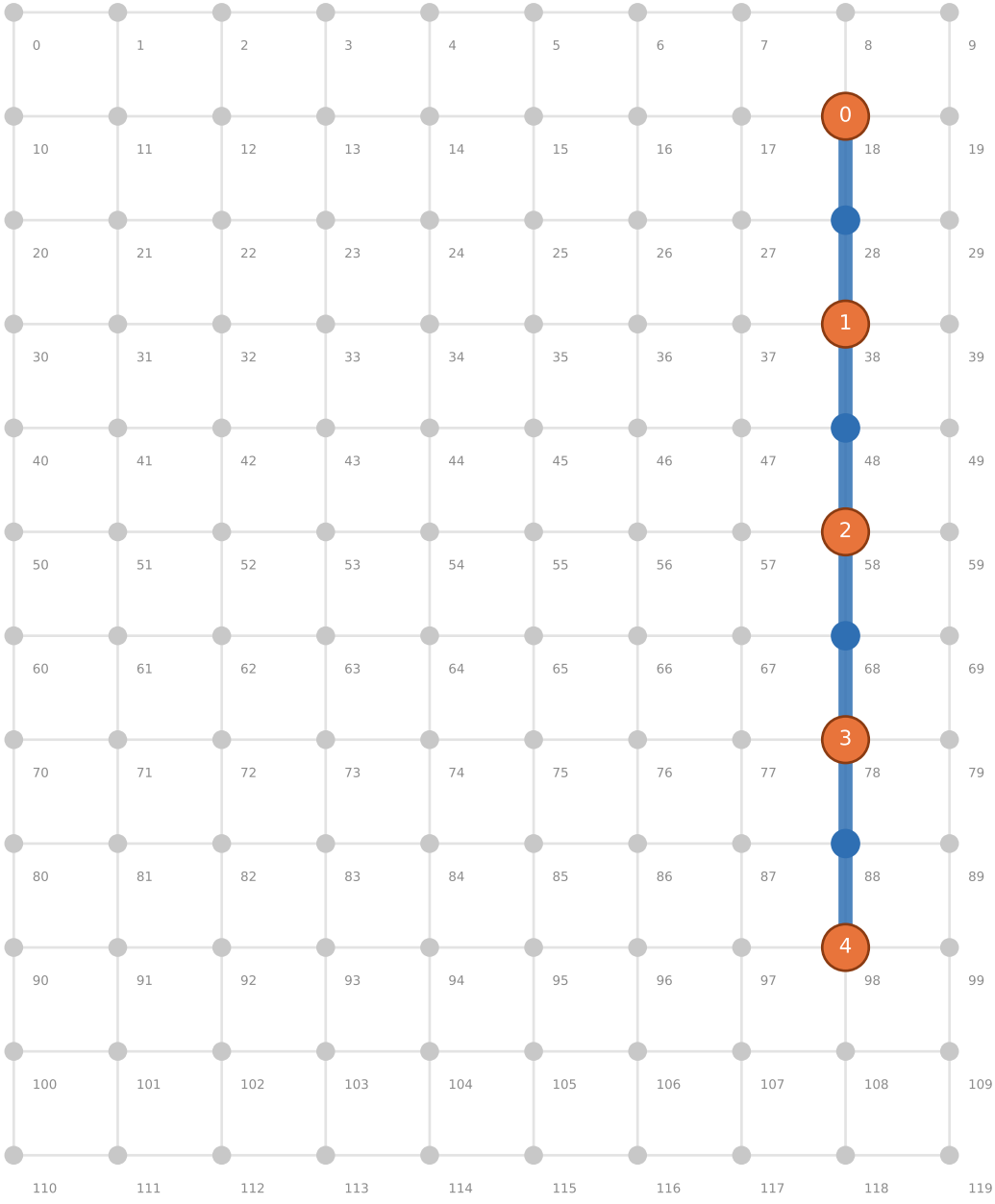
[XZZZX+YZZZY] DYNAMIC circuit layout, NISQ compact (θ gate, XX gate, merged XX+Z readout), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers




[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 86, 96, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
1
2
0
3
4

[XZZZX+YZZZY] DYNAMIC circuit (nisq_compact) transpiled for ibm_phoenix: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 8, 2q gates 32